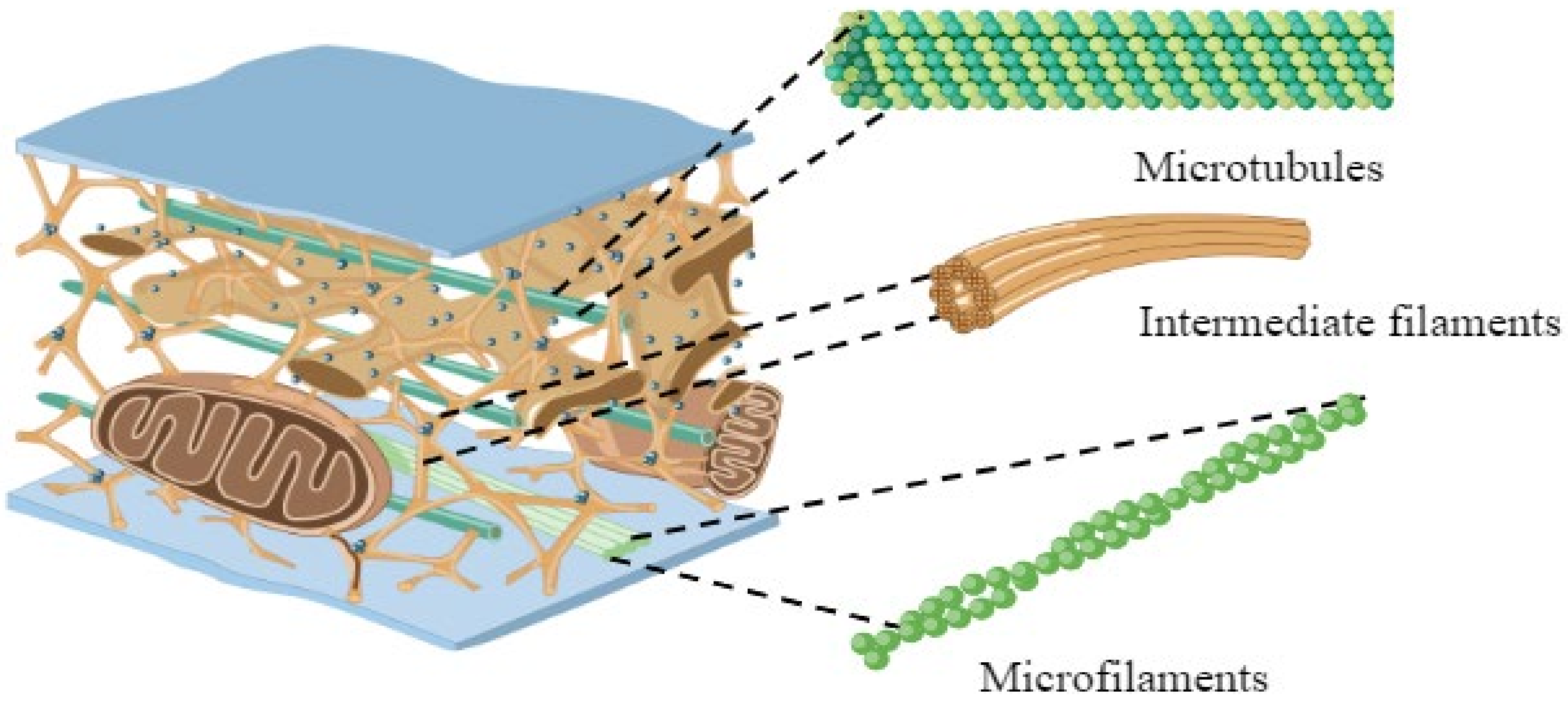
بسم الله الرحمن الرحيم

# Cytoskeleton: structure and function



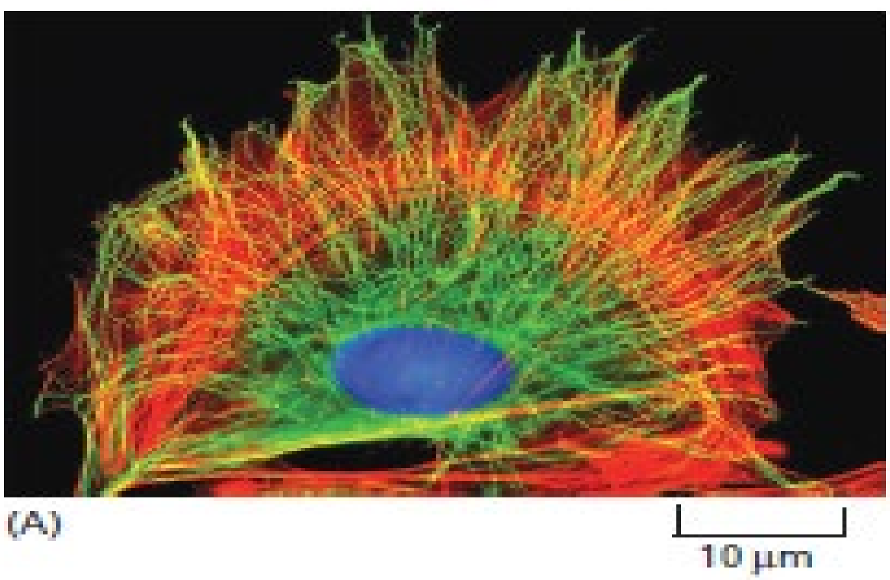
# Lecture Outline

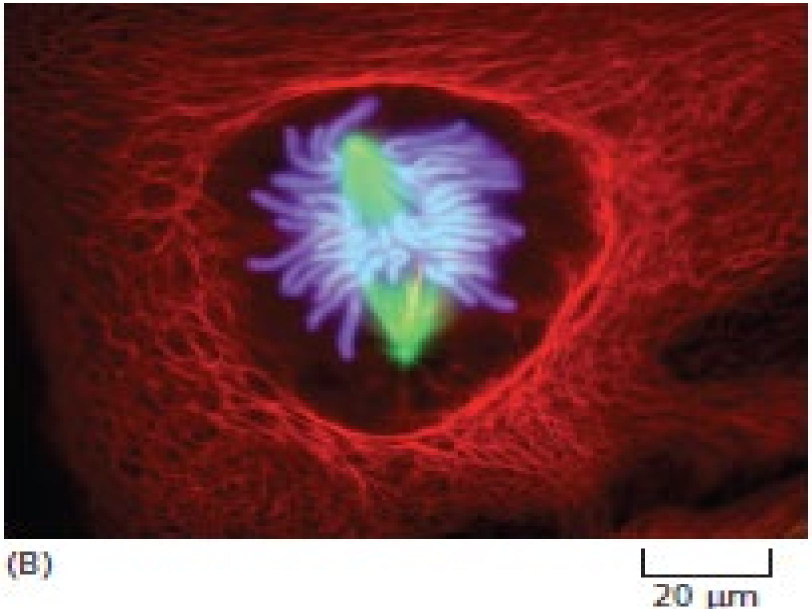
1. Introduction and Definition
2. The Three Types of Fibers
    - A. Microfilaments (Actin filaments)
    - B. Intermediate Filaments
    - C. Microtubules
3. Conclusion and summary

# 1. Introduction and definition

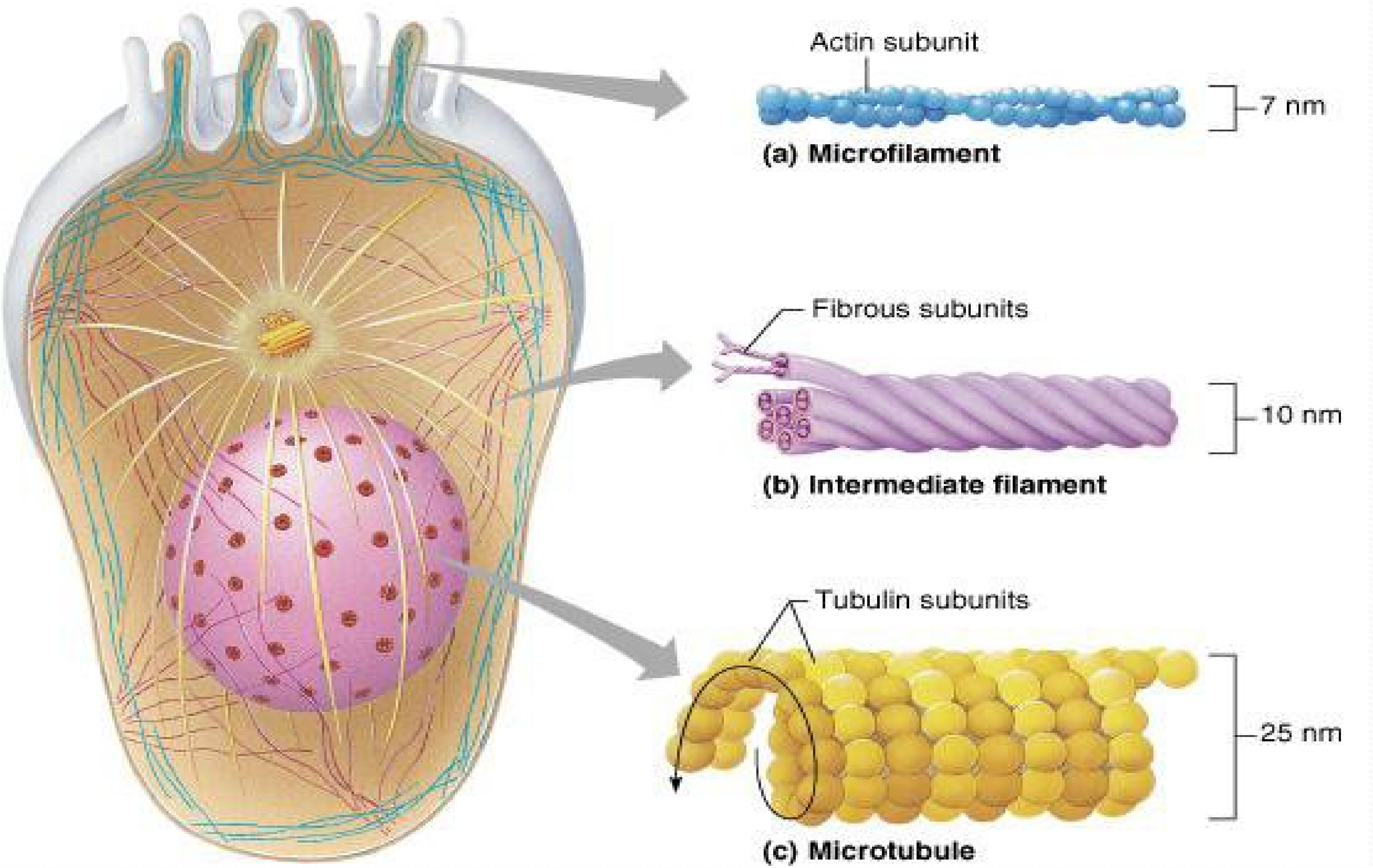
**Cytoskeleton:**

- Is a dynamic network of protein filaments.
- Provides structural support.
- Determines cell shape.
- Organizes intracellular transport.



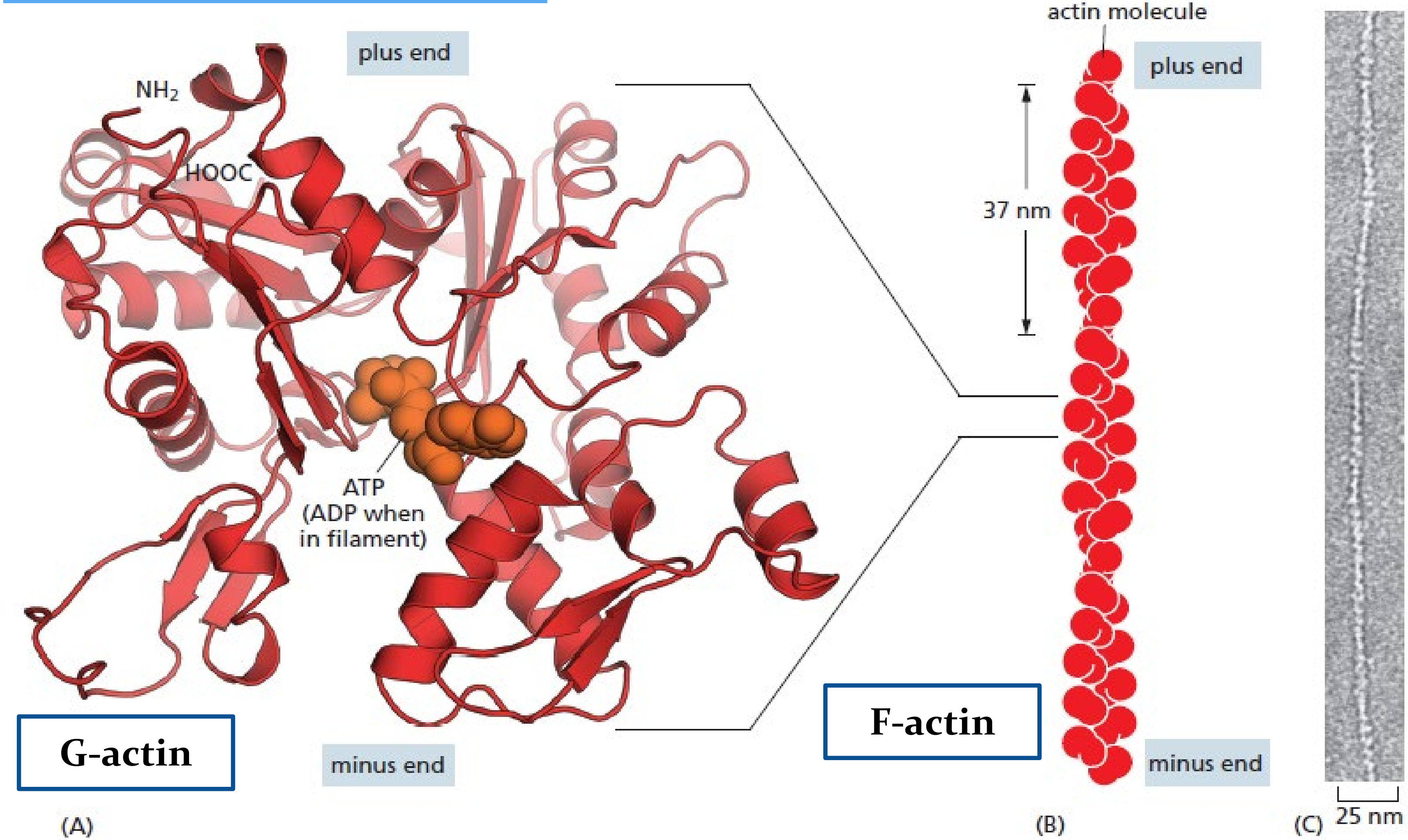



# The cytoskeleton is composed of three types of fibrous polymers



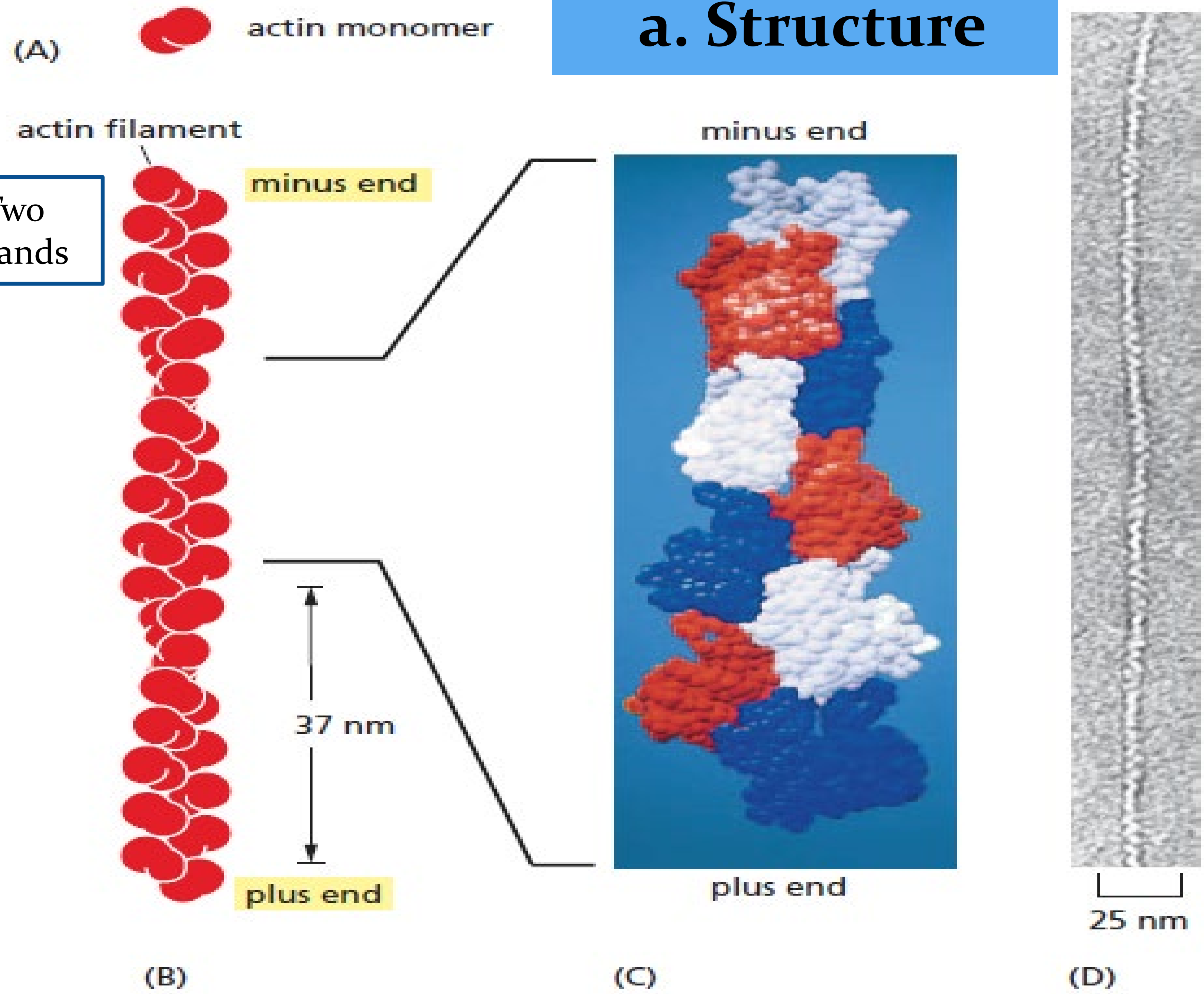
# 1. Actin filaments and their associated proteins

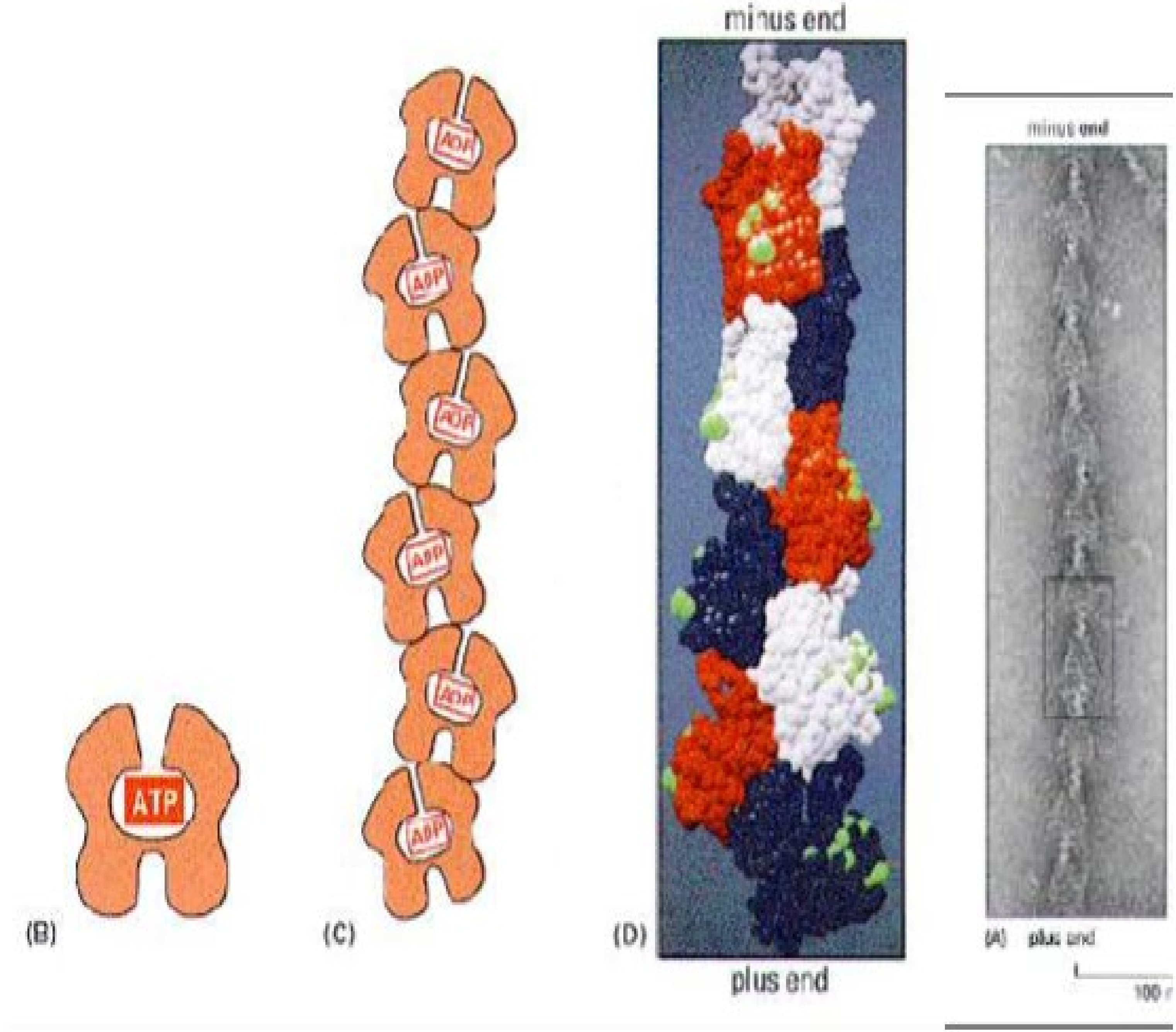
## a. Structure



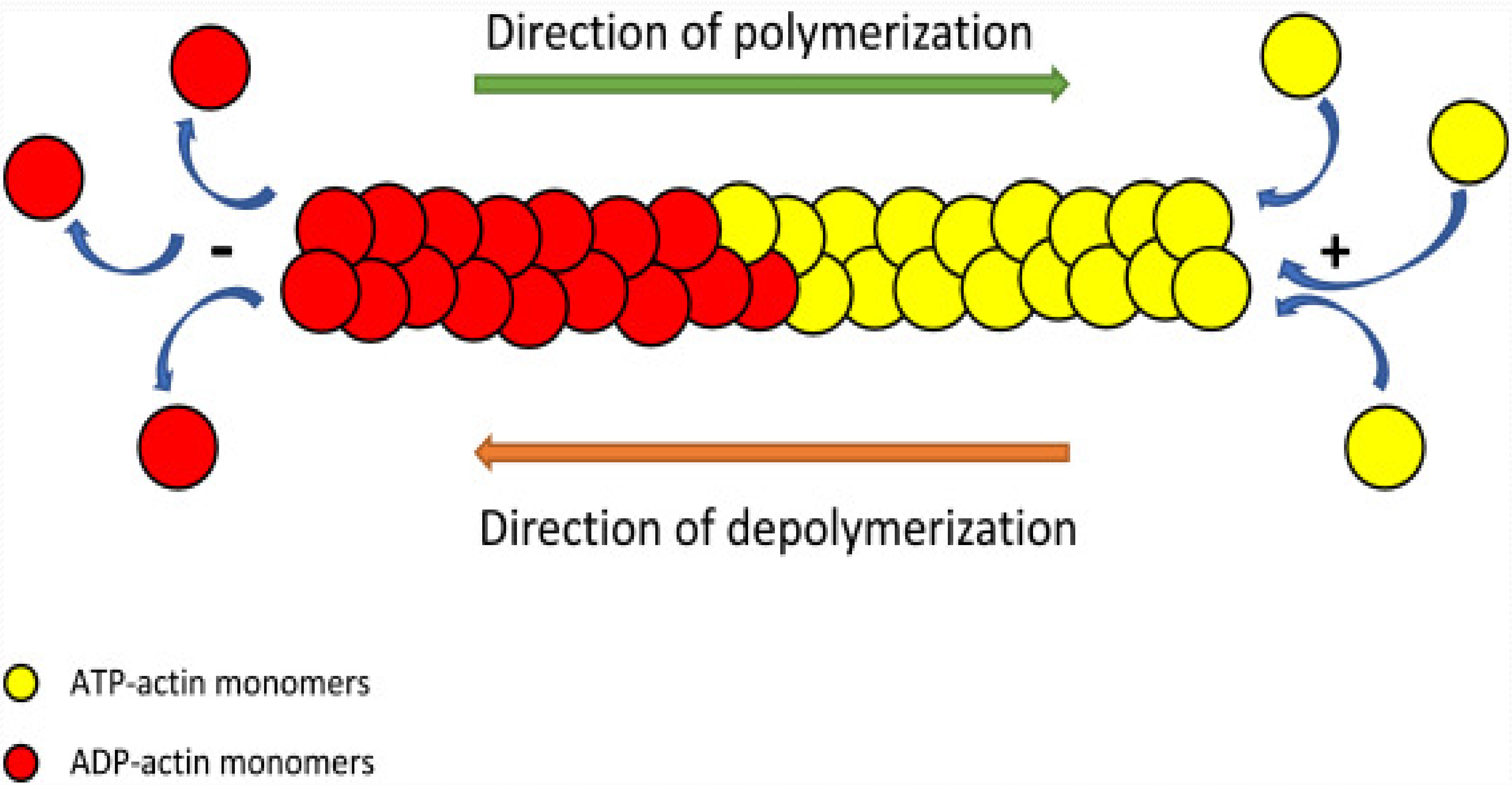
# a. Structure

Two strands

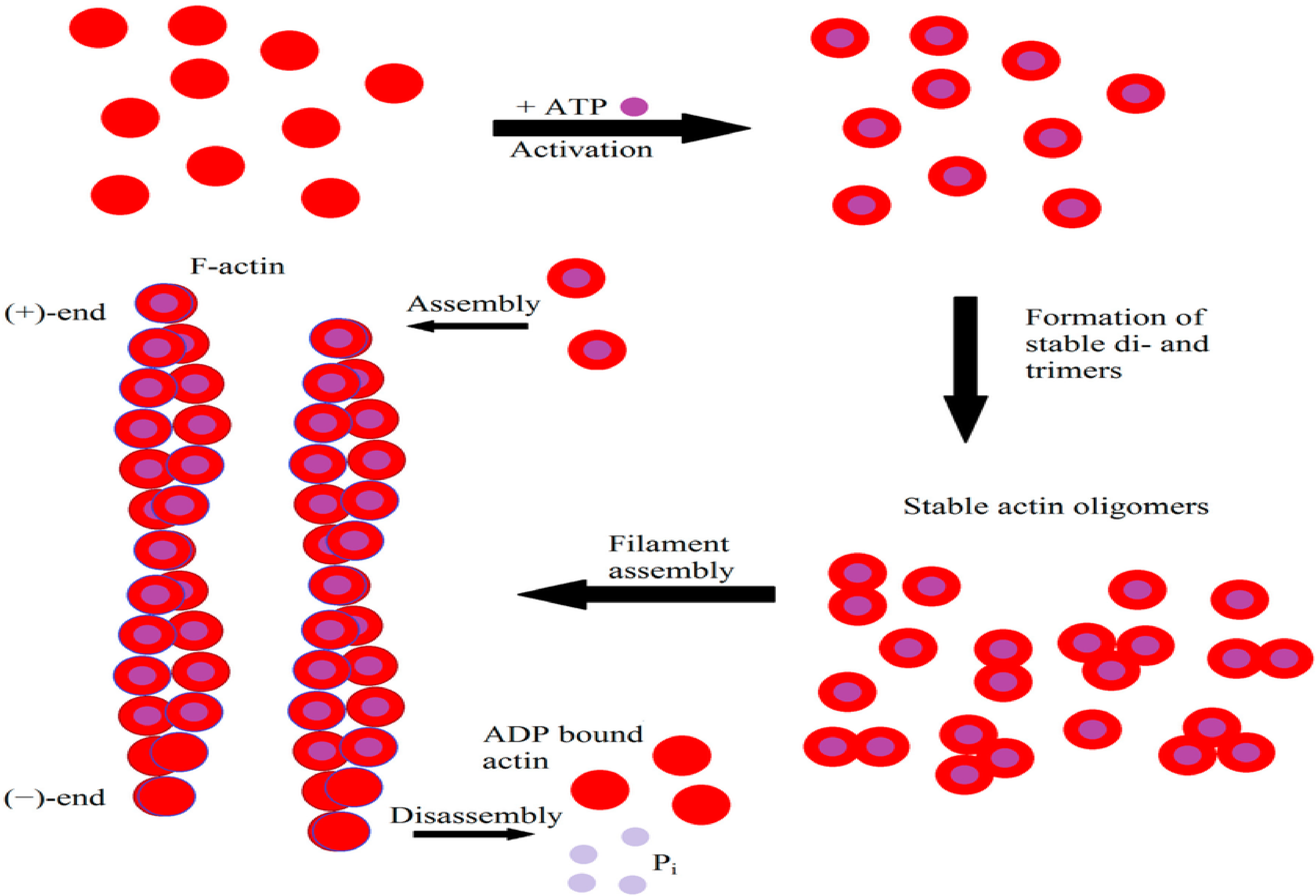



minus end
ATP
(B)
(C)
(D)
plus end
minus end
(A) plus end

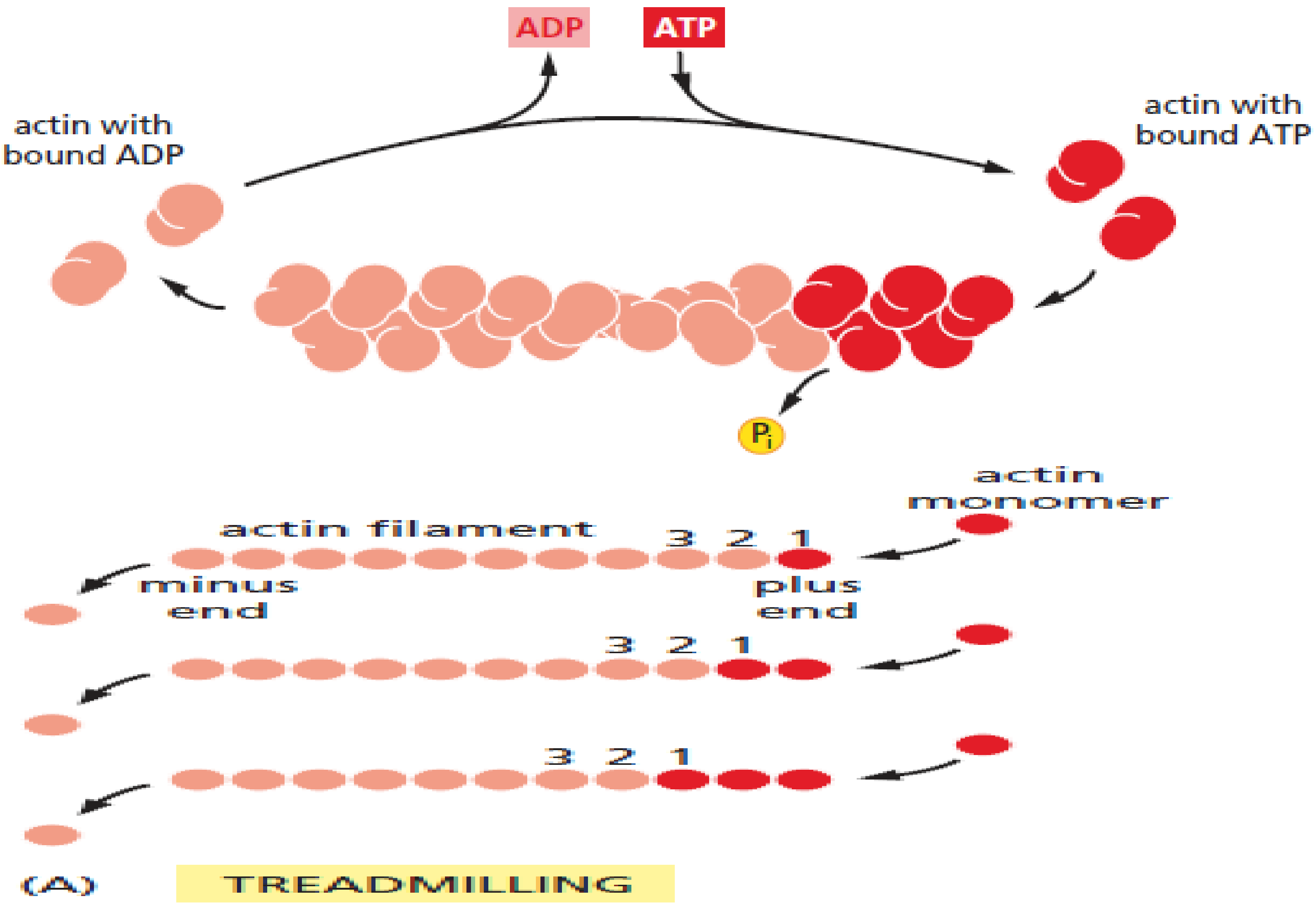
# b. Polymerization of G-actin into F-actin



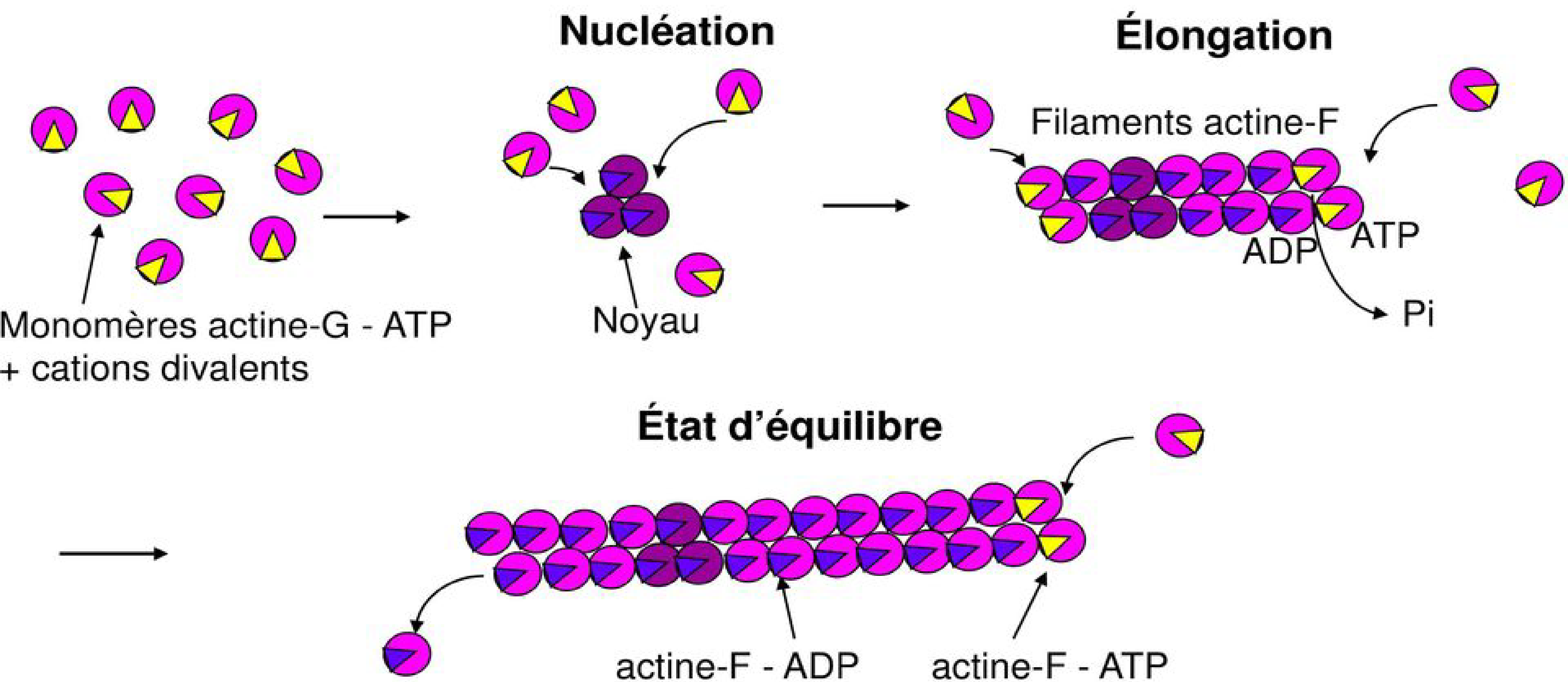
# b. Polymerization of G-actin into F-actin



# b. Polymerization of G-actin into F-actin

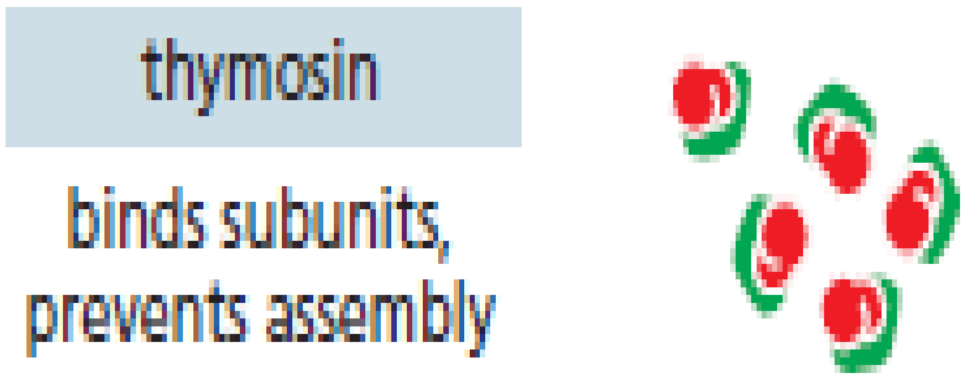


# The polymerization process takes place in three stages

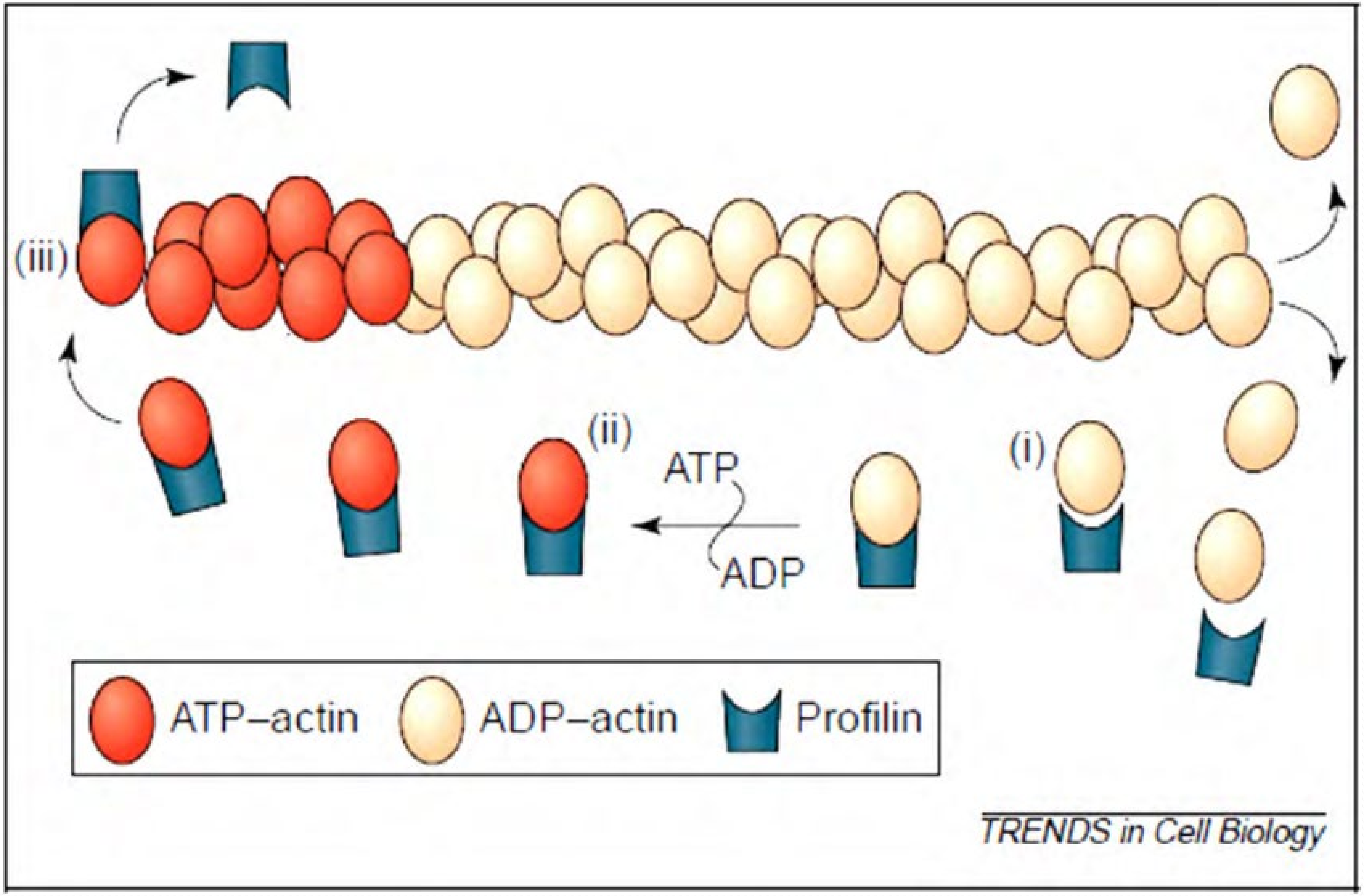


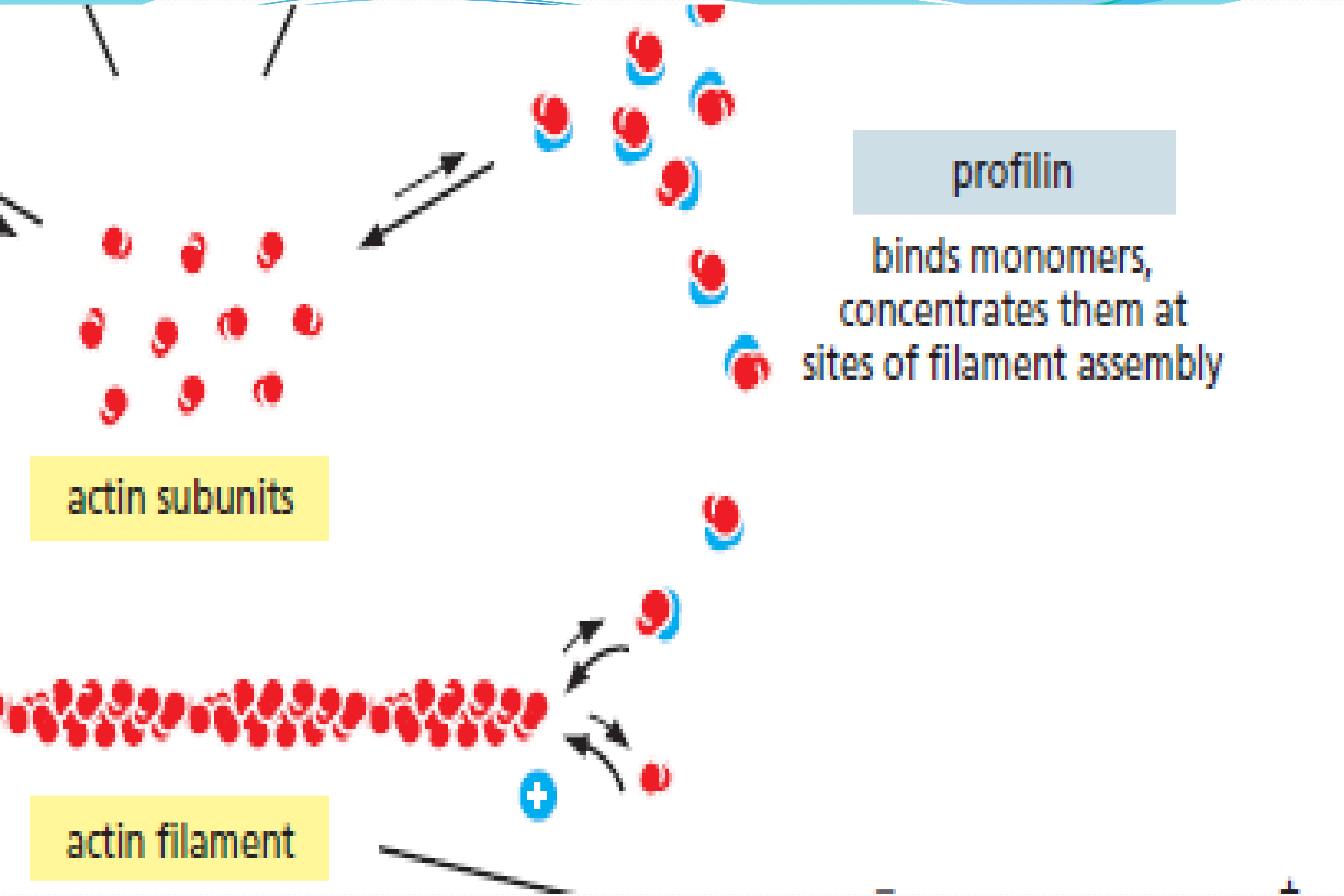
# C. - Actin-Binding Proteins (ABP)

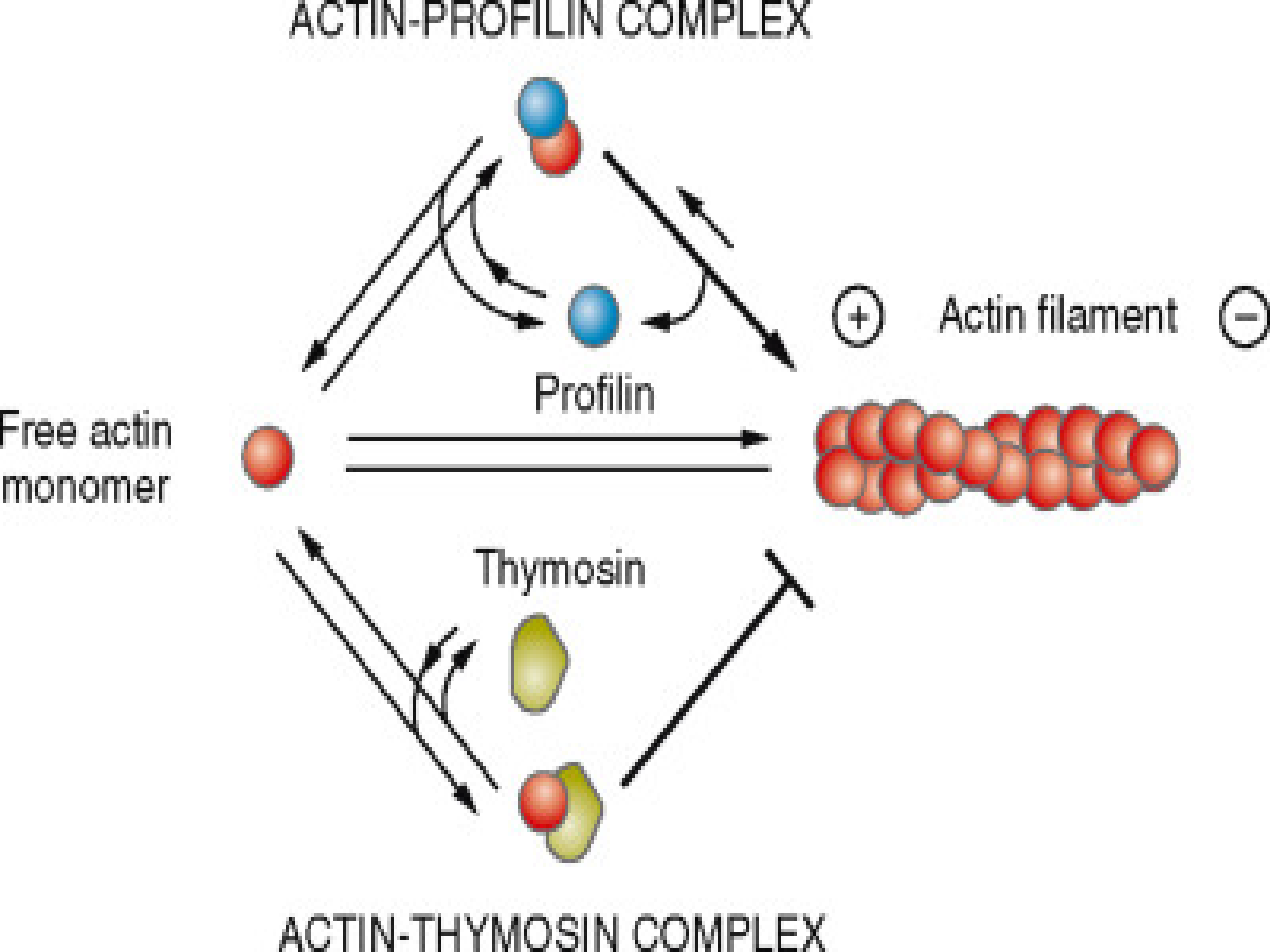
## c.1. Thymosin β4



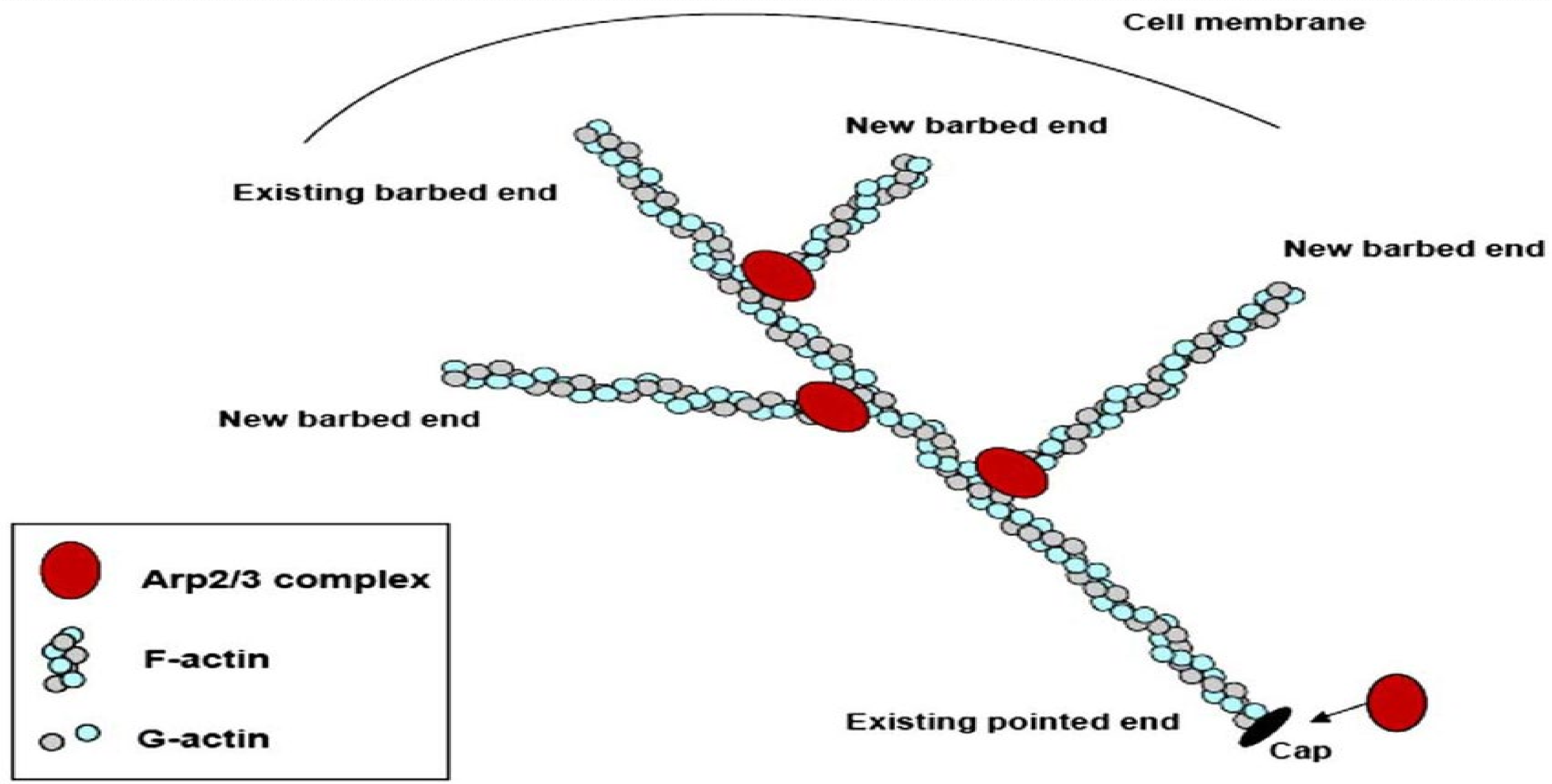
# c.2. Profilin


(iii)
(ii)
ATP
ADP
(i)
ATP–actin
ADP–actin
Profilin
TRENDS in Cell Biology

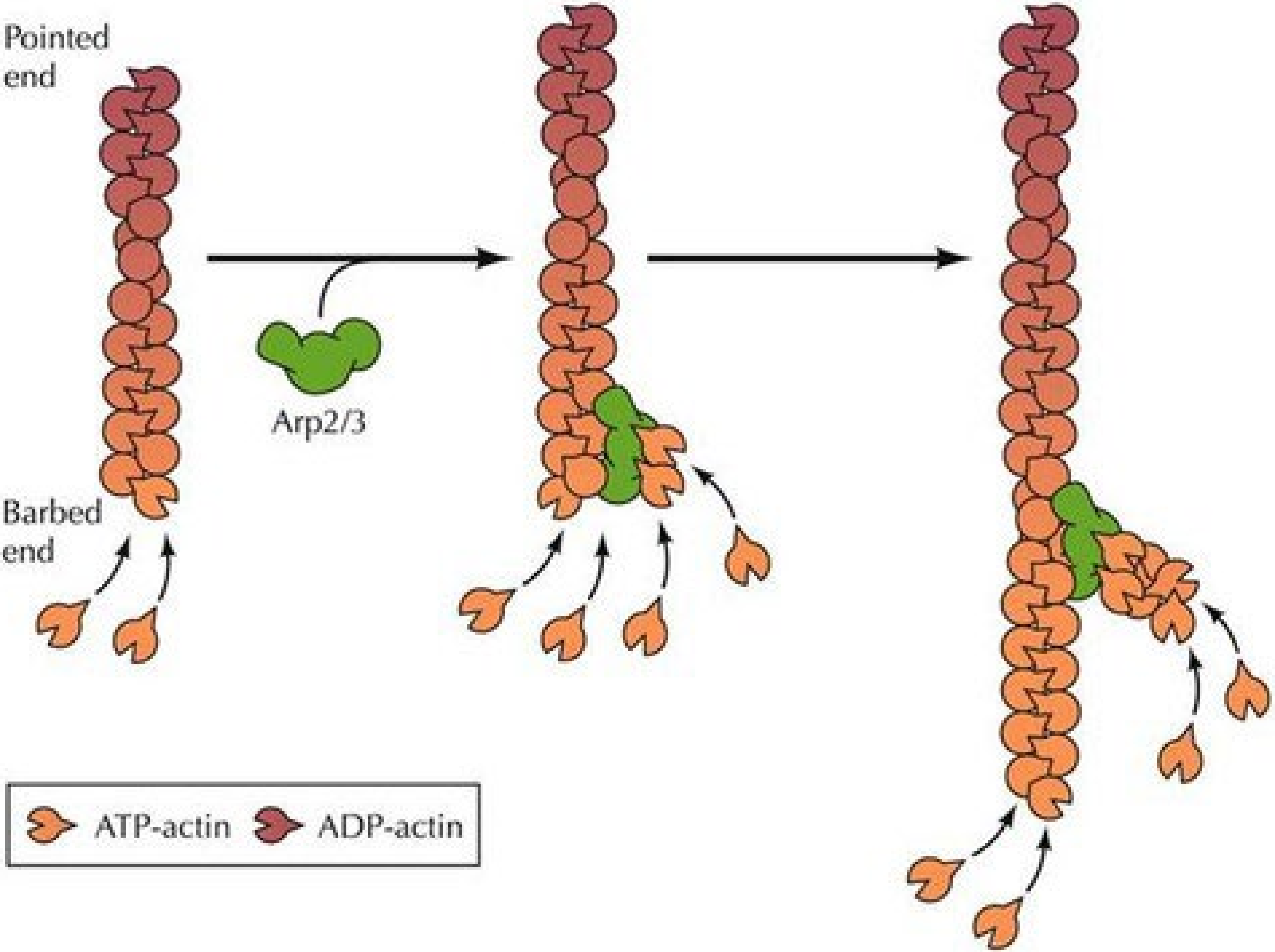

profilin
binds monomers, concentrates them at sites of filament assembly
actin subunits
actin filament


ACTIN-PROFILIN COMPLEX
Actin filament
Profilin
Free actin
monomer
Thymosin
ACTIN-THYMOSIN COMPLEX

# C.3. ARP2/3

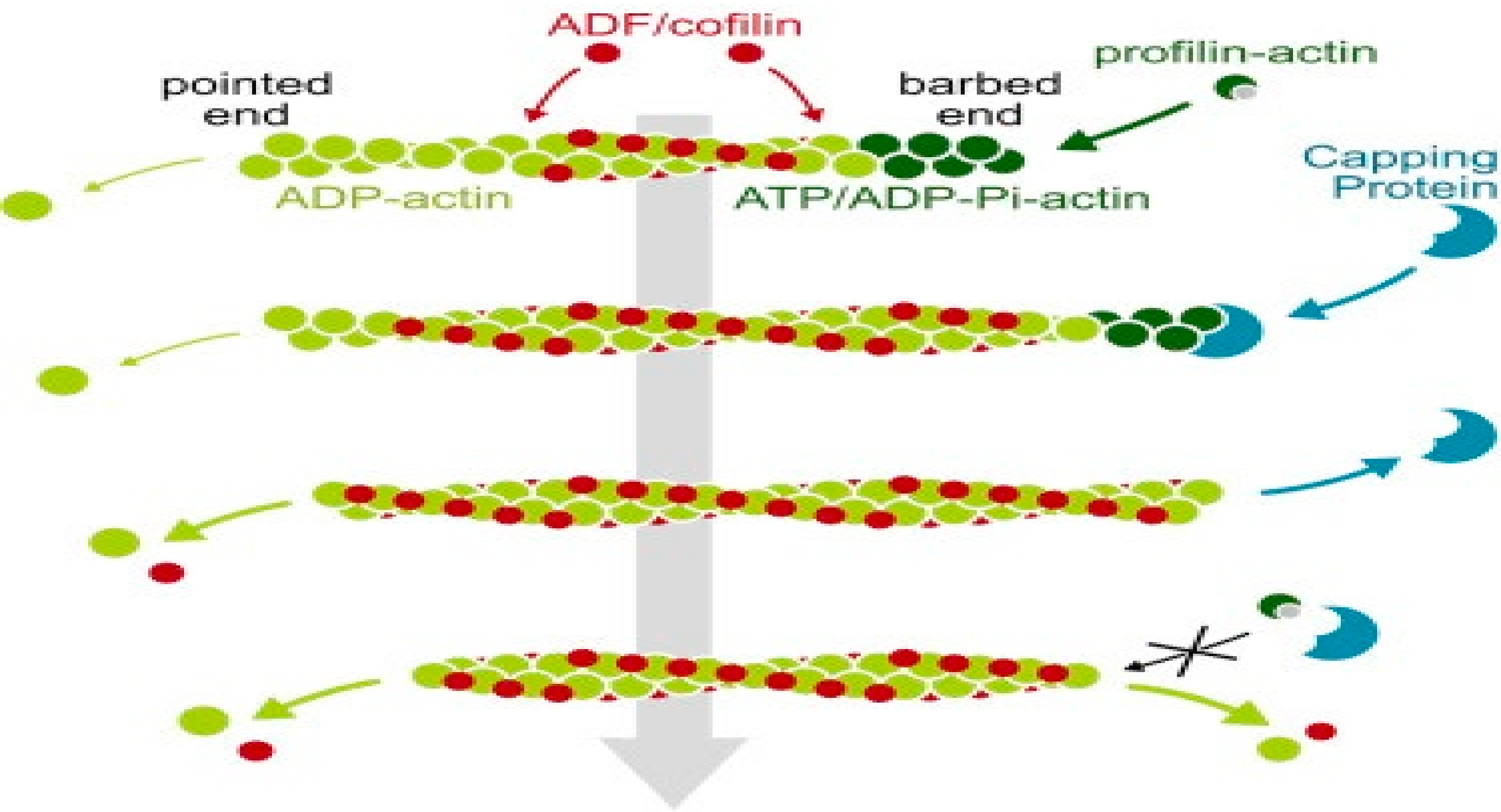



**Arp2/3 nucleates new branches from existing actin filaments, creating fast-growing barbed ends**

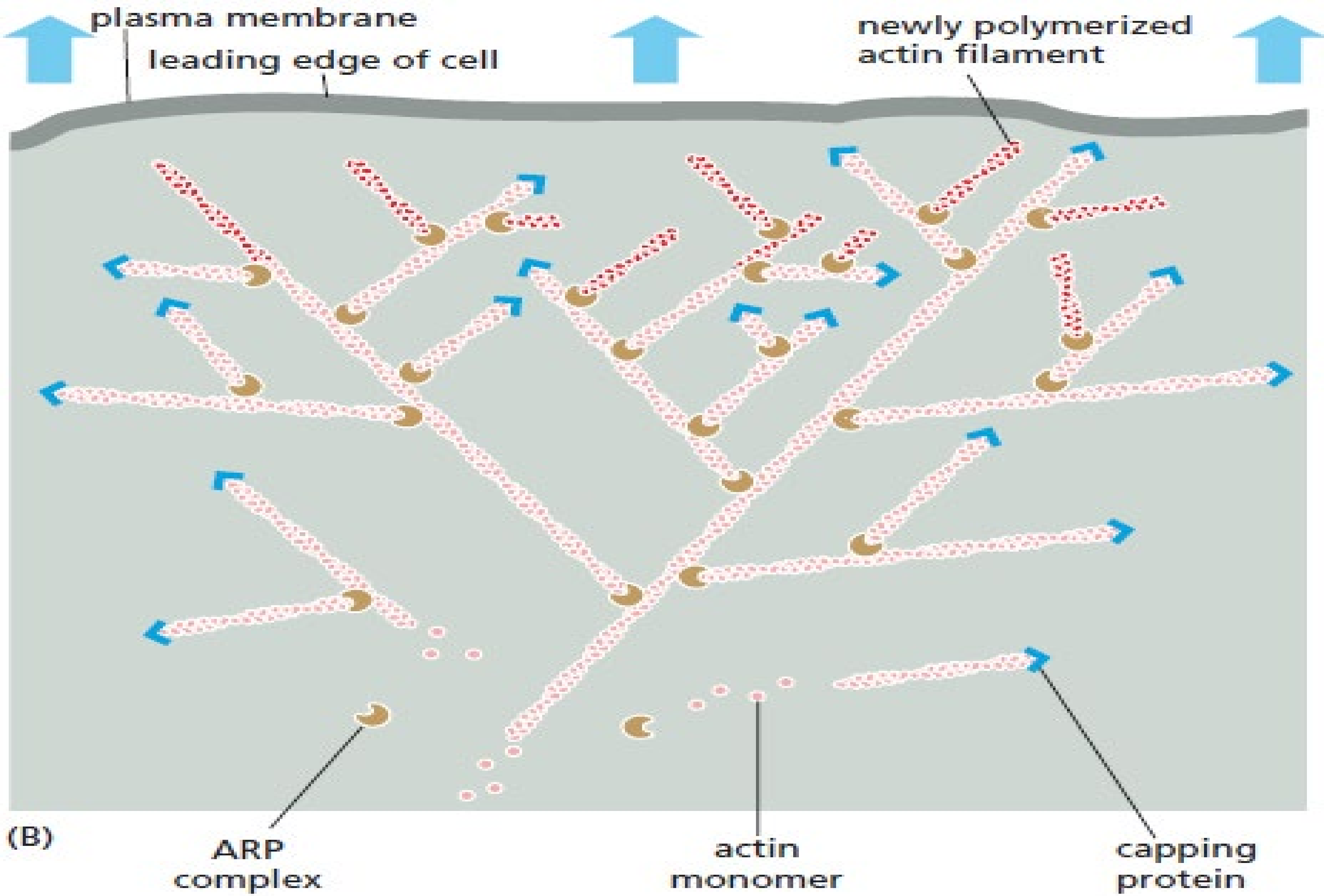


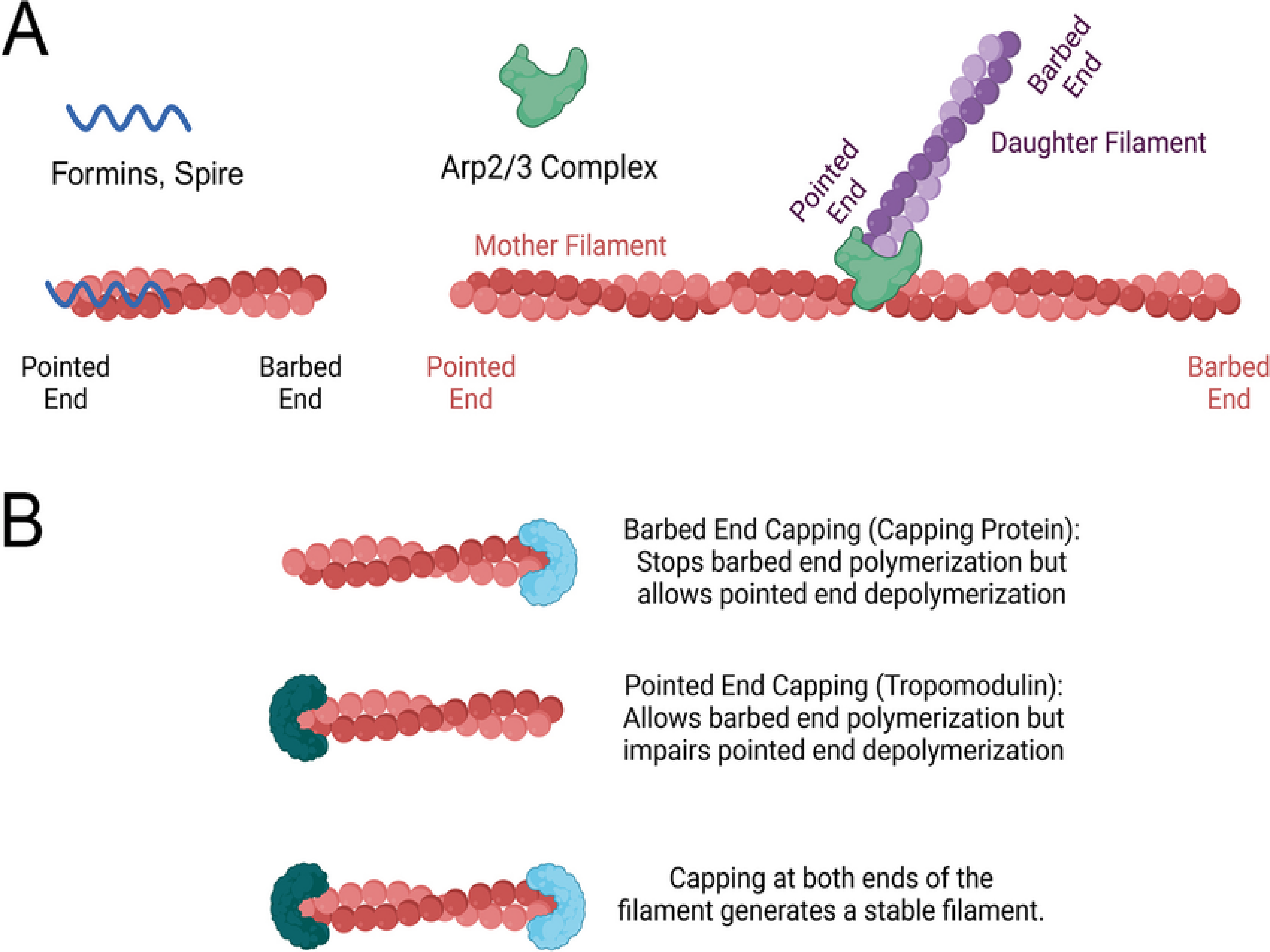

THE CELL 5e, Figure 12.7

# c.4. Capping Proteins



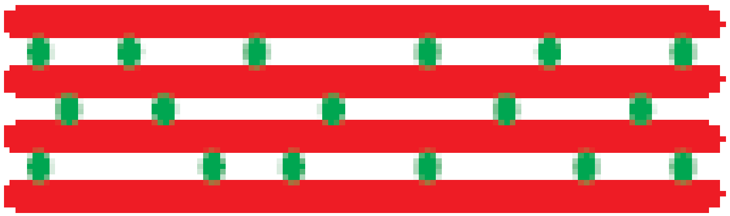
# c.4. Capping Proteins



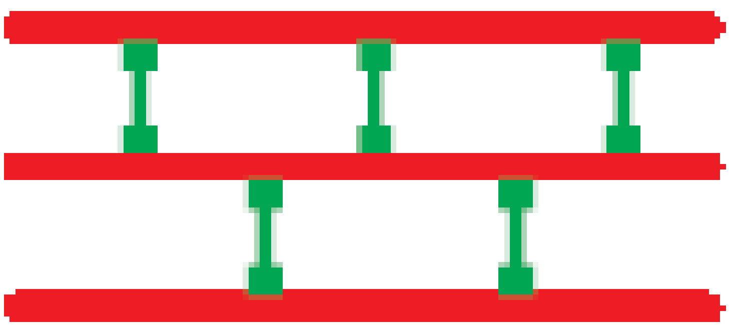

A
Formins, Spire
Arp2/3 Complex
Barbed End
Daughter Filament
Pointed End
Mother Filament
Pointed End
Barbed End
Pointed End
Barbed End
B
Barbed End Capping (Capping Protein):
Stops barbed end polymerization but allows pointed end depolymerization
Pointed End Capping (Tropomodulin):
Allows barbed end polymerization but impairs pointed end depolymerization
Capping at both ends of the filament generates a stable filament.

# c.5. Stabilization of actin filaments

**Faisceaux sérés**

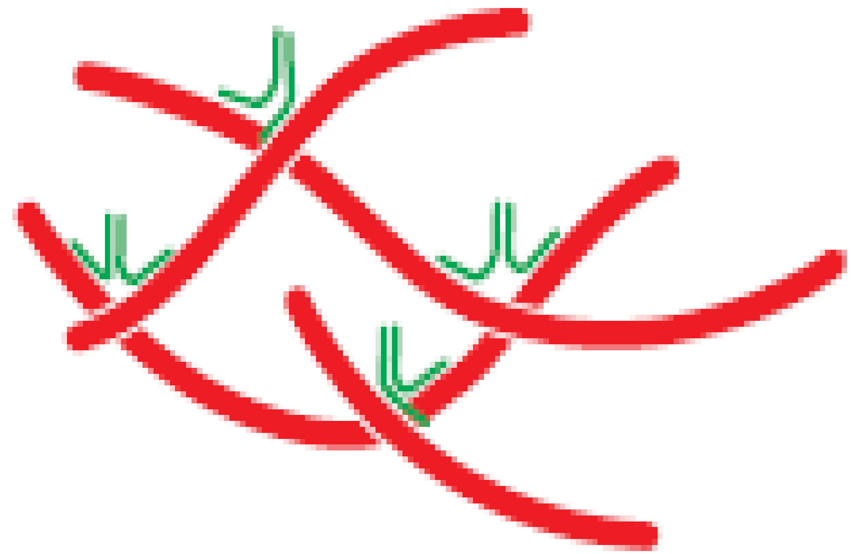


fimbrin



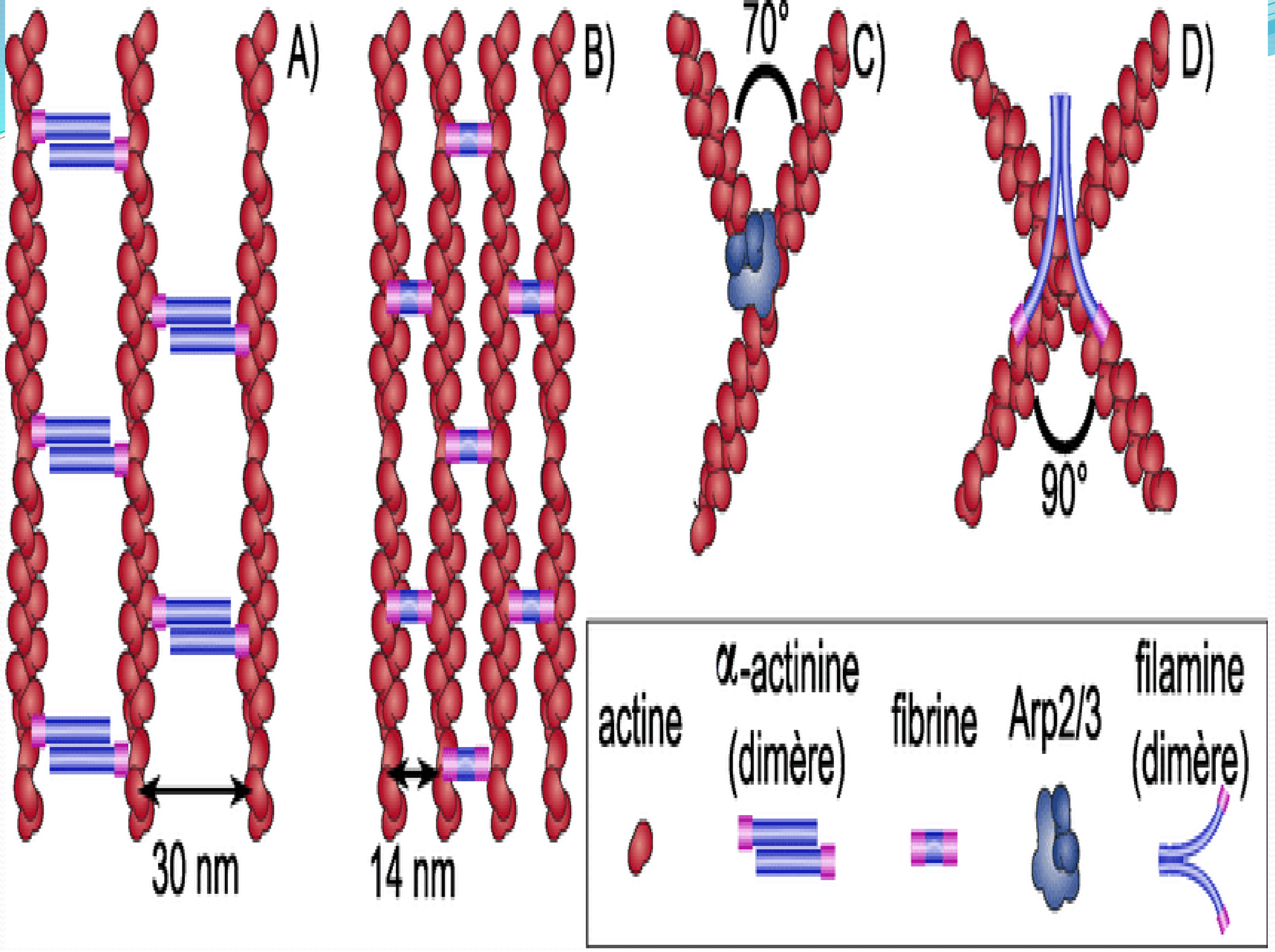
α-actinin

**Faisceaux larges**



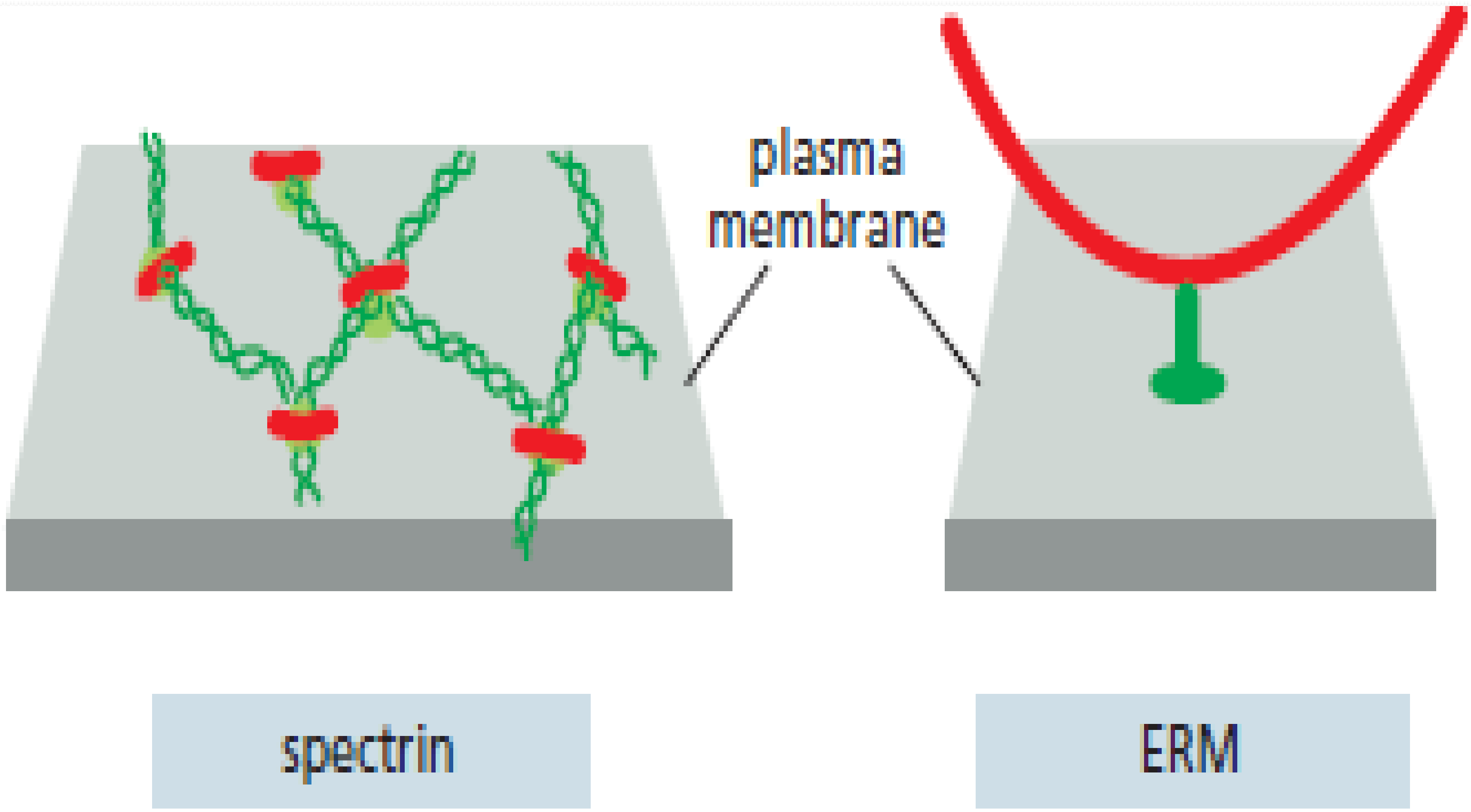
filamin

**stabilisation**



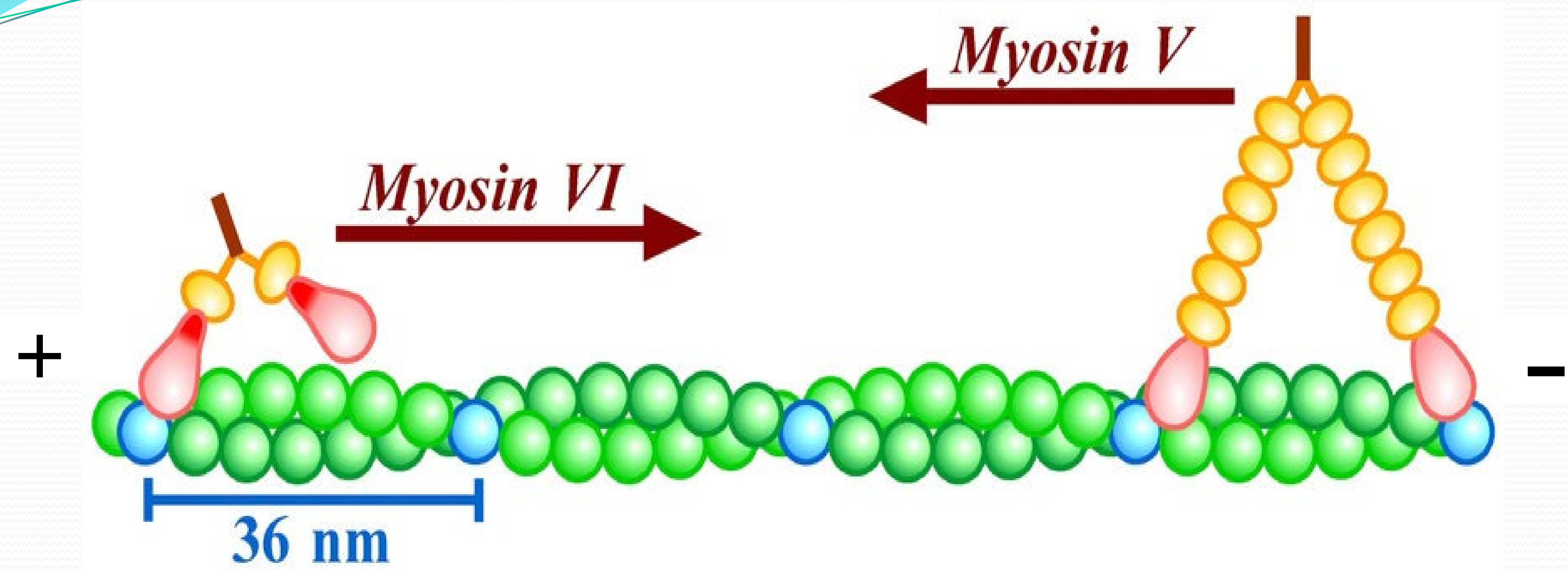
A)
B)
C)
D)
70°
90°
30 nm
14 nm
actine
α-actinine
(dimère)
fibrine
Arp2/3
filamine
(dimère)

# c.6. Actin-linking proteins to the cell membrane



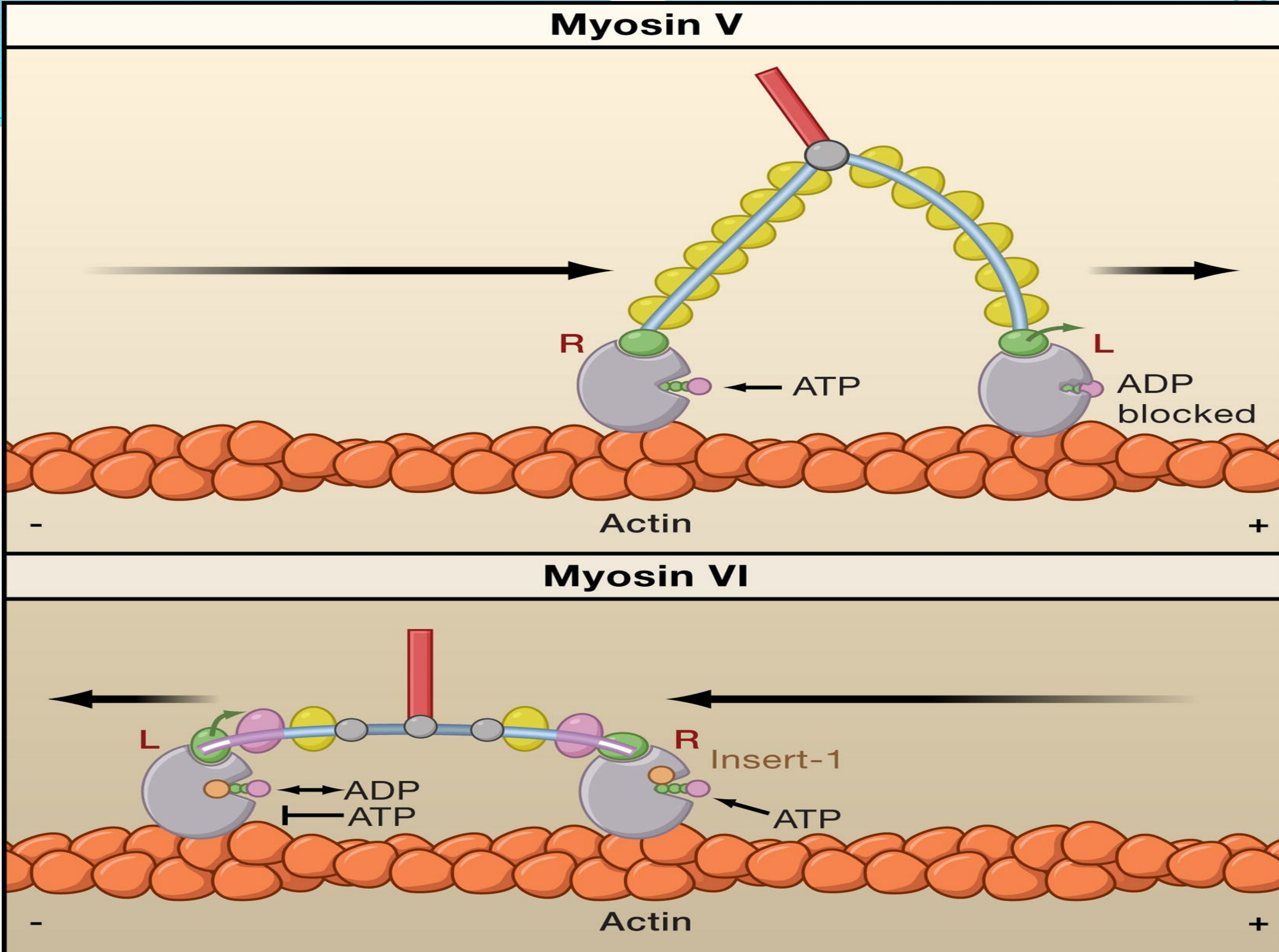

**Ezrine, Radixine et Moesine**

# c.7. Motor Proteins (Movement)



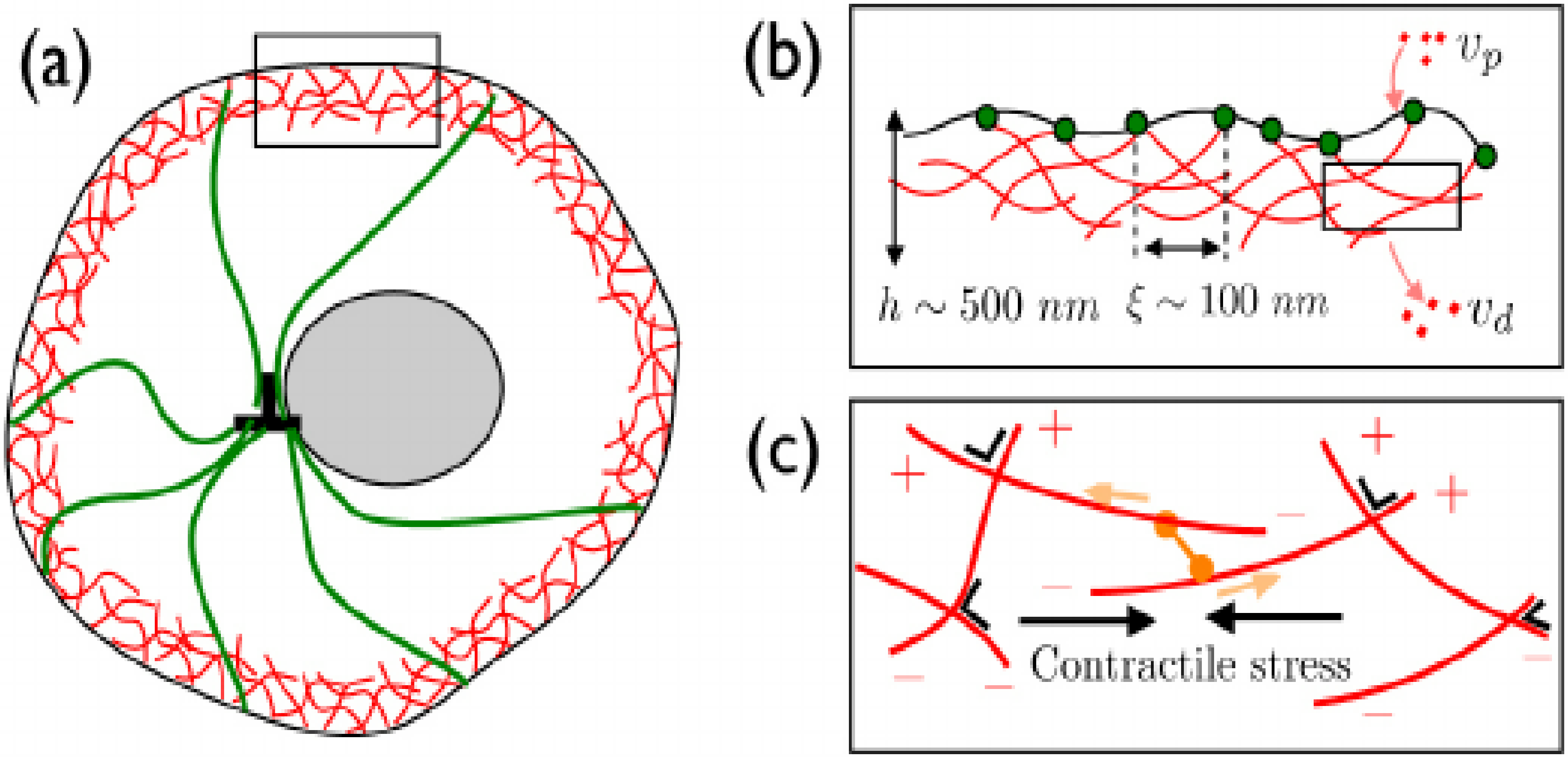

**Myosine VI: elle se déplace vers l'extrémité "moins" (le pôle pointu) des filaments d'actine (endocytose)**

**Myosine V : Récupère la vésicule à la fin du microtubule pour la transporter sur les filaments d'actine jusqu'à la membrane plasmique pour l'exocytose.**


Myosin V
R
ATP
L
ADP
blocked
-
Actin
+
Myosin VI
L
ADP
ATP
R
Insert-1
ATP
-
Actin
+

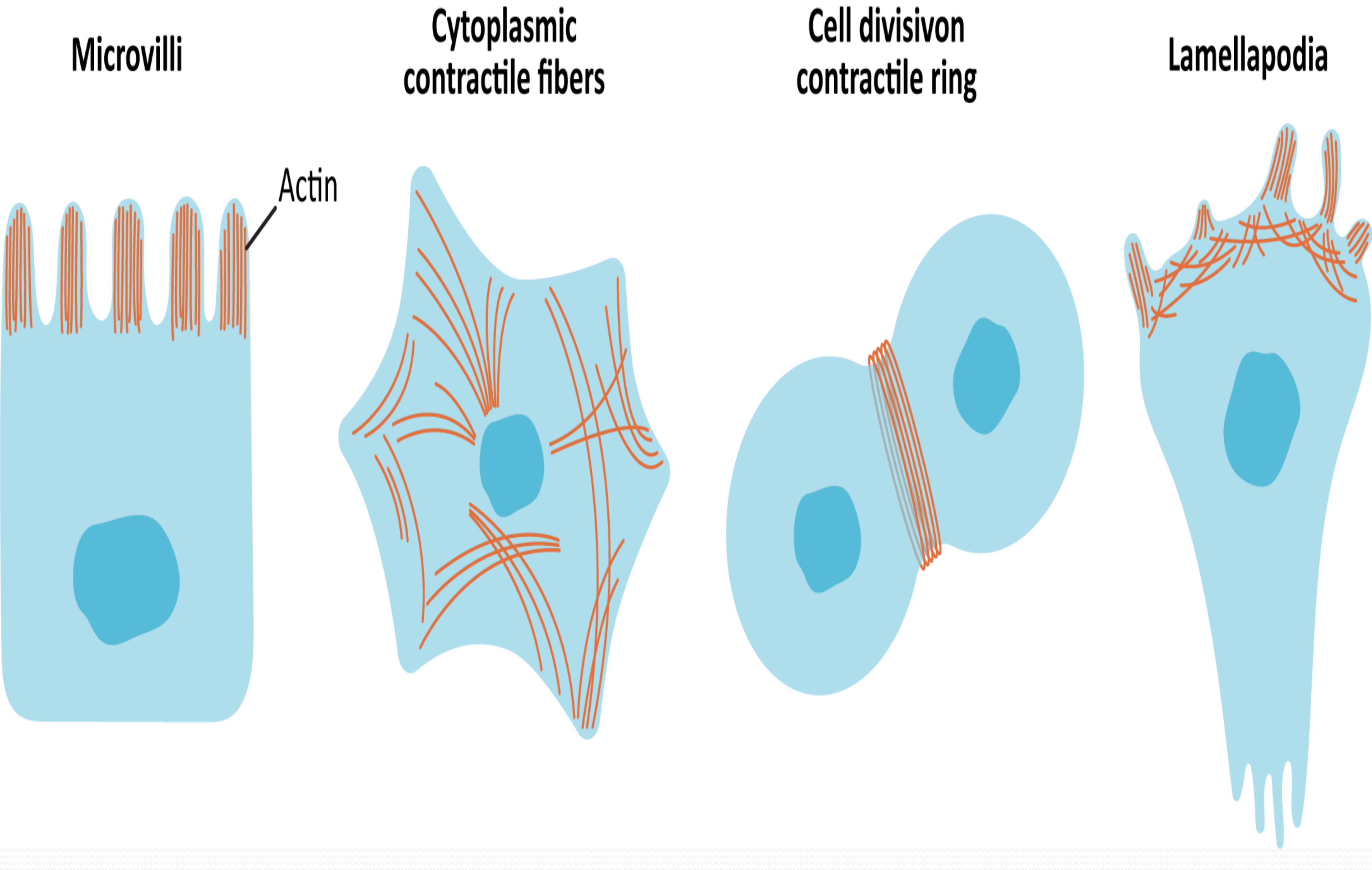
# D. Functions of actin in the cell

## D.1. Structural Support

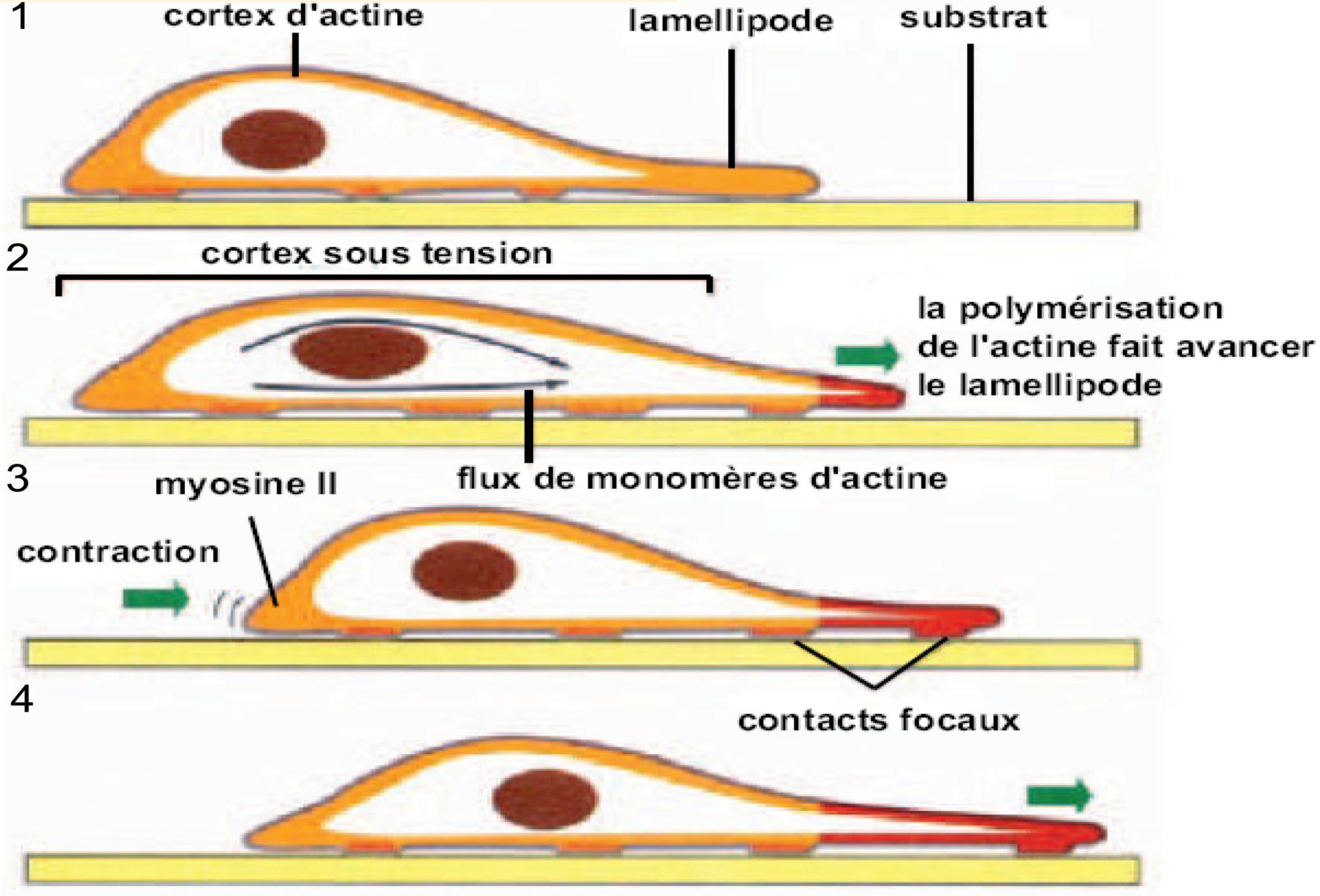


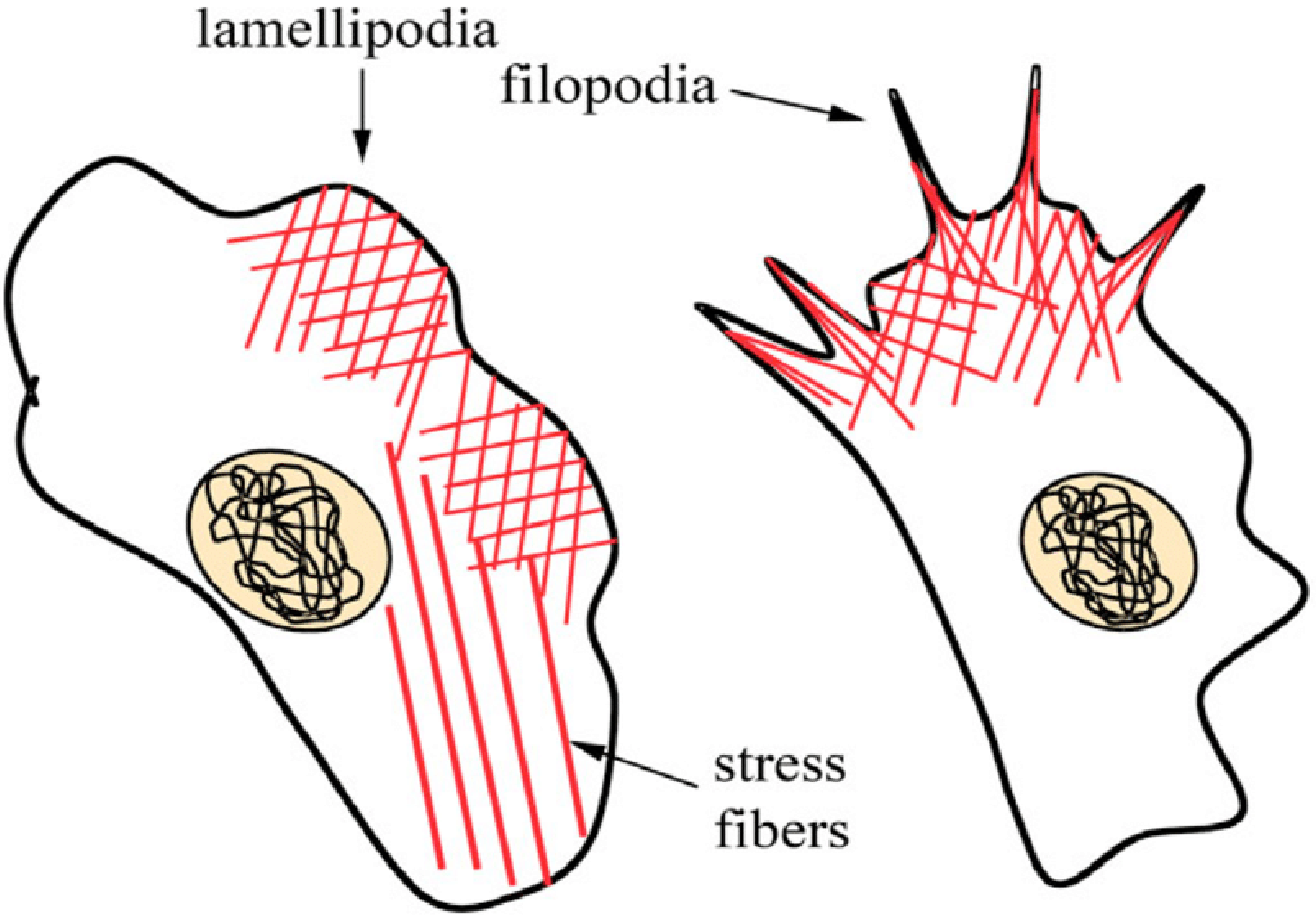

Cell cortex

# D.2. Shape and division

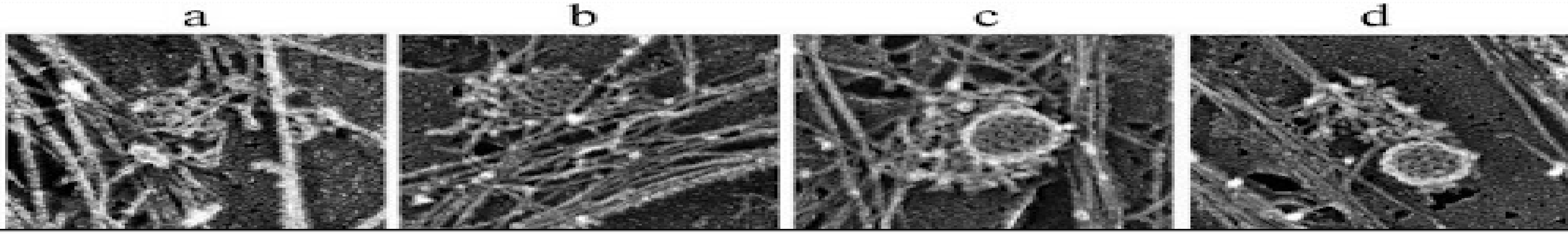


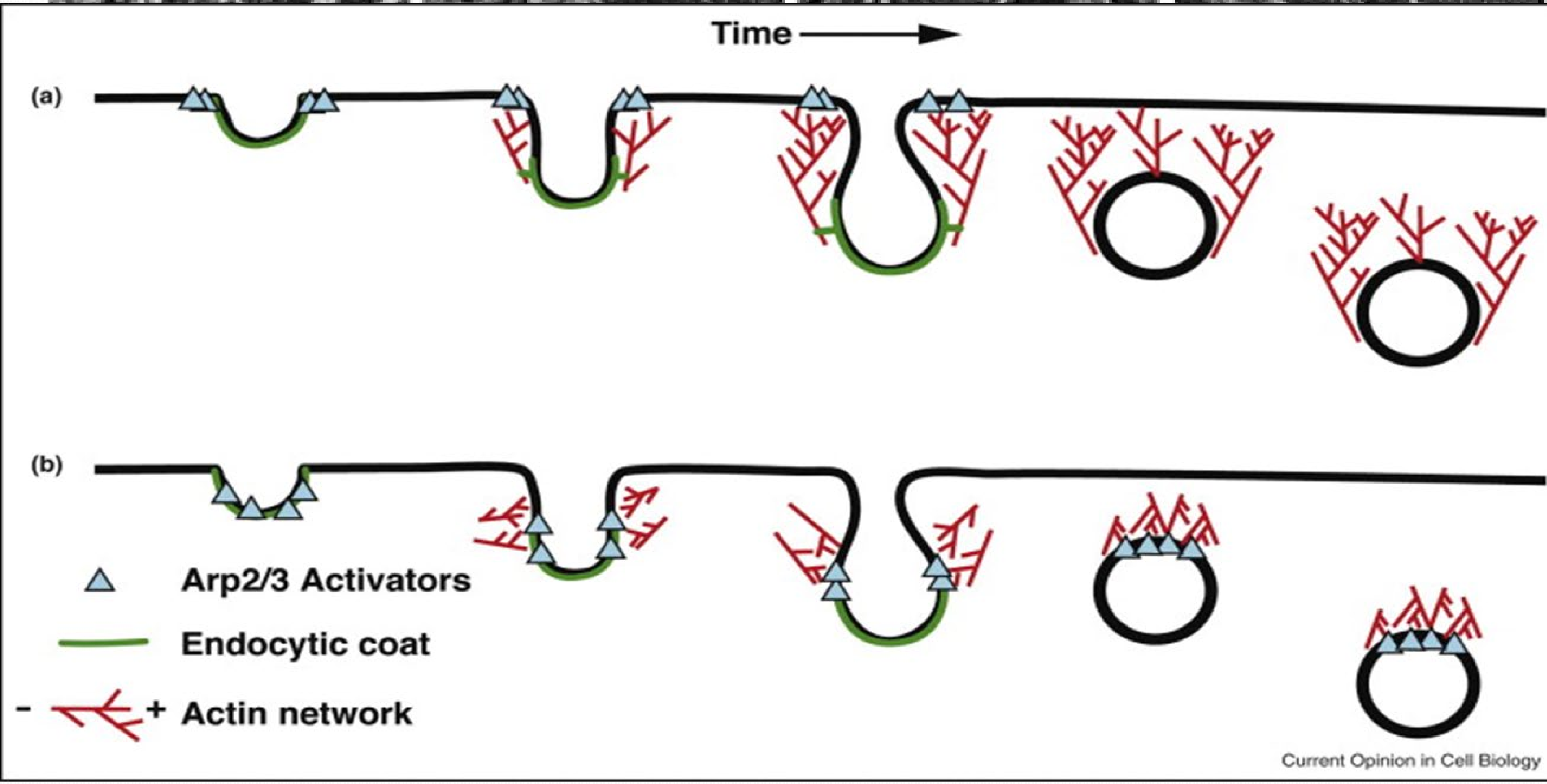
# D.3. Cell movements

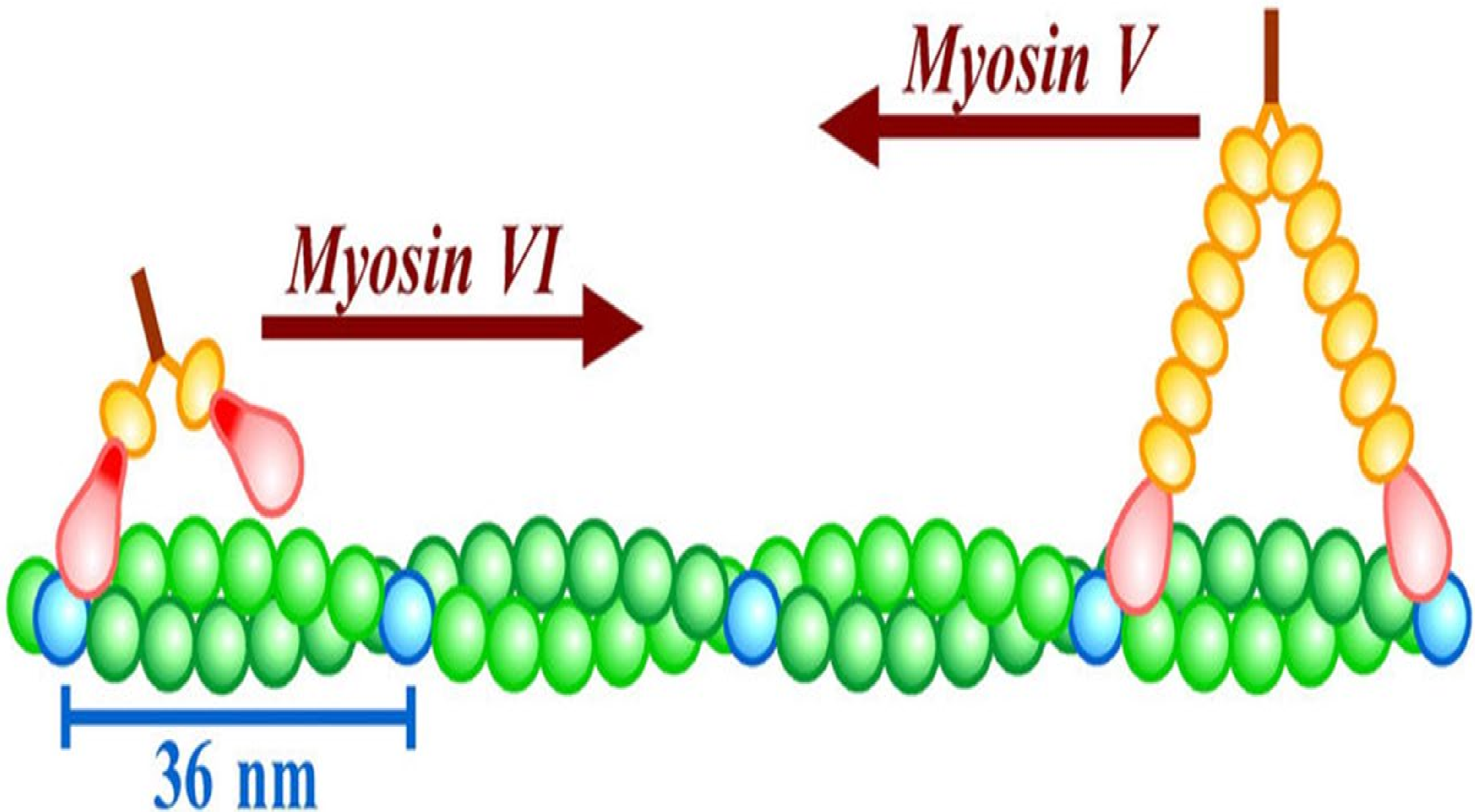



lamellipodia
filopodia
stress
fibers

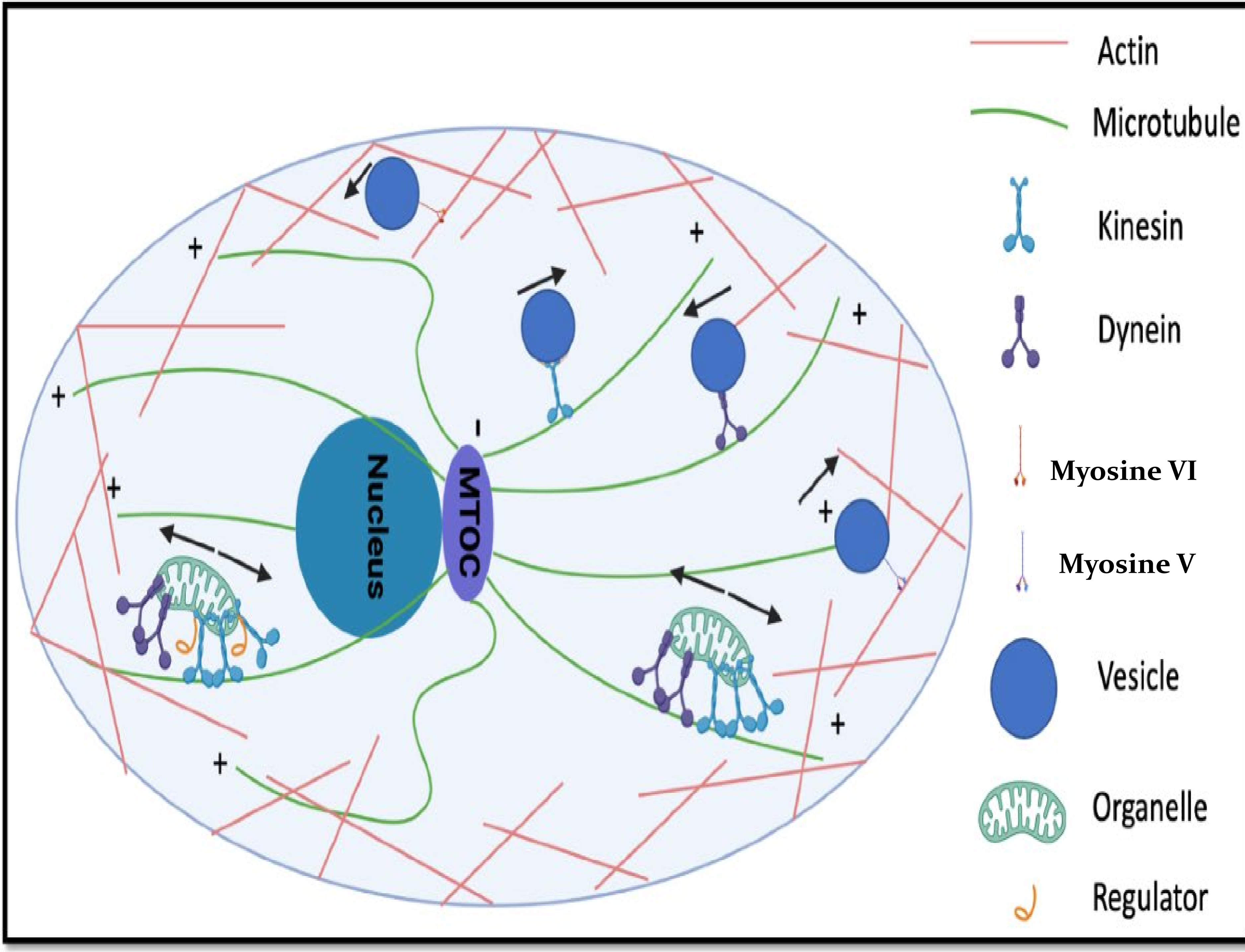
# D.4. Endocytosis

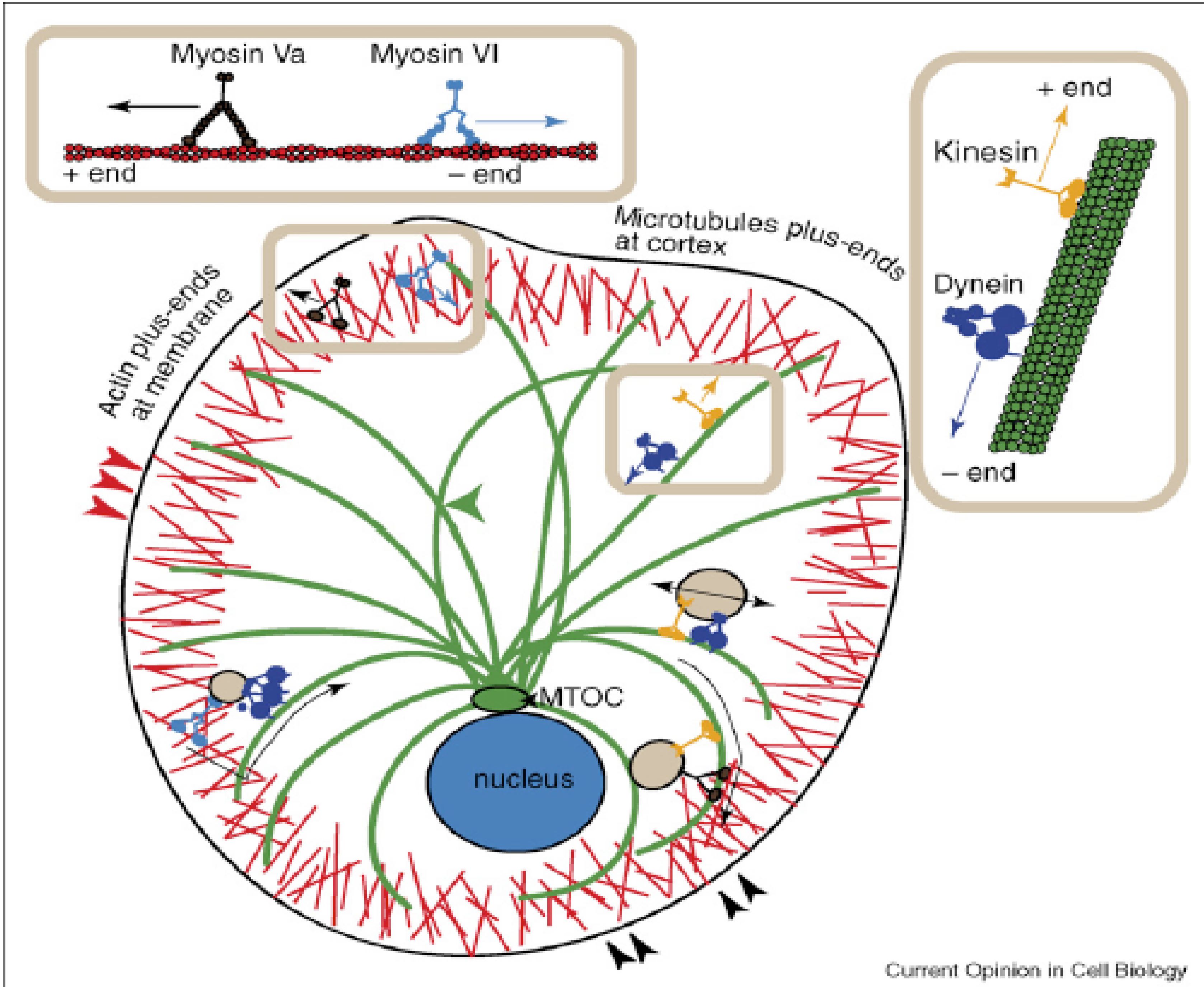


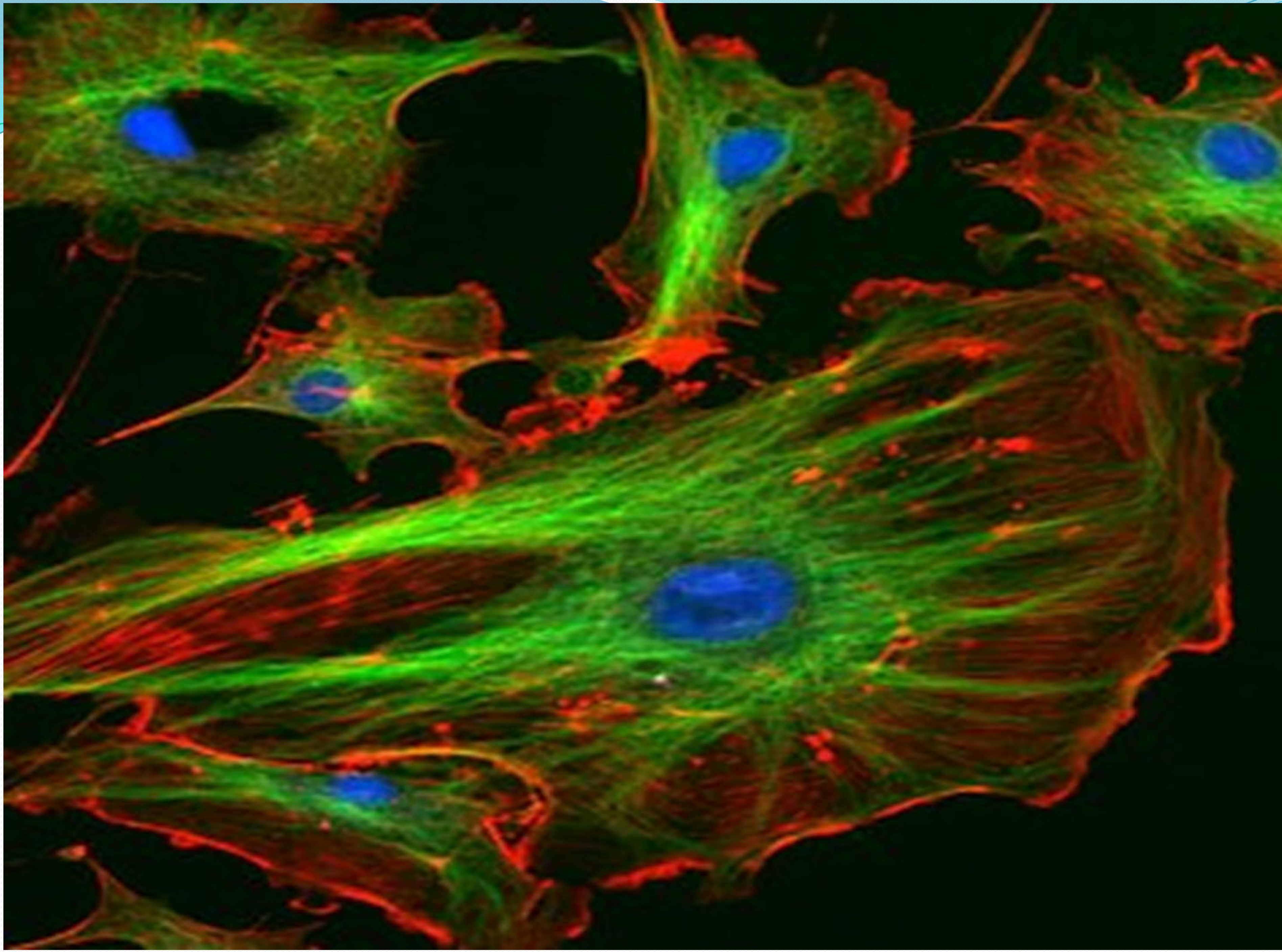



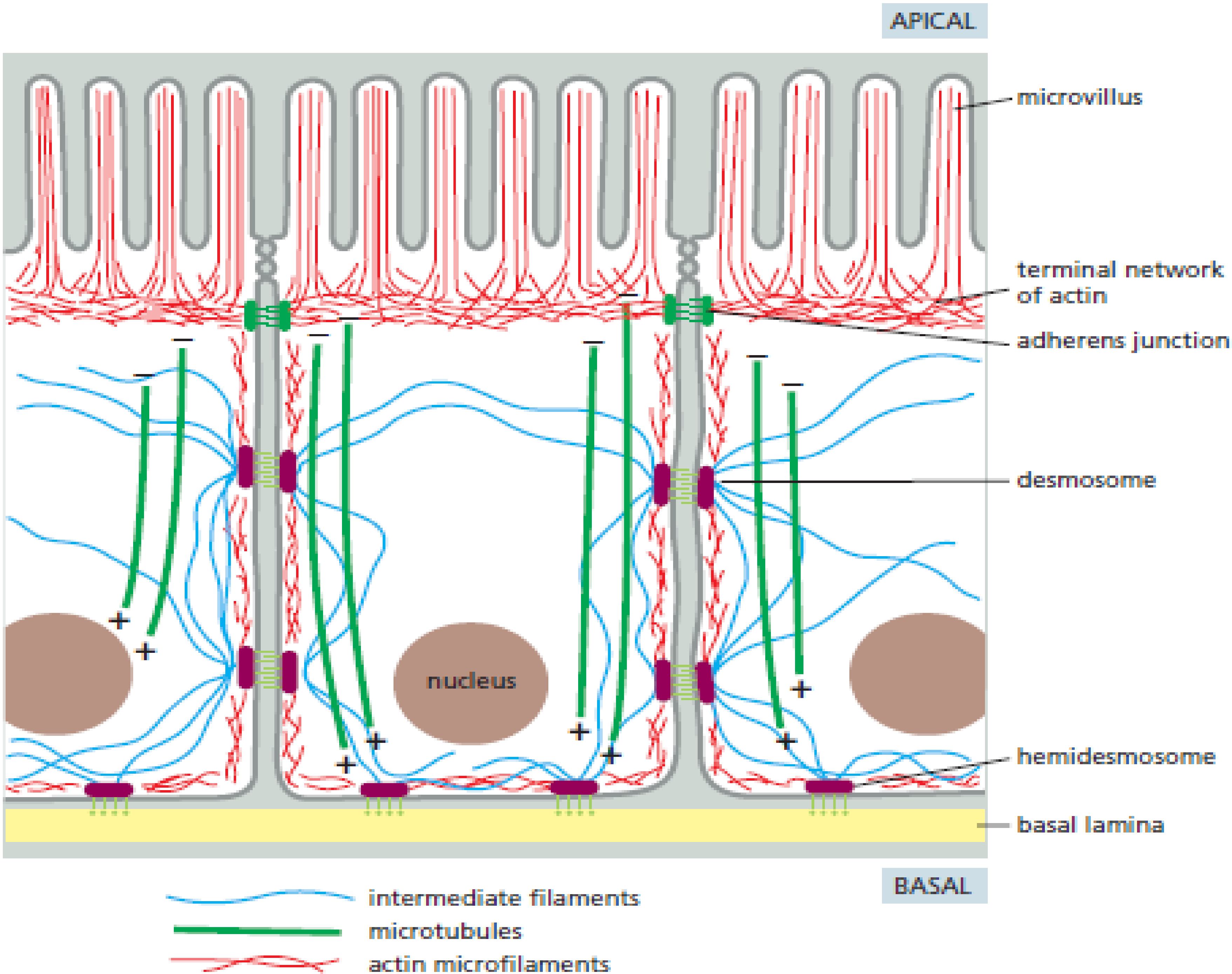


Myosin V
Myosin VI
36 nm

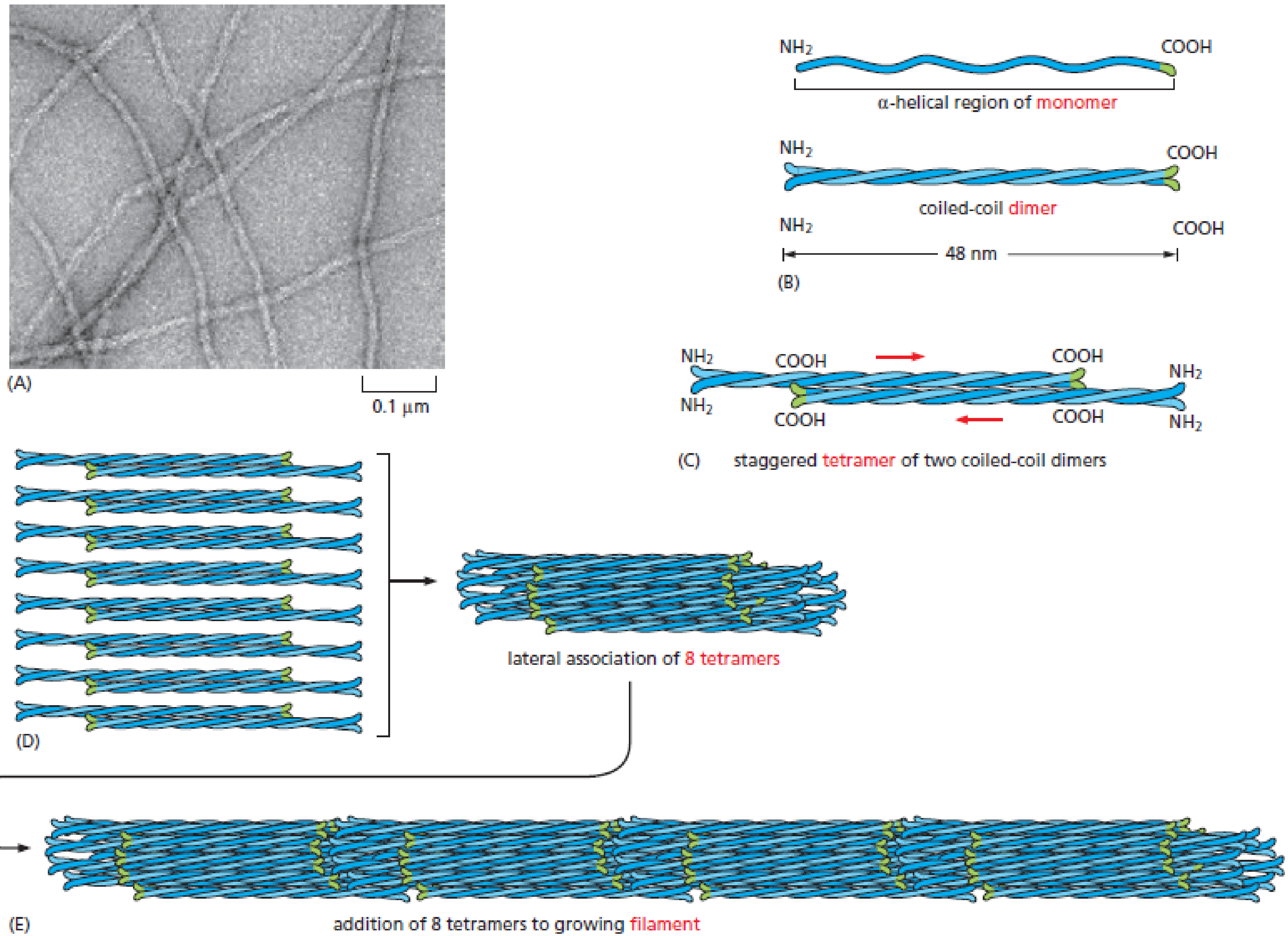

Actin
Microtubule
Kinesin
Dynein
Myosine VI
Myosine V
Vesicle
Organelle
Regulator
Nucleus
MTOC


Myosin Va
Myosin VI
+ end
– end
+ end
Kinesin
Dynein
– end
Microtubules plus-ends at cortex
Actin plus-ends at membrane
MTOC
nucleus
Current Opinion in Cell Biology




APICAL
microvillus
terminal network of actin
adherens junction
desmosome
nucleus
hemidesmosome
basal lamina
BASAL
intermediate filaments
microtubules
actin microfilaments

# 2. Intermediate filaments



# IF proteins

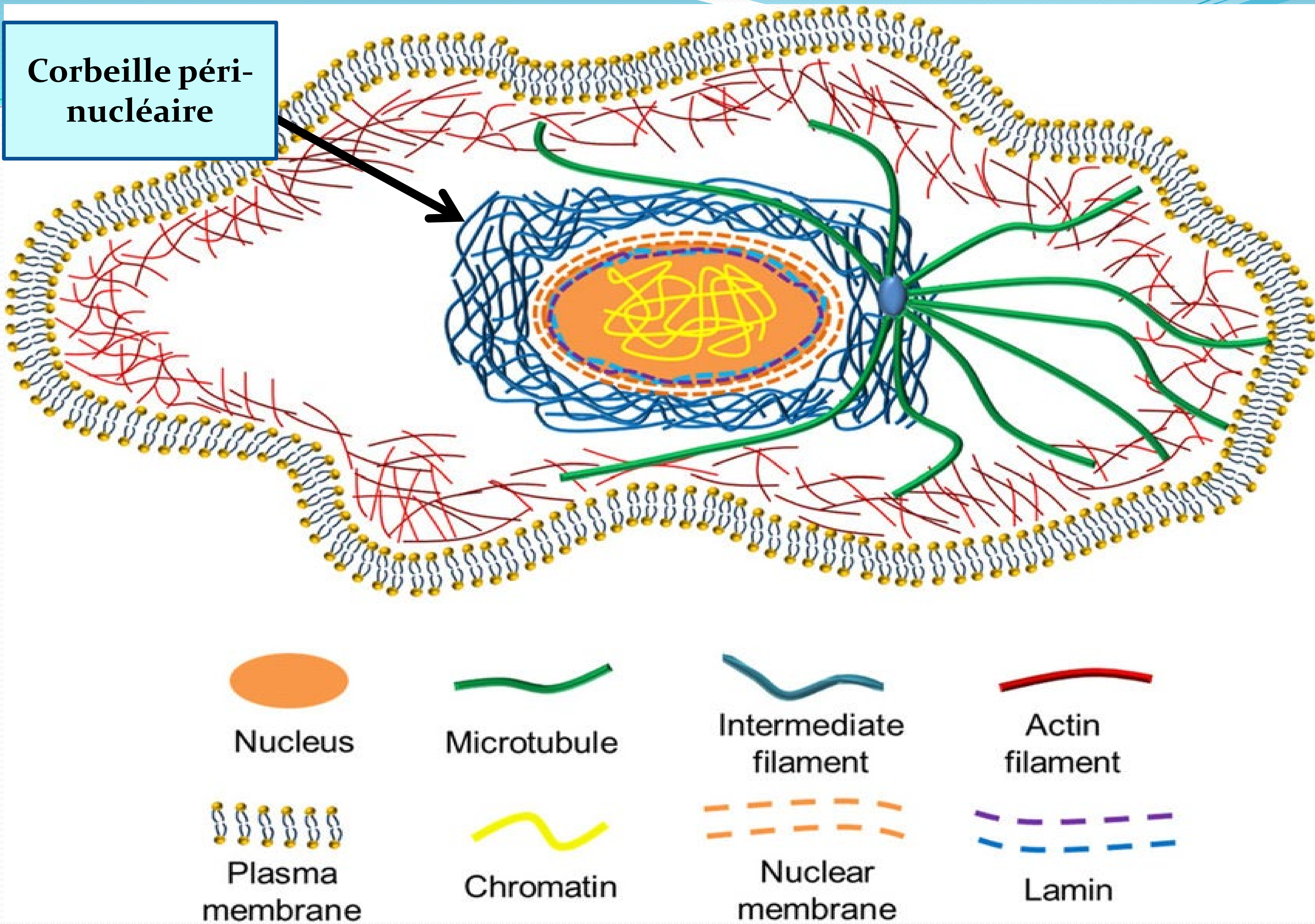
Type I IF: Cytokeratin: all epithelial cells.

Type II IF: Vimentin: in mesenchymal cells. Desmin: in muscle cells.

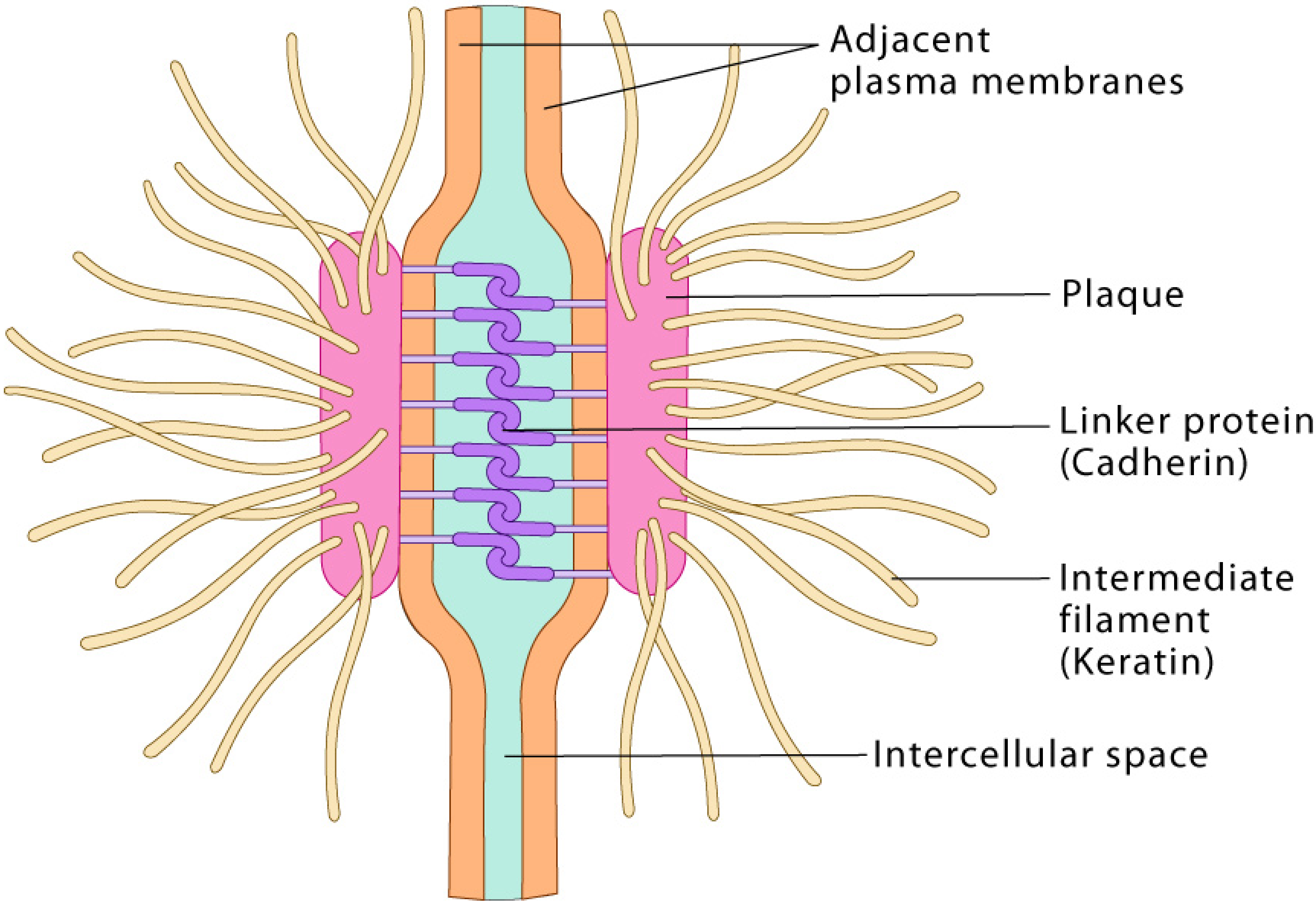
Type III IF: Neurofilaments: neurons (central and peripheral nervous system).

Type IV IF: Lamins A, B and C: found in the nucleus of all animal cells.

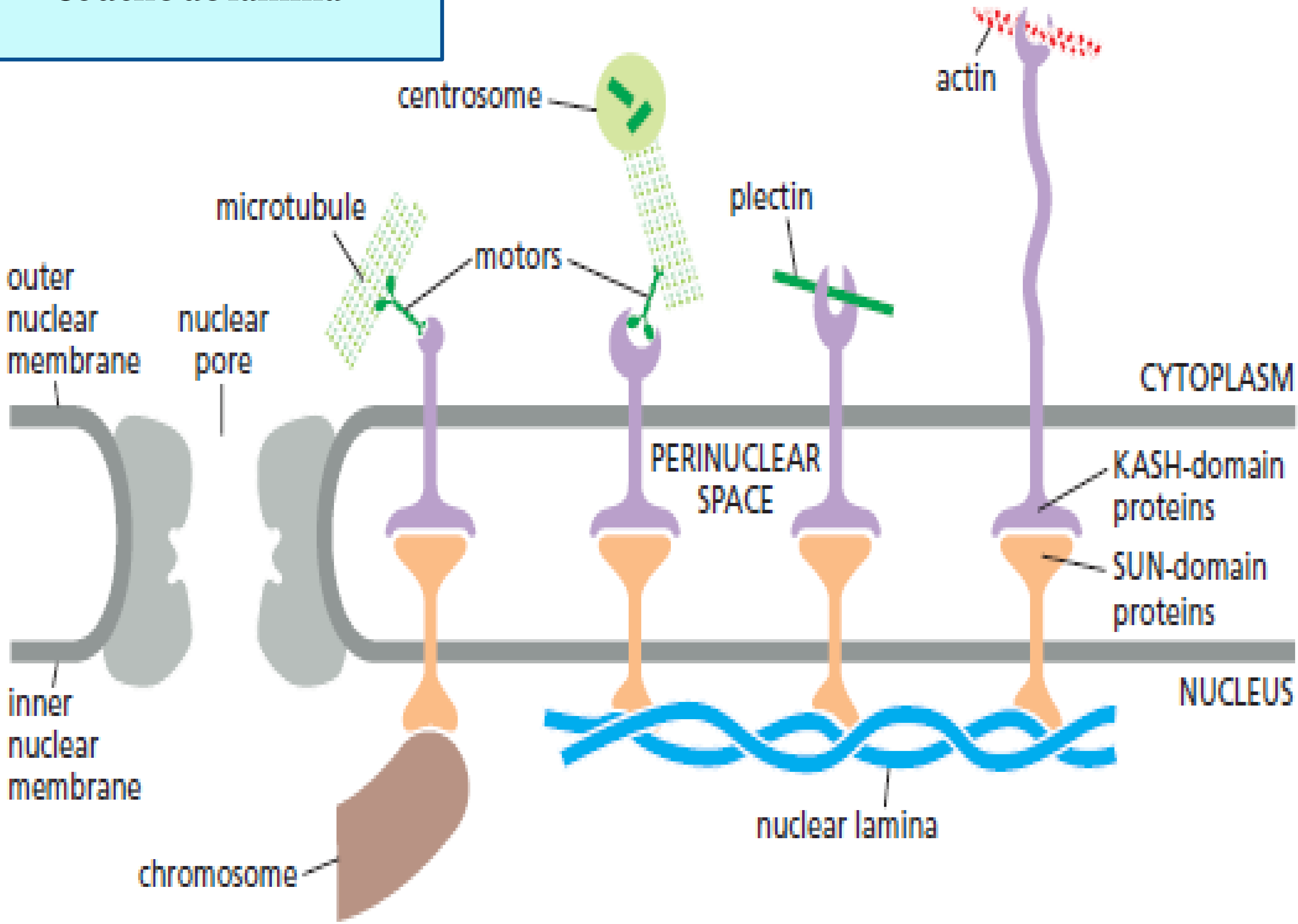
**Corbeille péri-nucléaire**

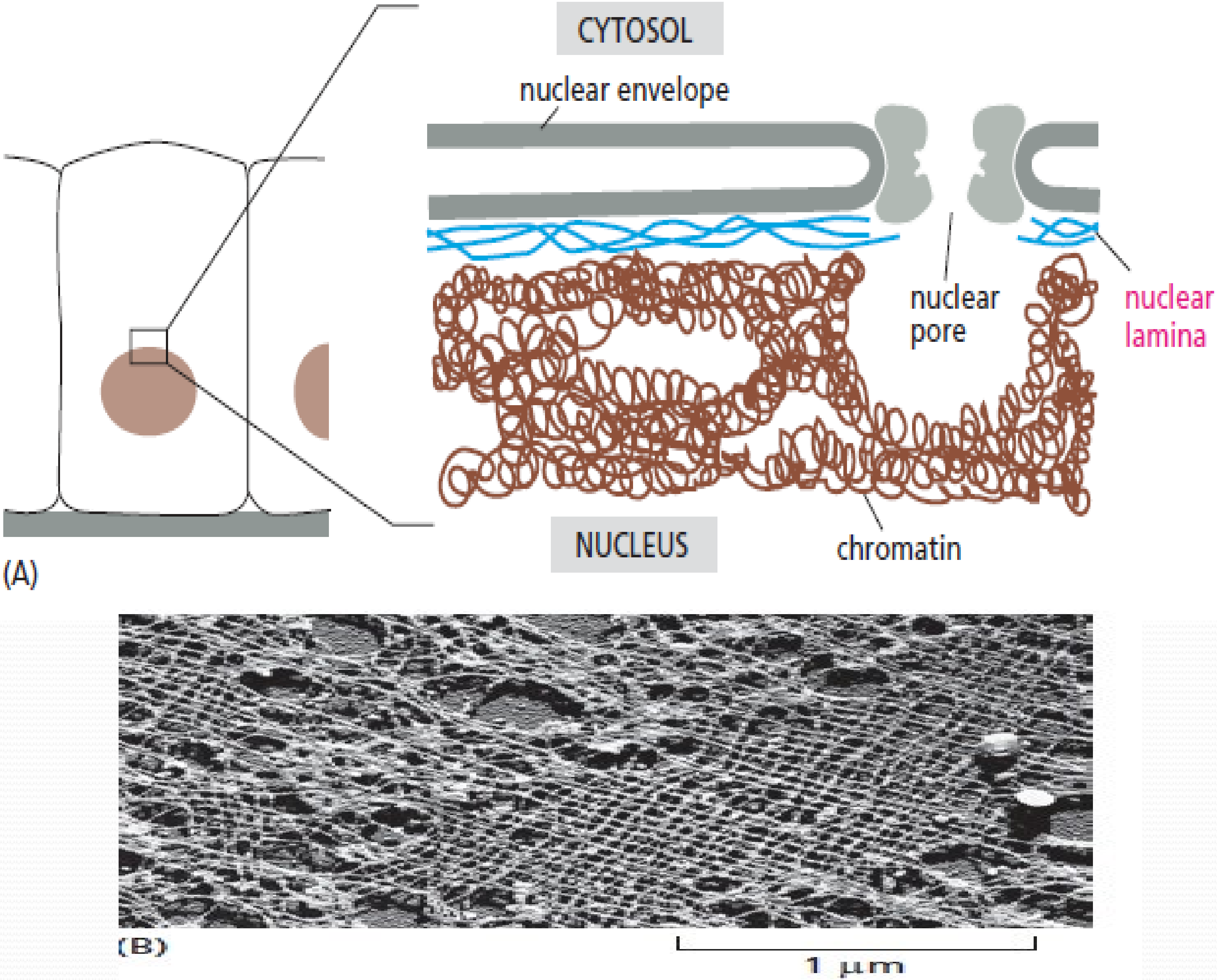


# Desmosome

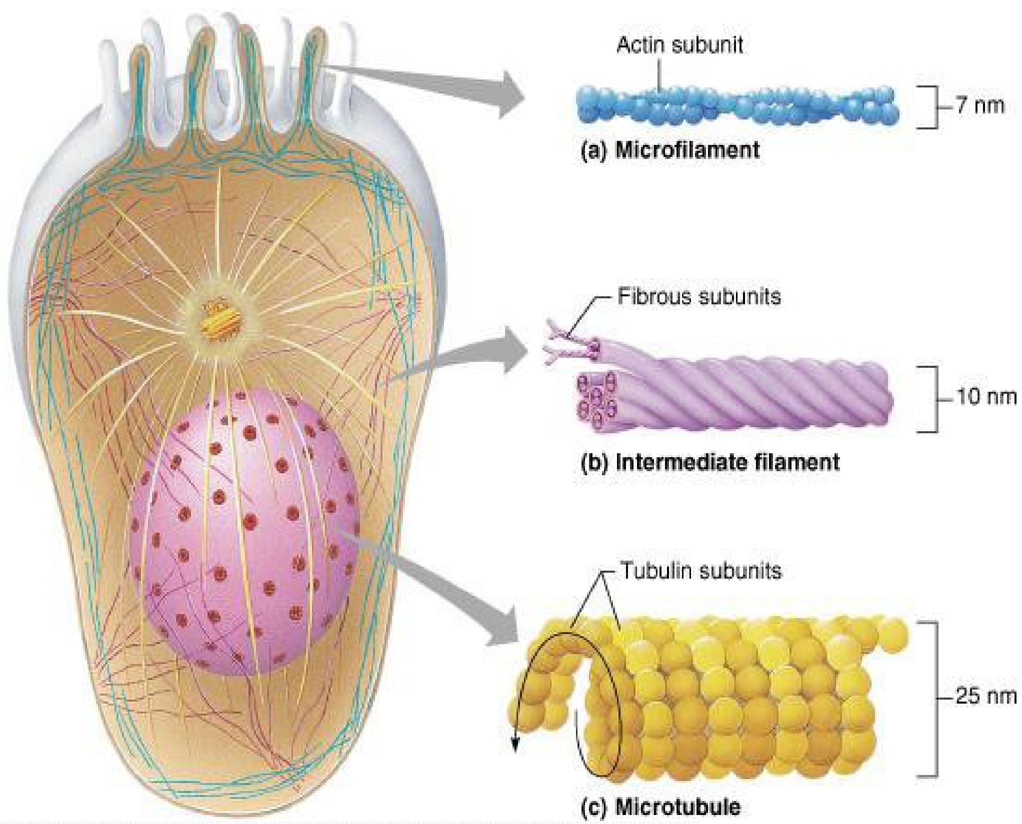

Adjacent plasma membranes
Plaque
Linker protein (Cadherin)
Intermediate filament (Keratin)
Intercellular space

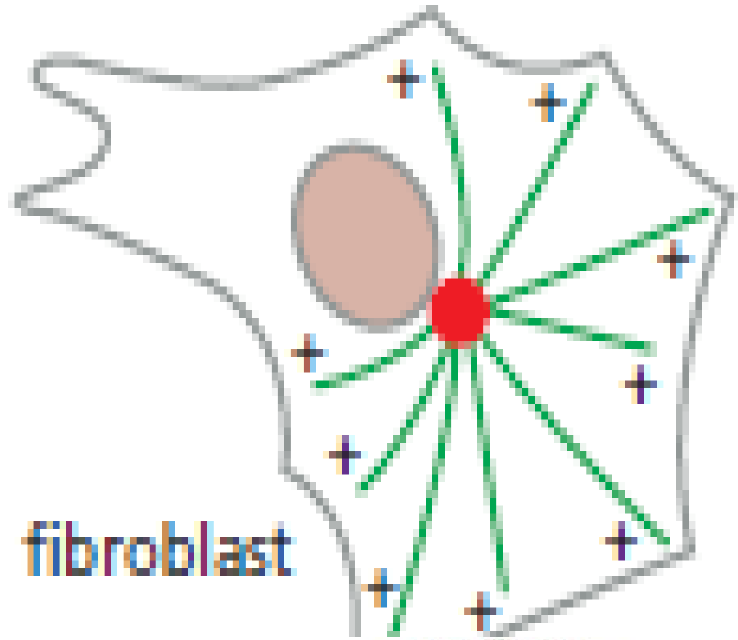
# Couche de lamina

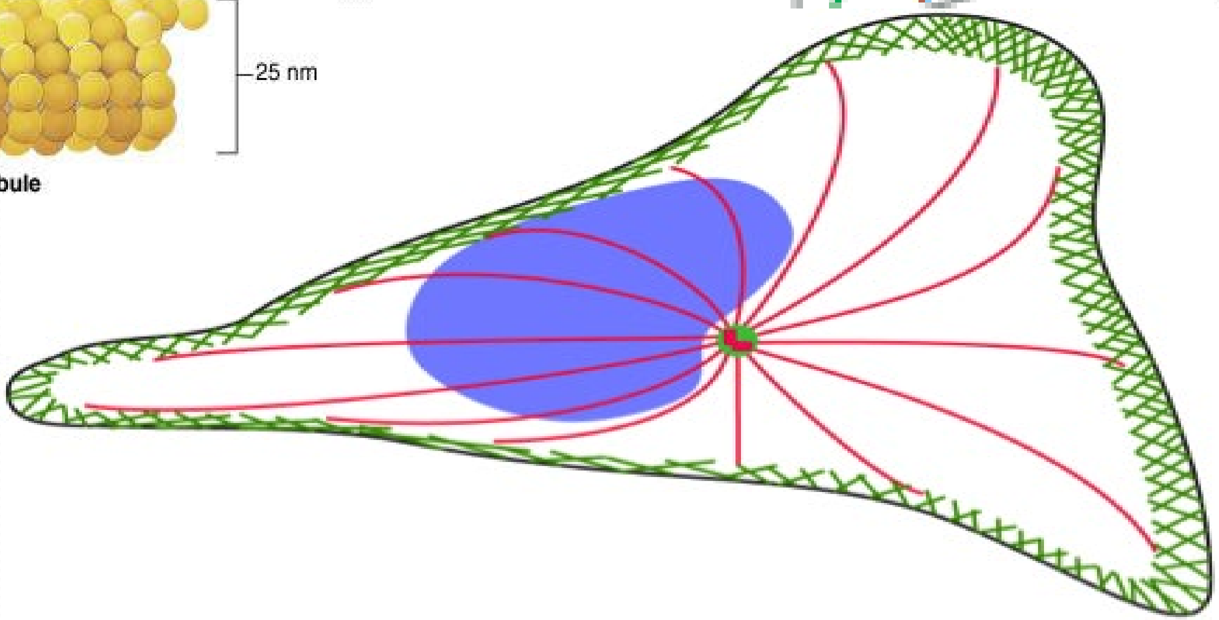



CYTOSOL
nuclear envelope
nuclear pore
nuclear lamina
chromatin
NUCLEUS
(A)
(B)
1 μm

# 3. Microtubules

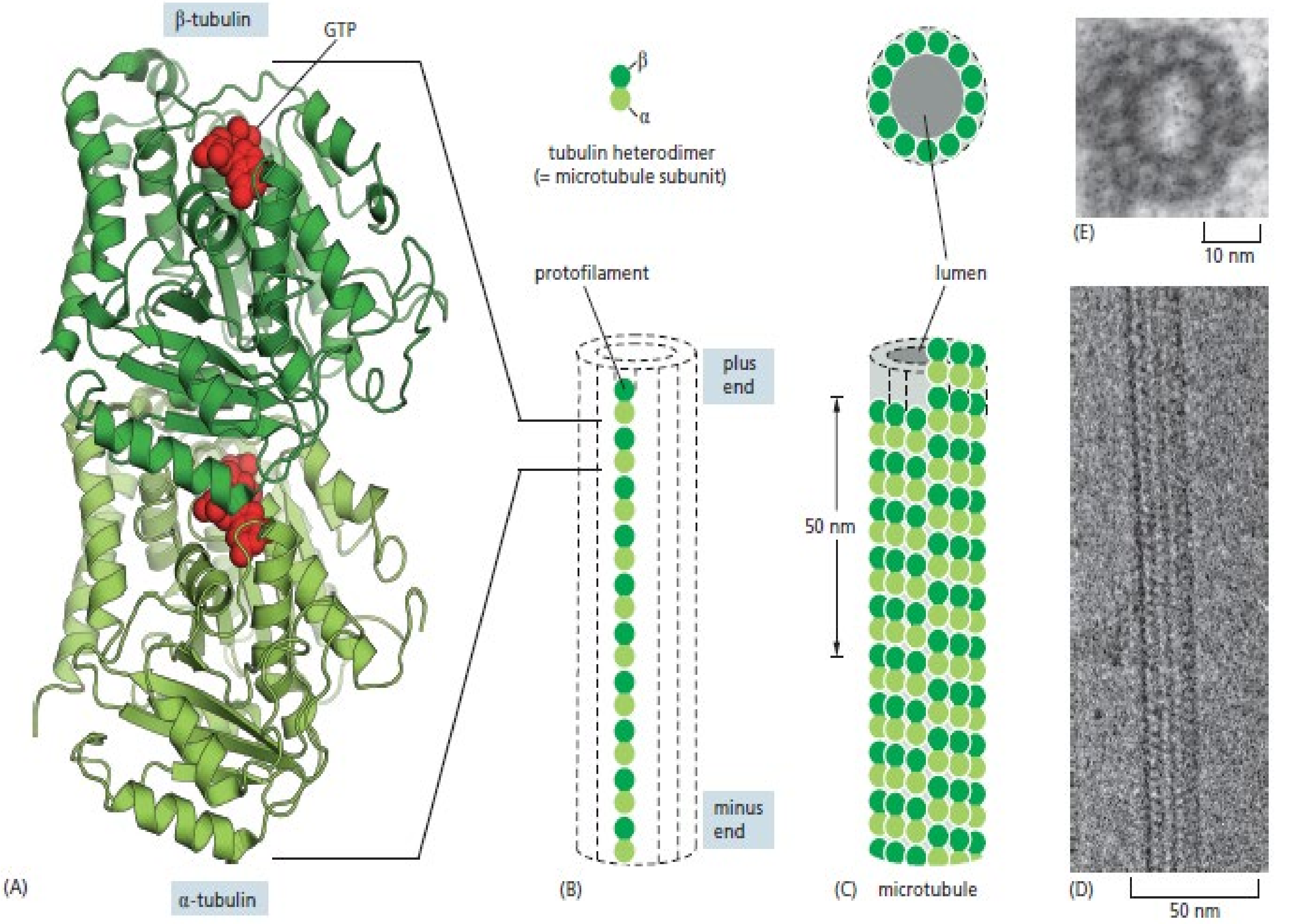


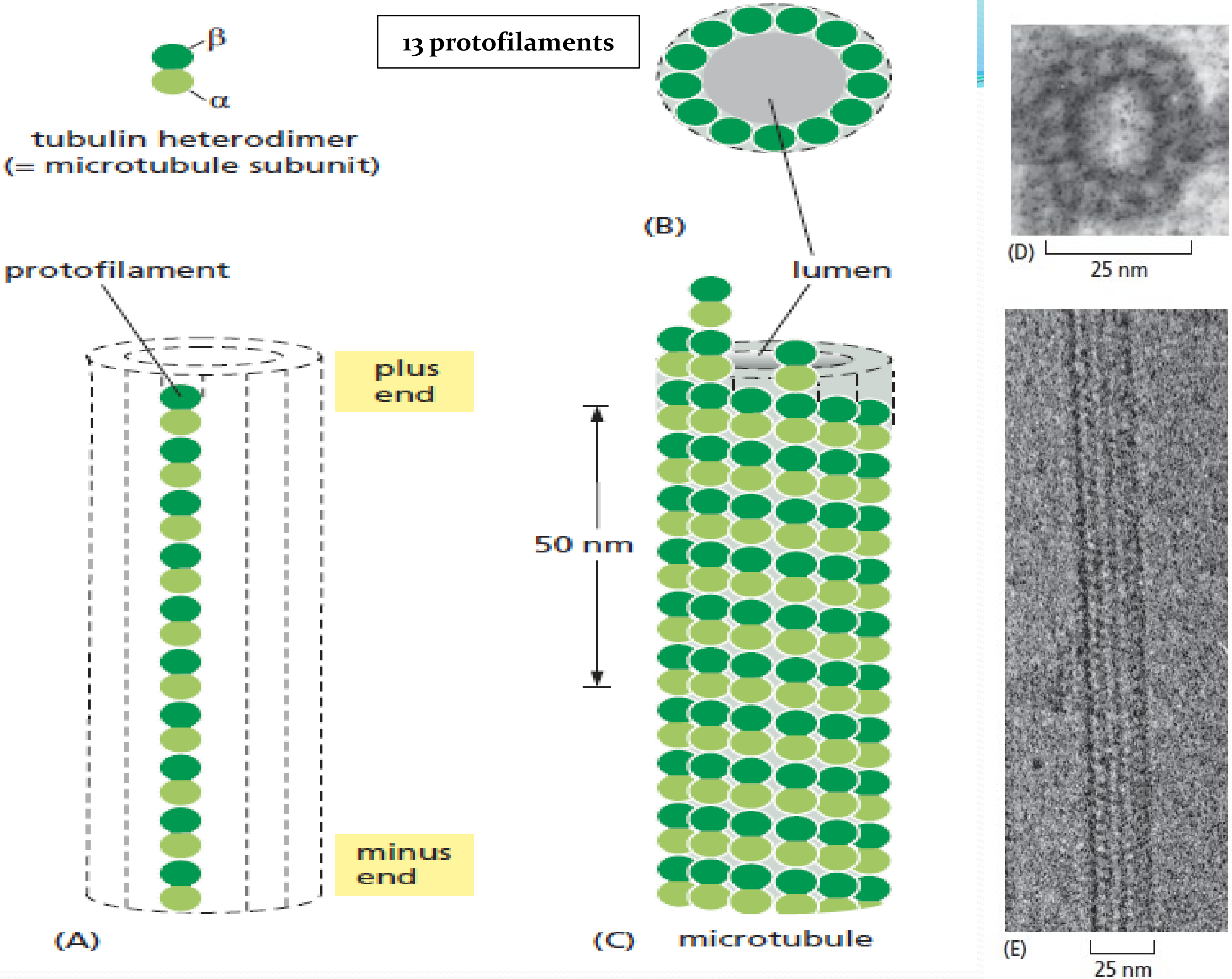



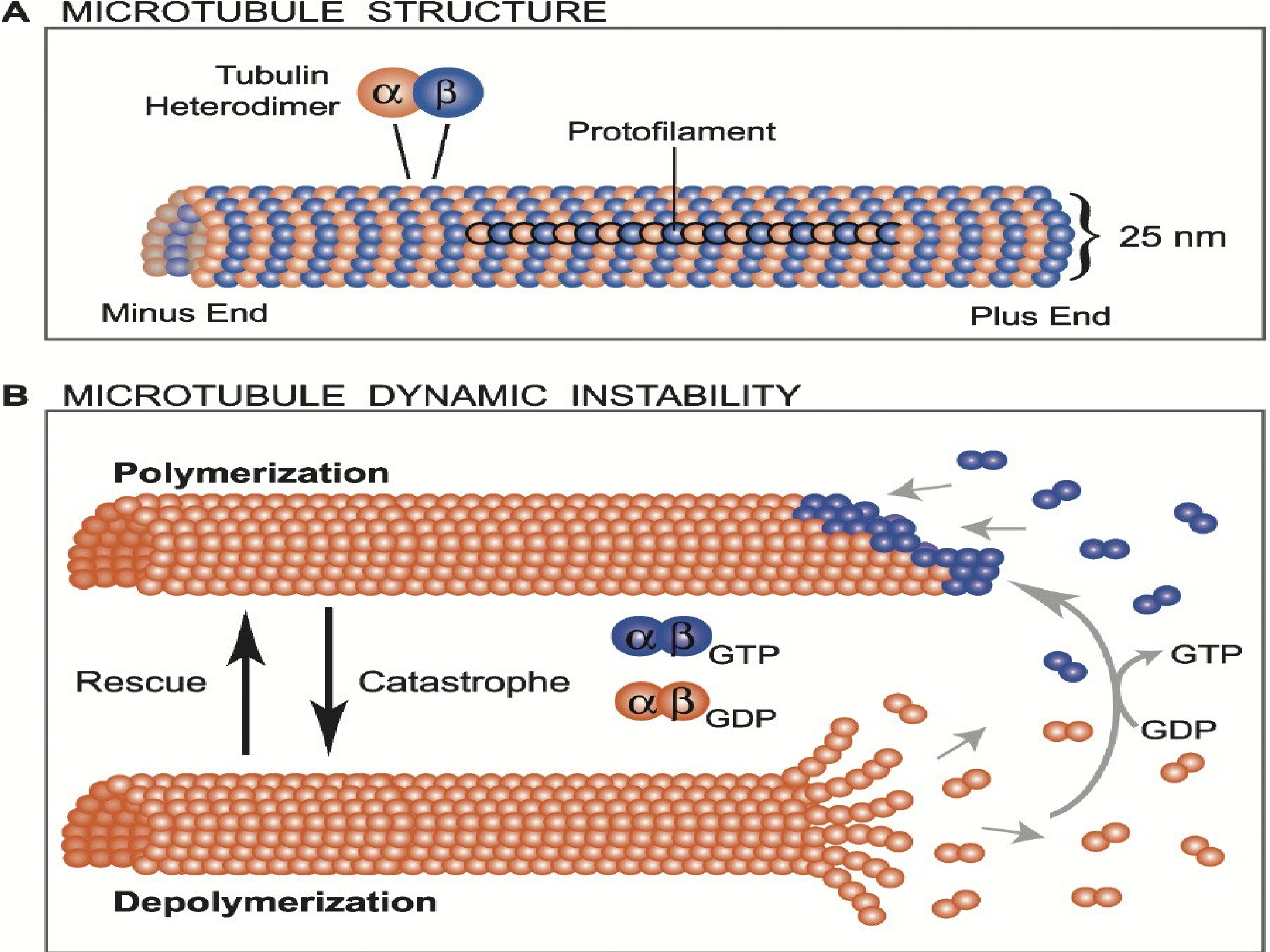



# 3. Microtubules

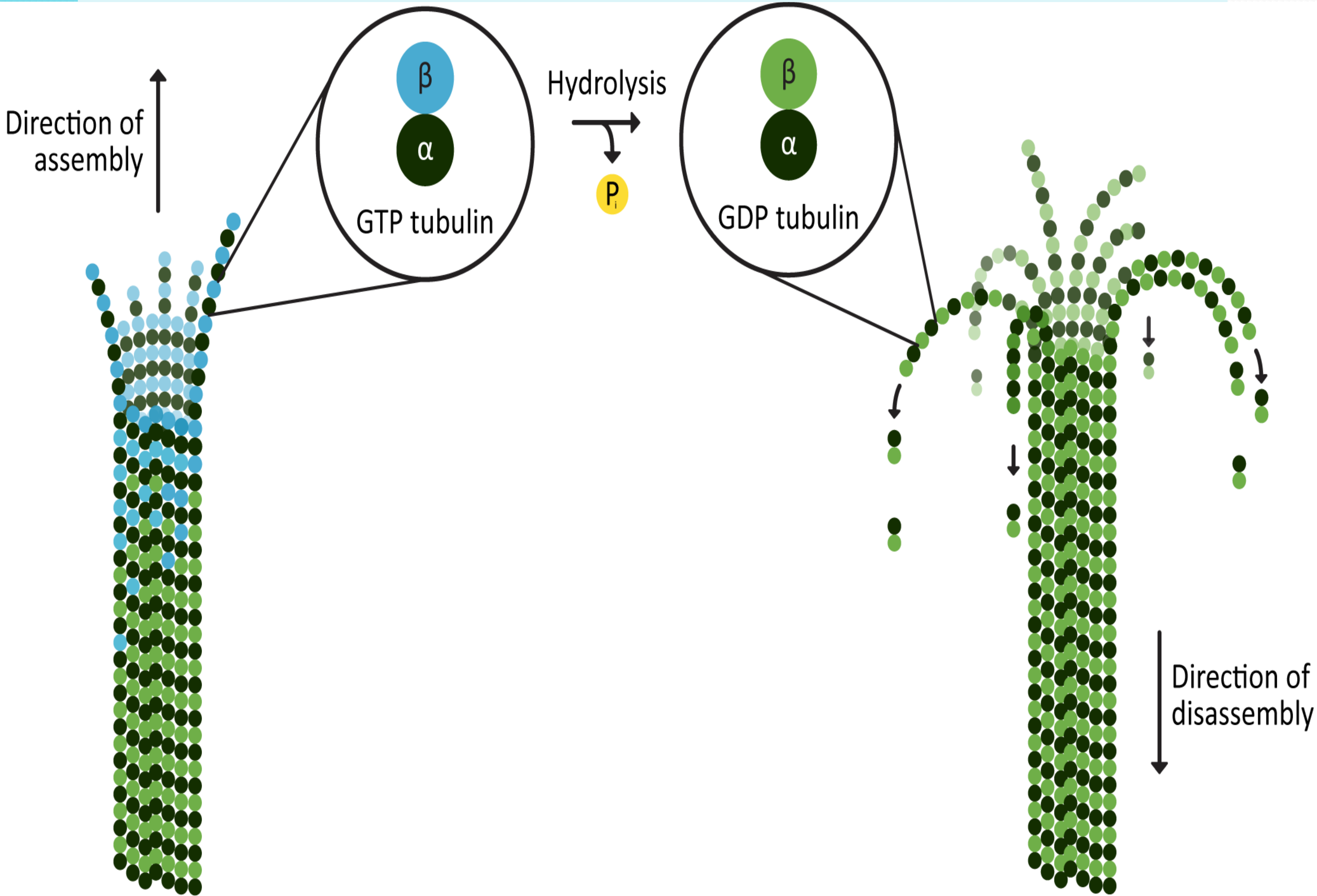
## 3.a. Structure



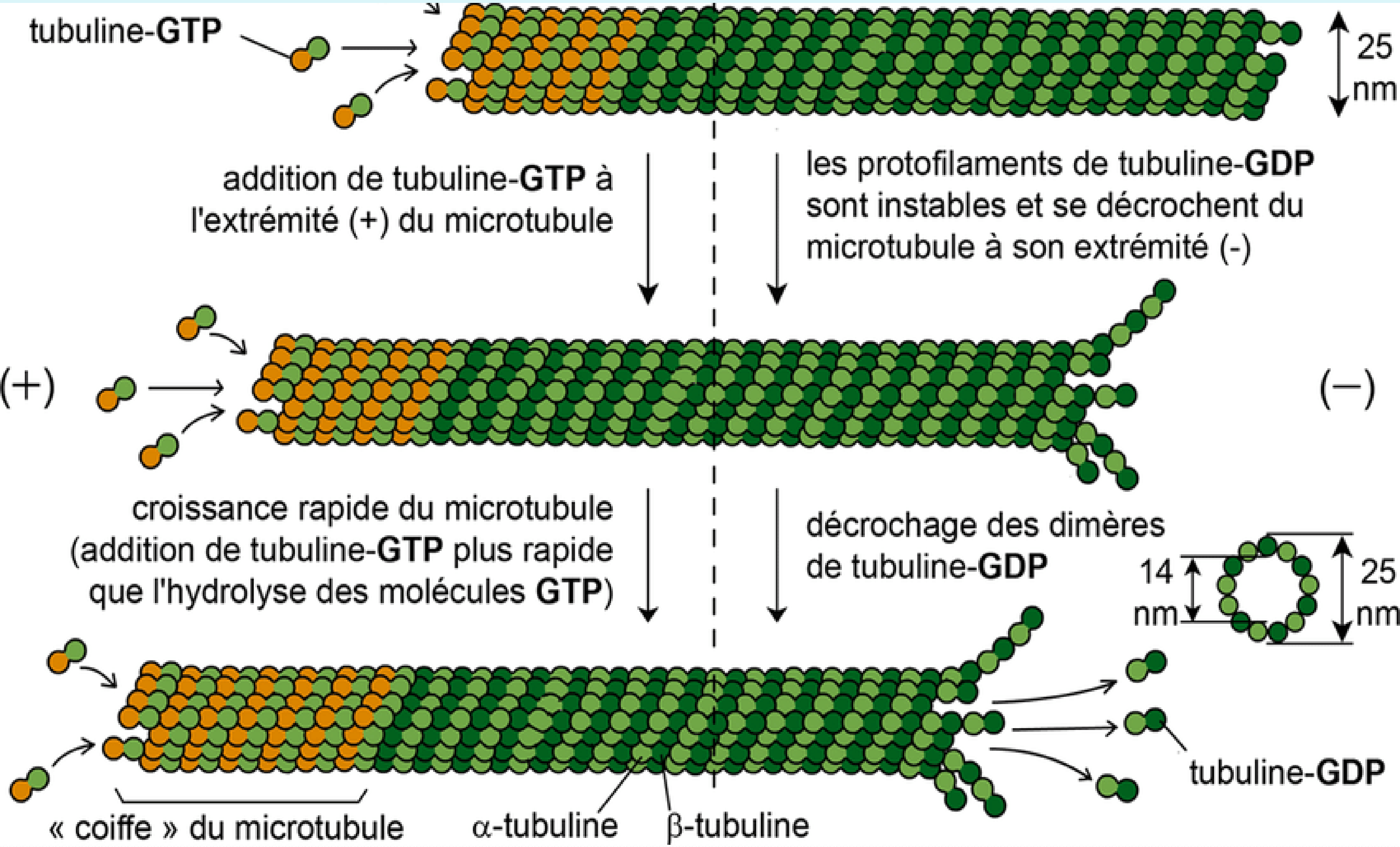

13 protofilaments
β
α
tubulin heterodimer
(= microtubule subunit)
(B)
protofilament
lumen
plus end
50 nm
minus end
(A)
(C)
microtubule
(D)
25 nm
(E)
25 nm

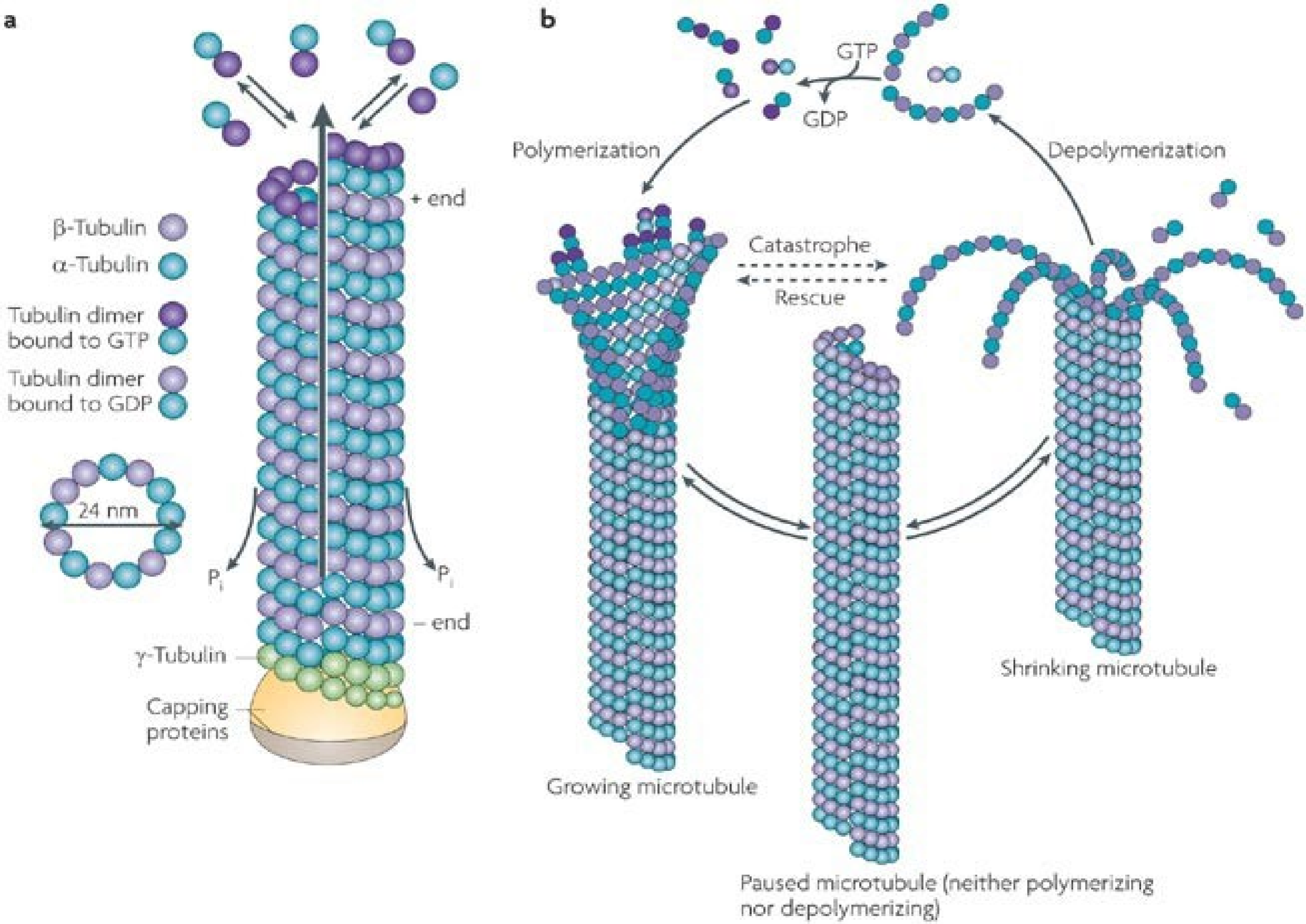

A MICROTUBULE STRUCTURE
Tubulin Heterodimer
α β
Protofilament
25 nm
Minus End
Plus End
B MICROTUBULE DYNAMIC INSTABILITY
Polymerization
Rescue
Catastrophe
α β GTP
α β GDP
GTP
GDP
Depolymerization

# 3.b. Polymérization of microtubules



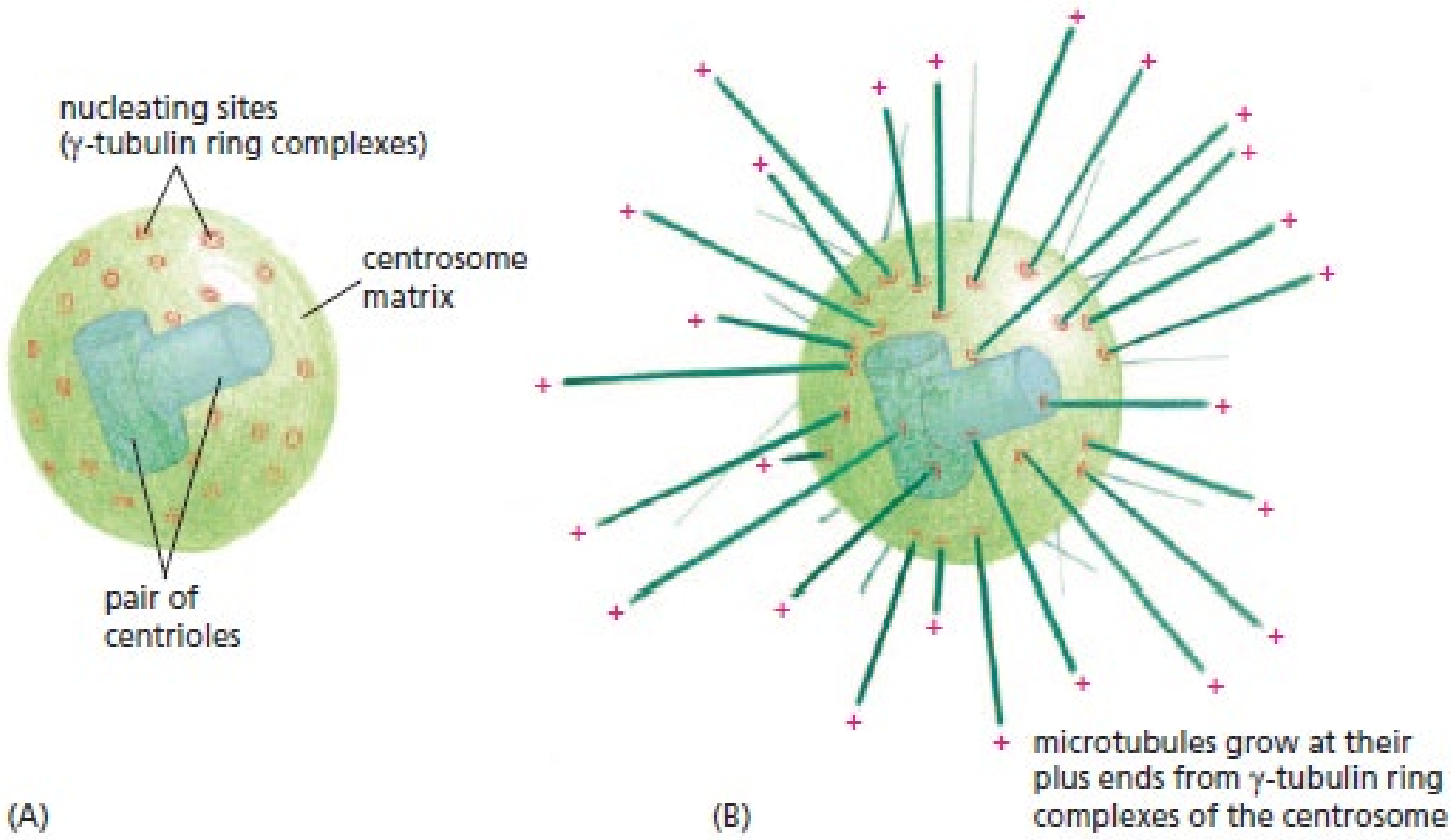
# 3.b. Polymerization and depolymerization of microtubules

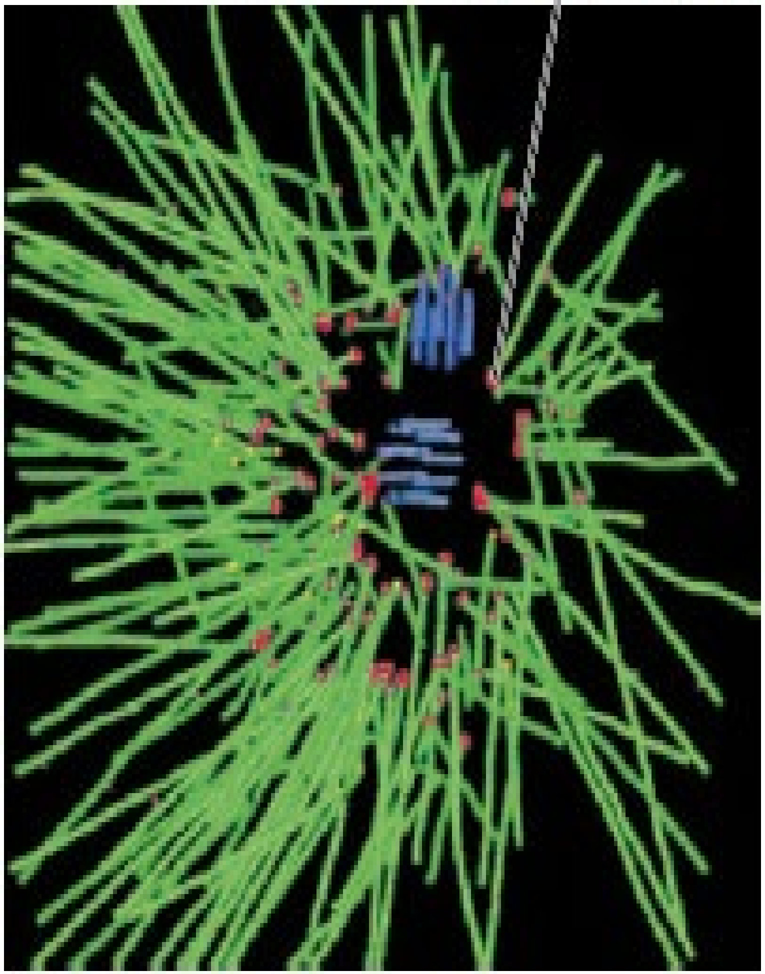


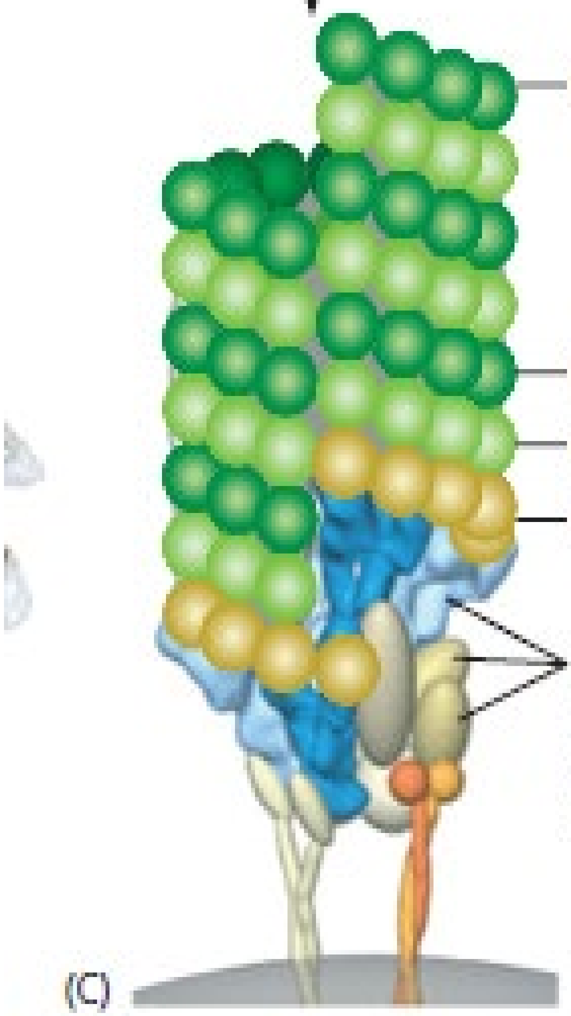


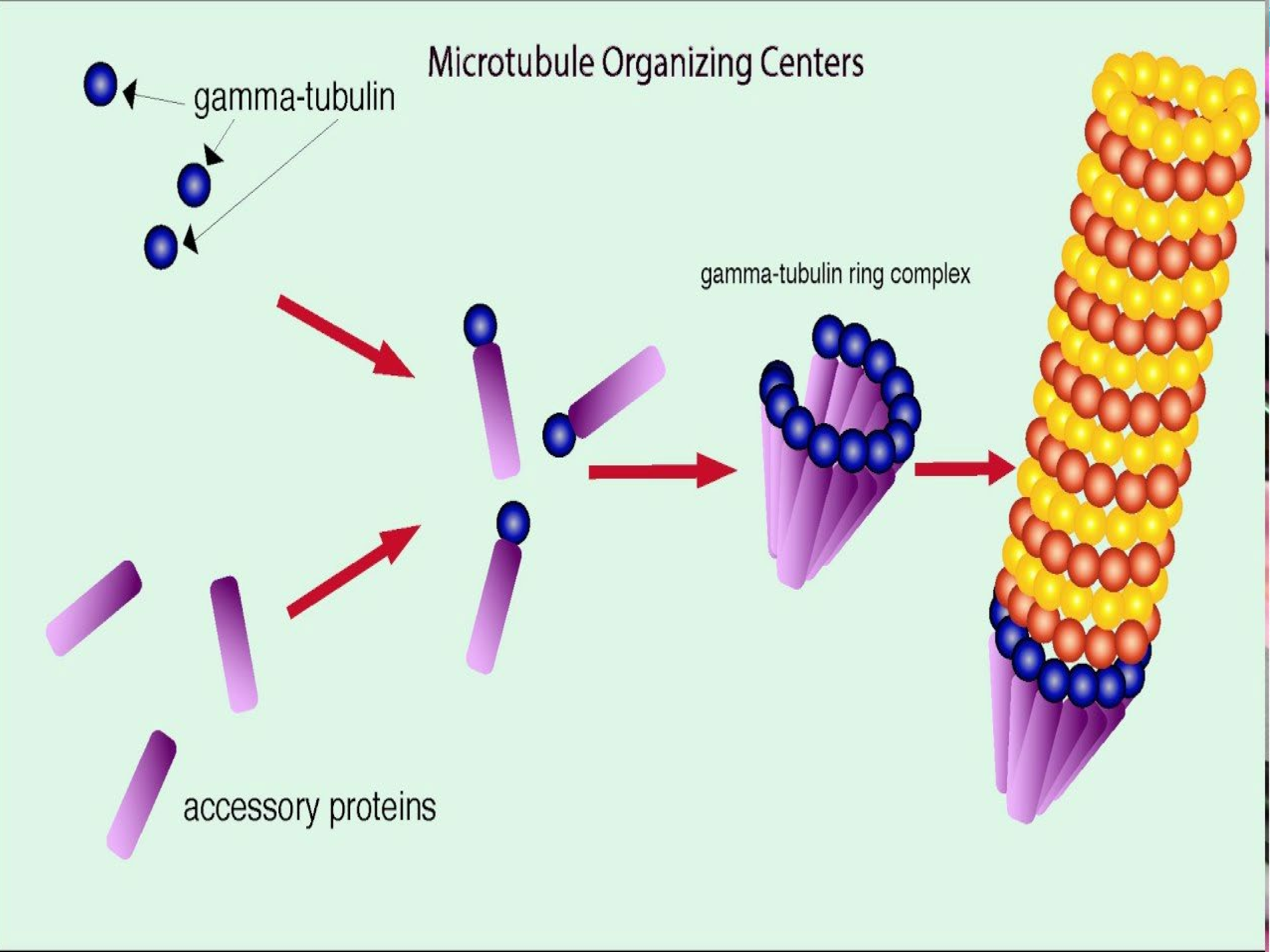

Nature Reviews | Neuroscience

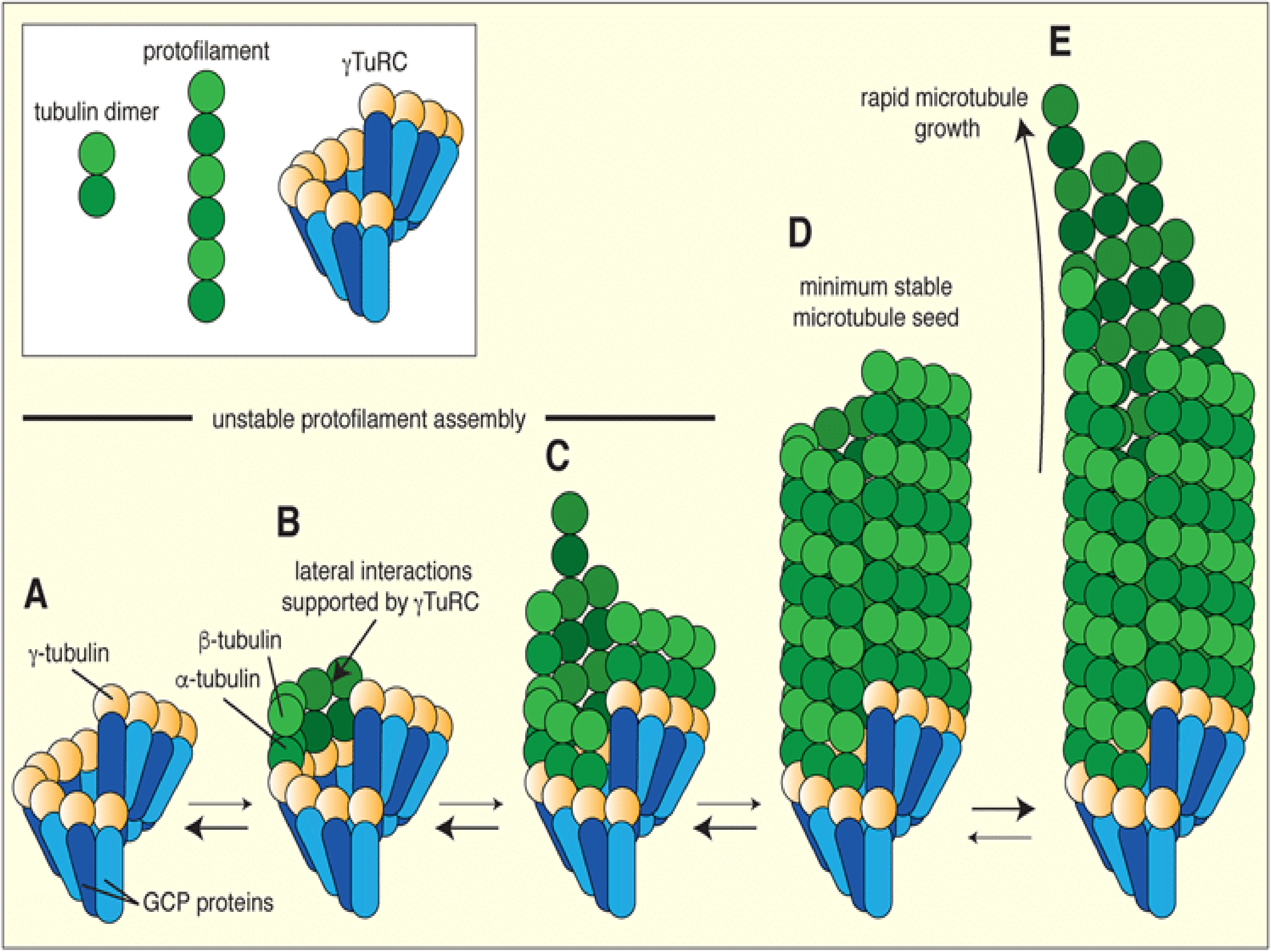
# 3.c.Tubulin polymerizes from nucleation sites on a centrosome.



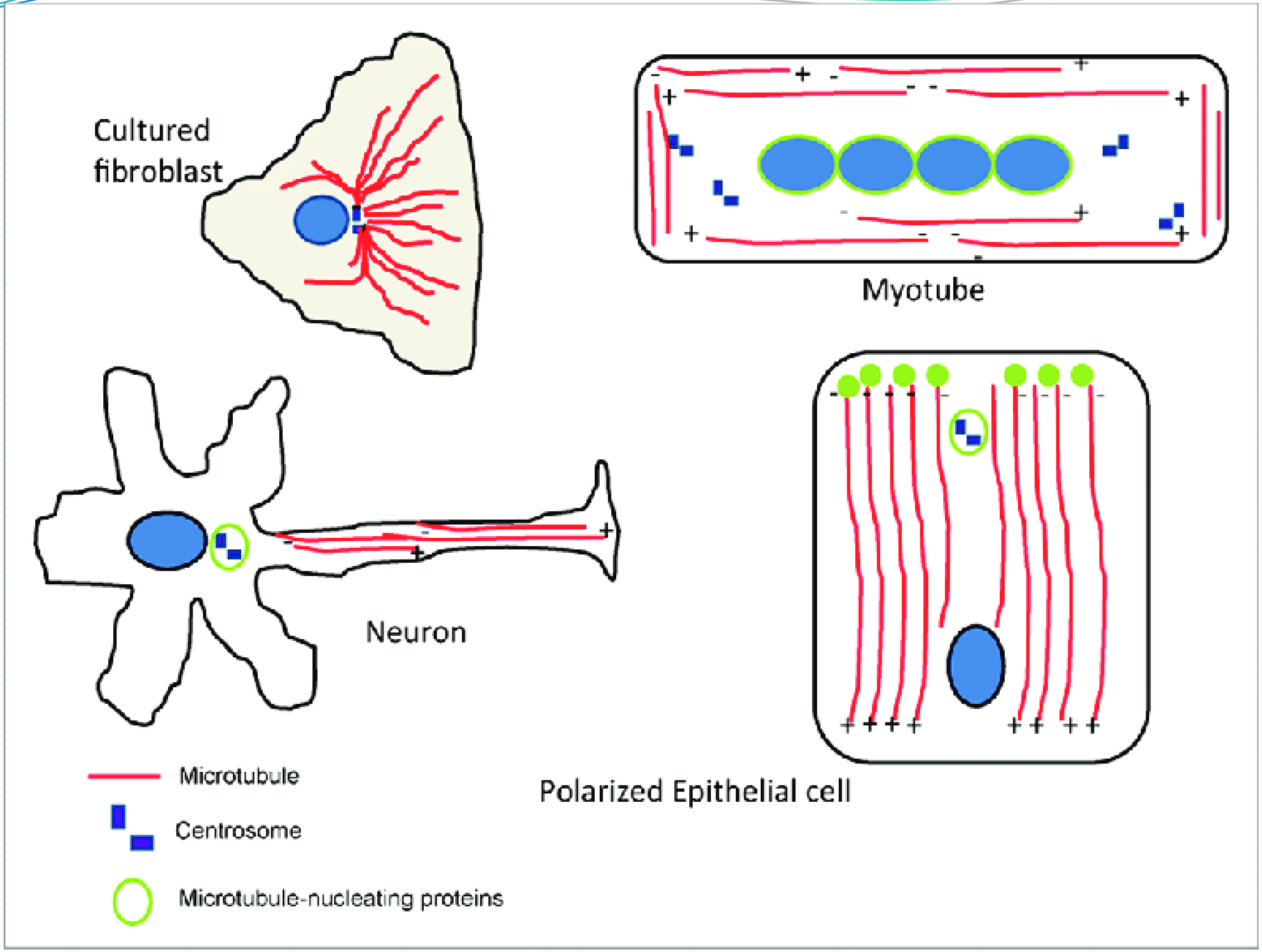

γ-tubulin ring complex (red)
(C)


microtubule "seam"
plus end
β-tubulin
α-tubulin
γ-tubulin
accessory proteins in γ-tubulin ring complex
(C)


Microtubule Organizing Centers
gamma-tubulin
gamma-tubulin ring complex
accessory proteins

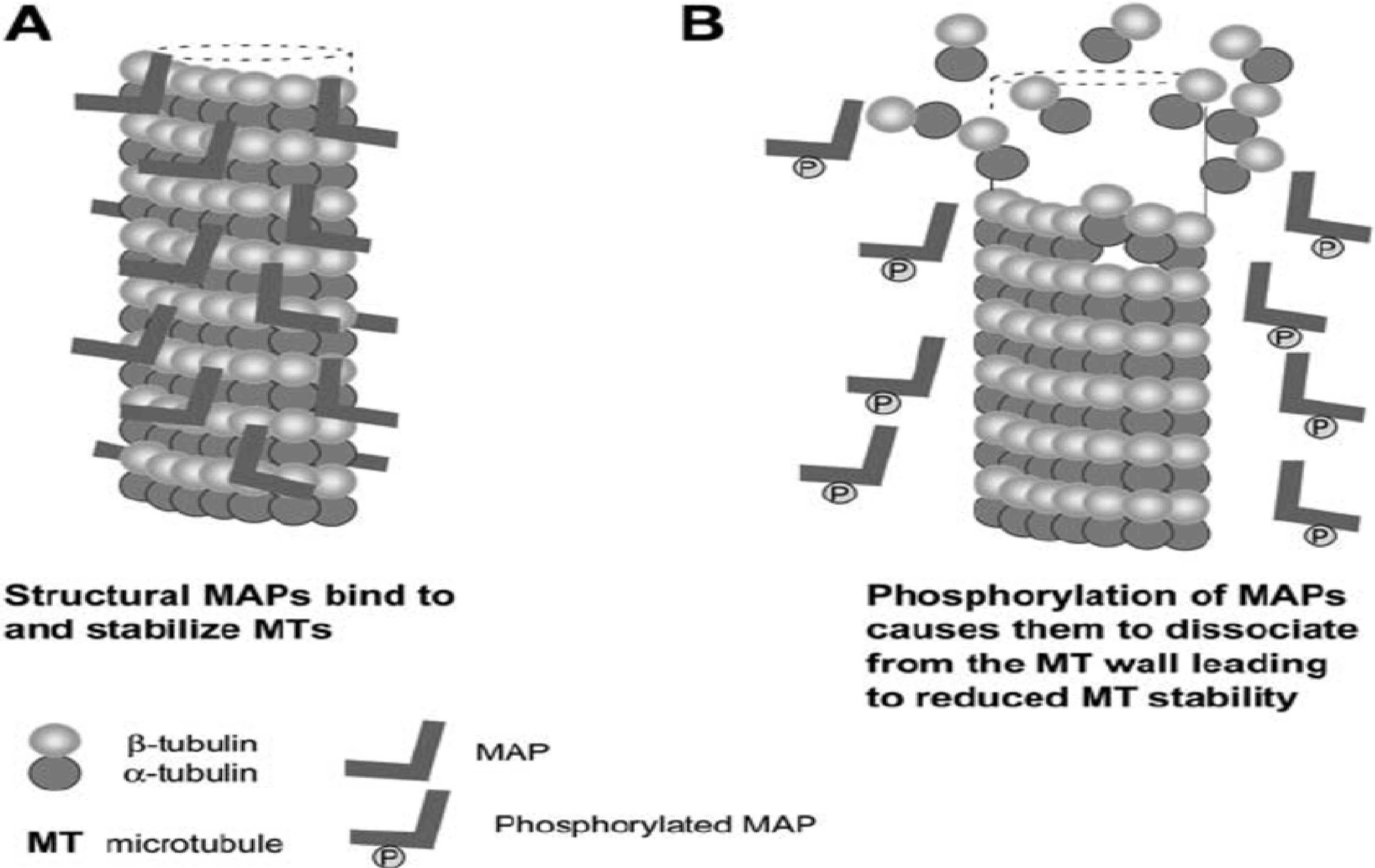


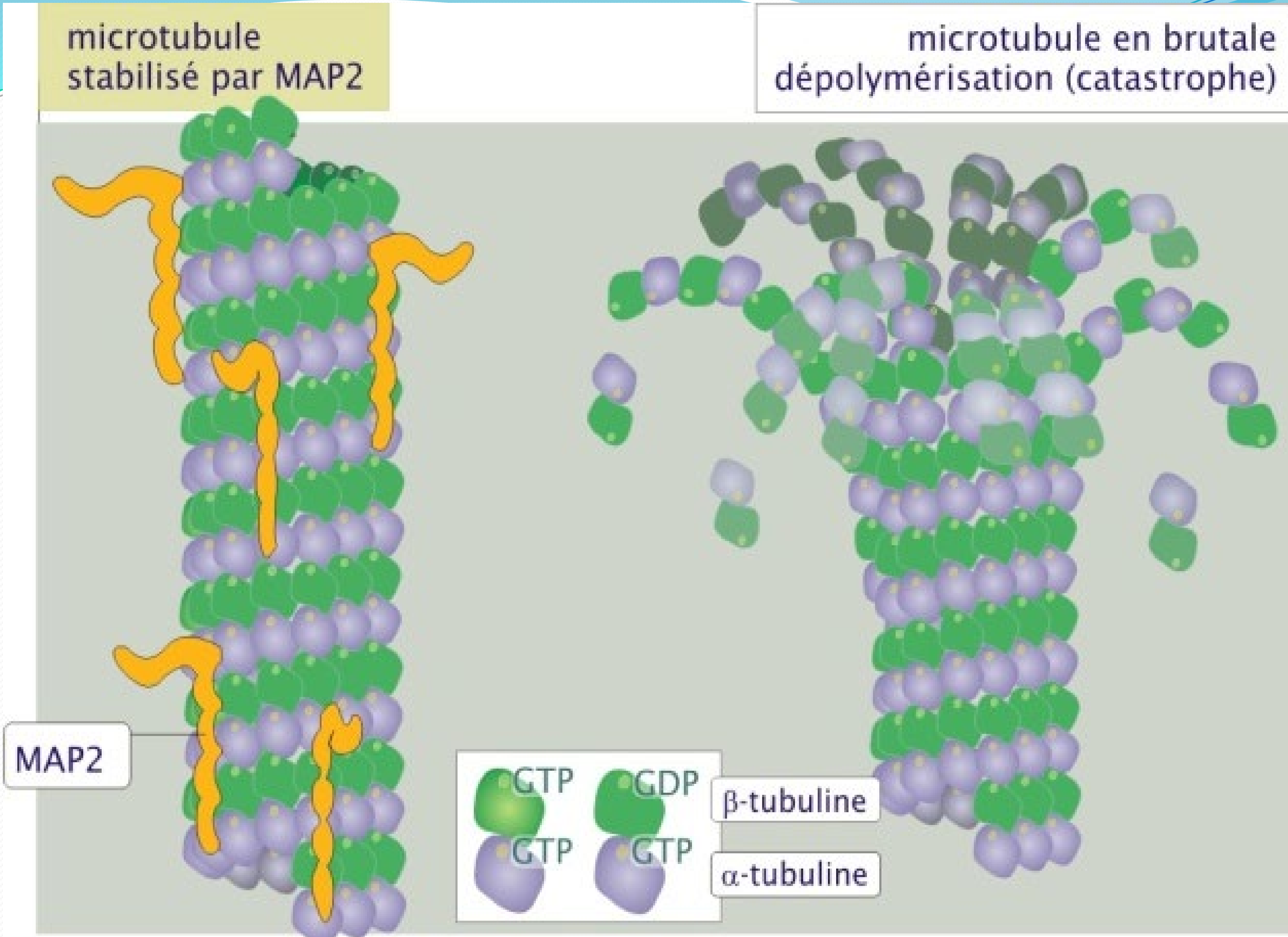
tubulin dimer
protofilament
γTuRC
E
rapid microtubule growth
D
minimum stable microtubule seed
unstable protofilament assembly
C
B
lateral interactions supported by γTuRC
A
γ-tubulin
β-tubulin
α-tubulin
GCP proteins

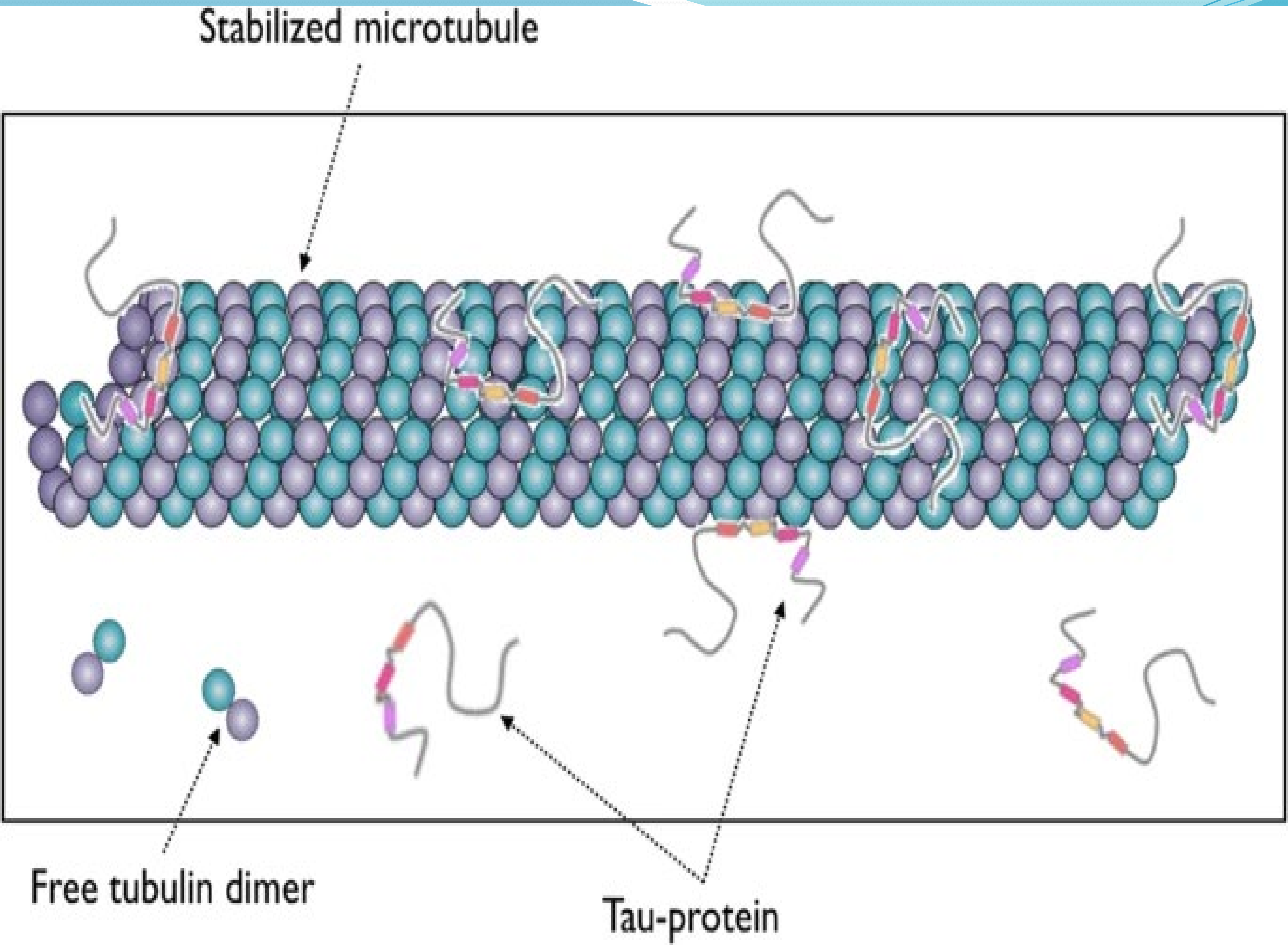


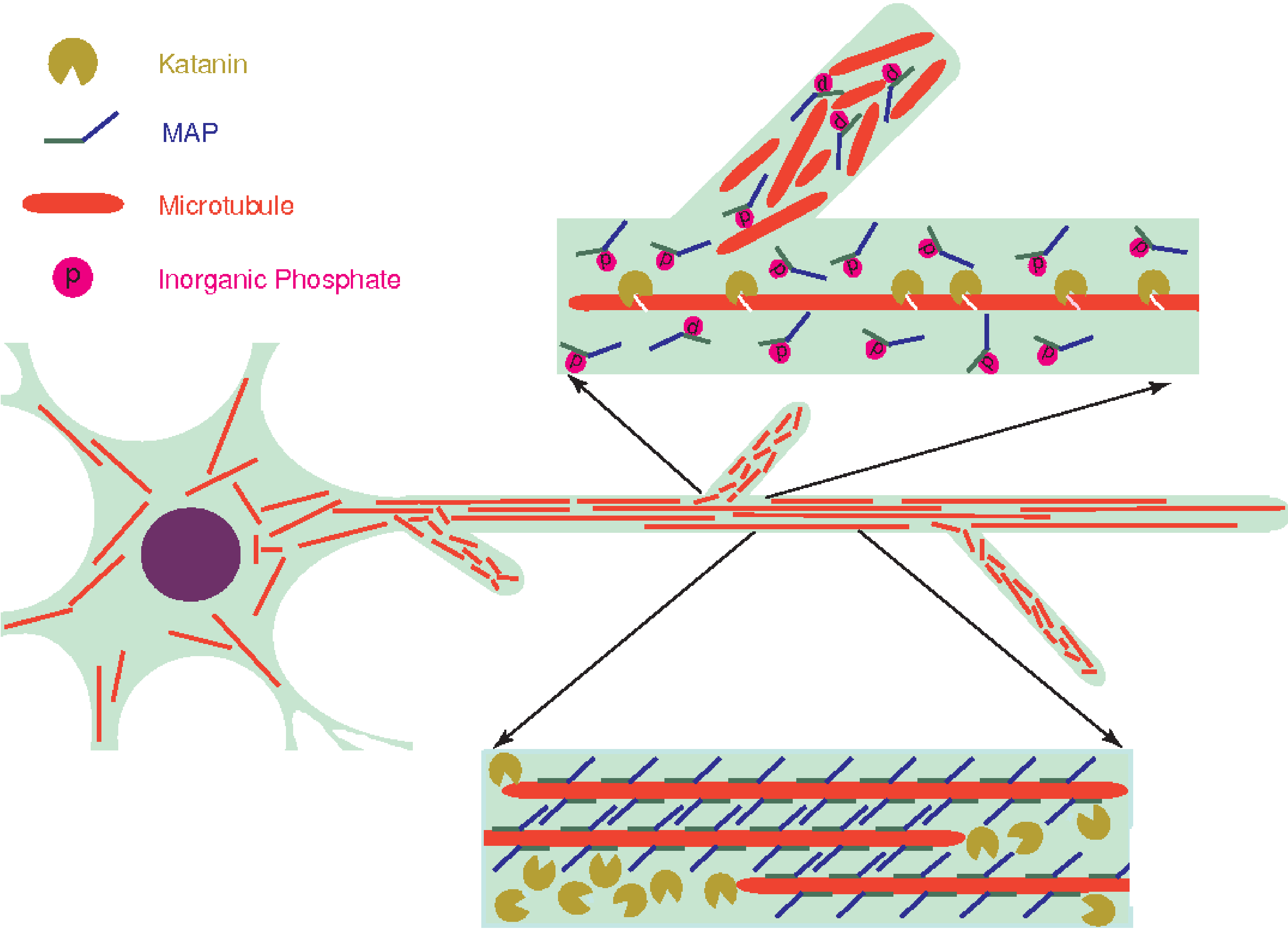
Cultured fibroblast
Myotube
Neuron
Polarized Epithelial cell
Microtubule
Centrosome
Microtubule-nucleating proteins

# 3.D. Microtubule associated proteins (MAP)

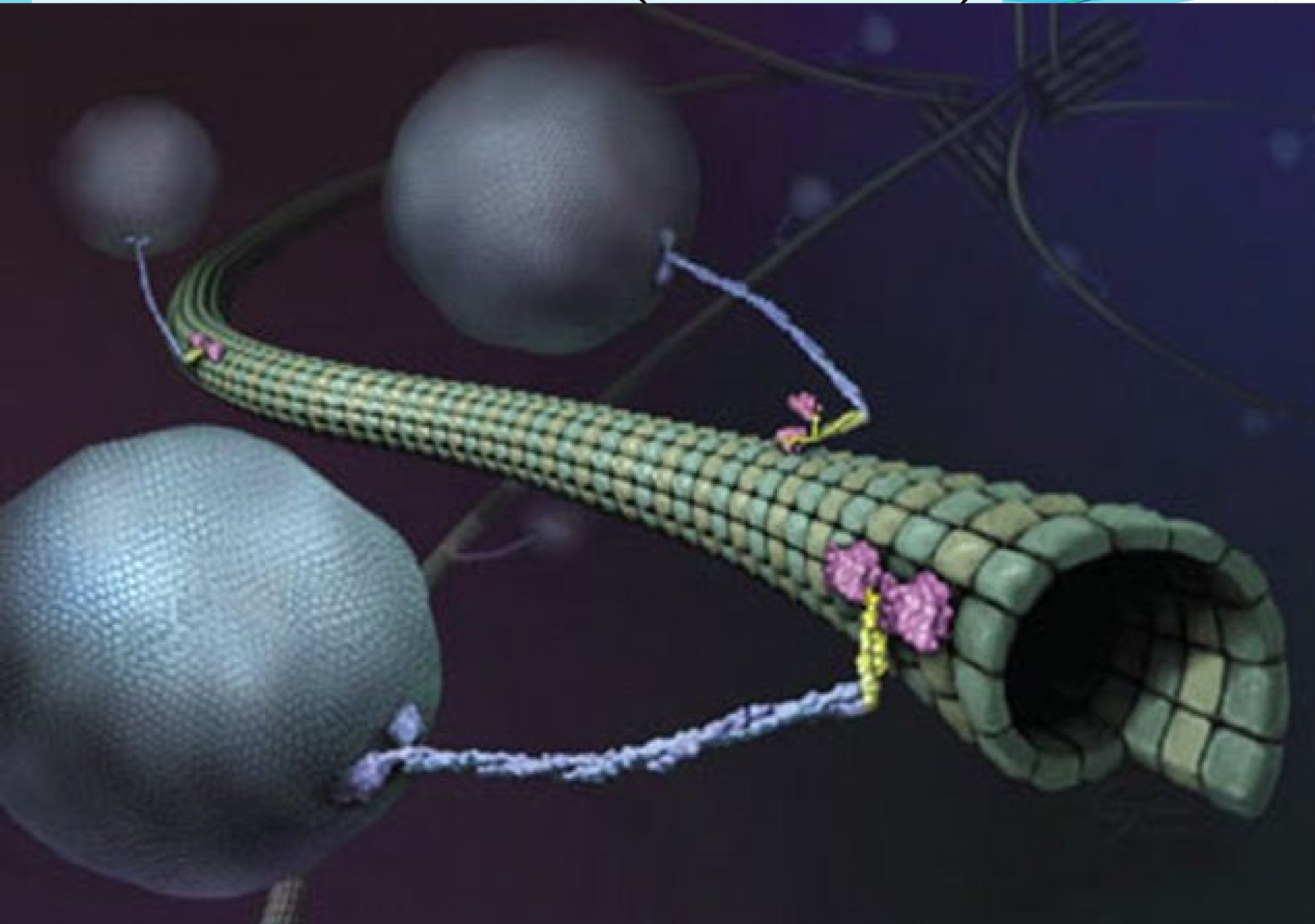


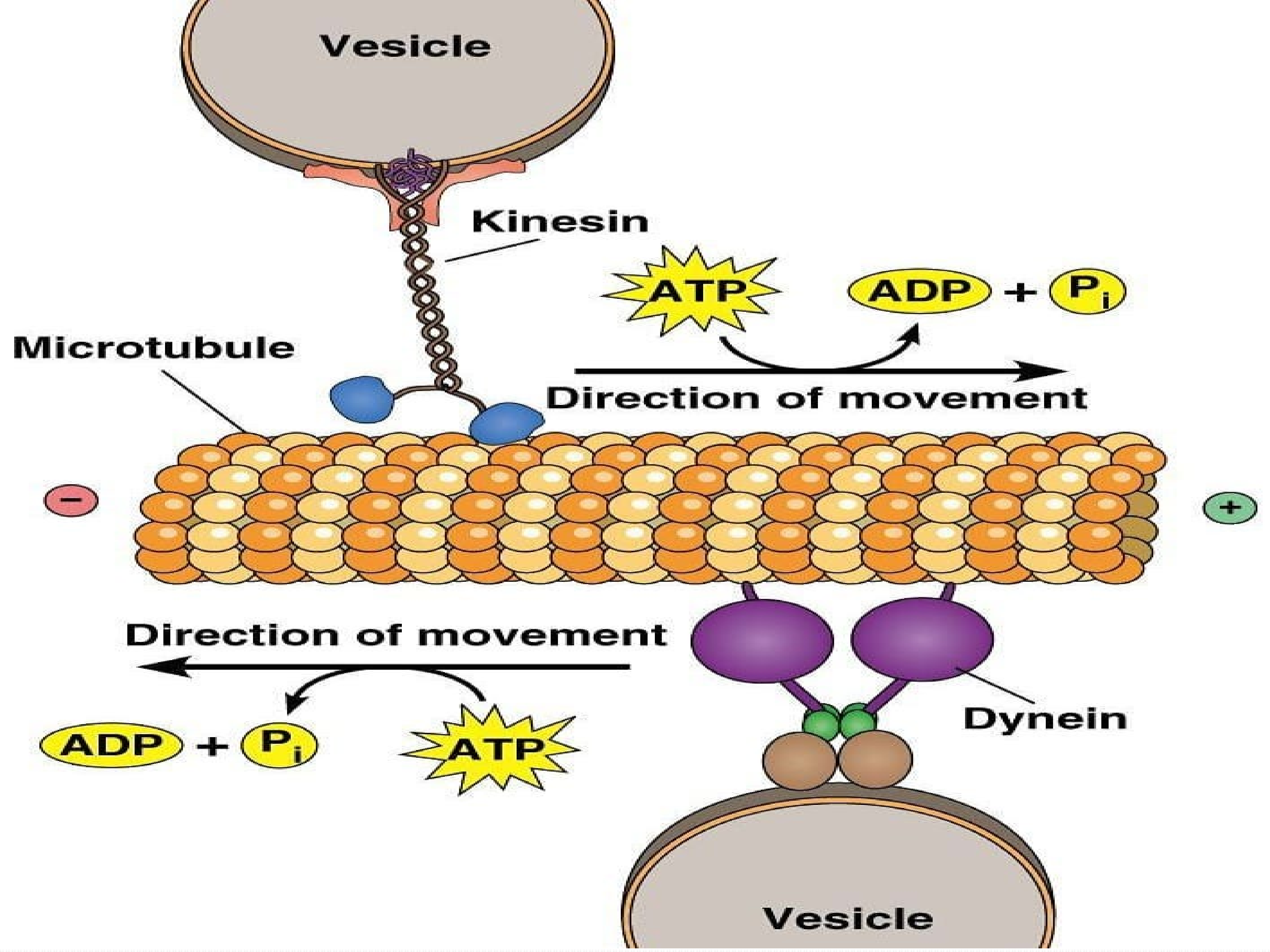

microtubule
stabilisé par MAP2
microtubule en brutale
dépolymérisation (catastrophe)
MAP2
GTP
GDP
GTP
GTP
β-tubuline
α-tubuline

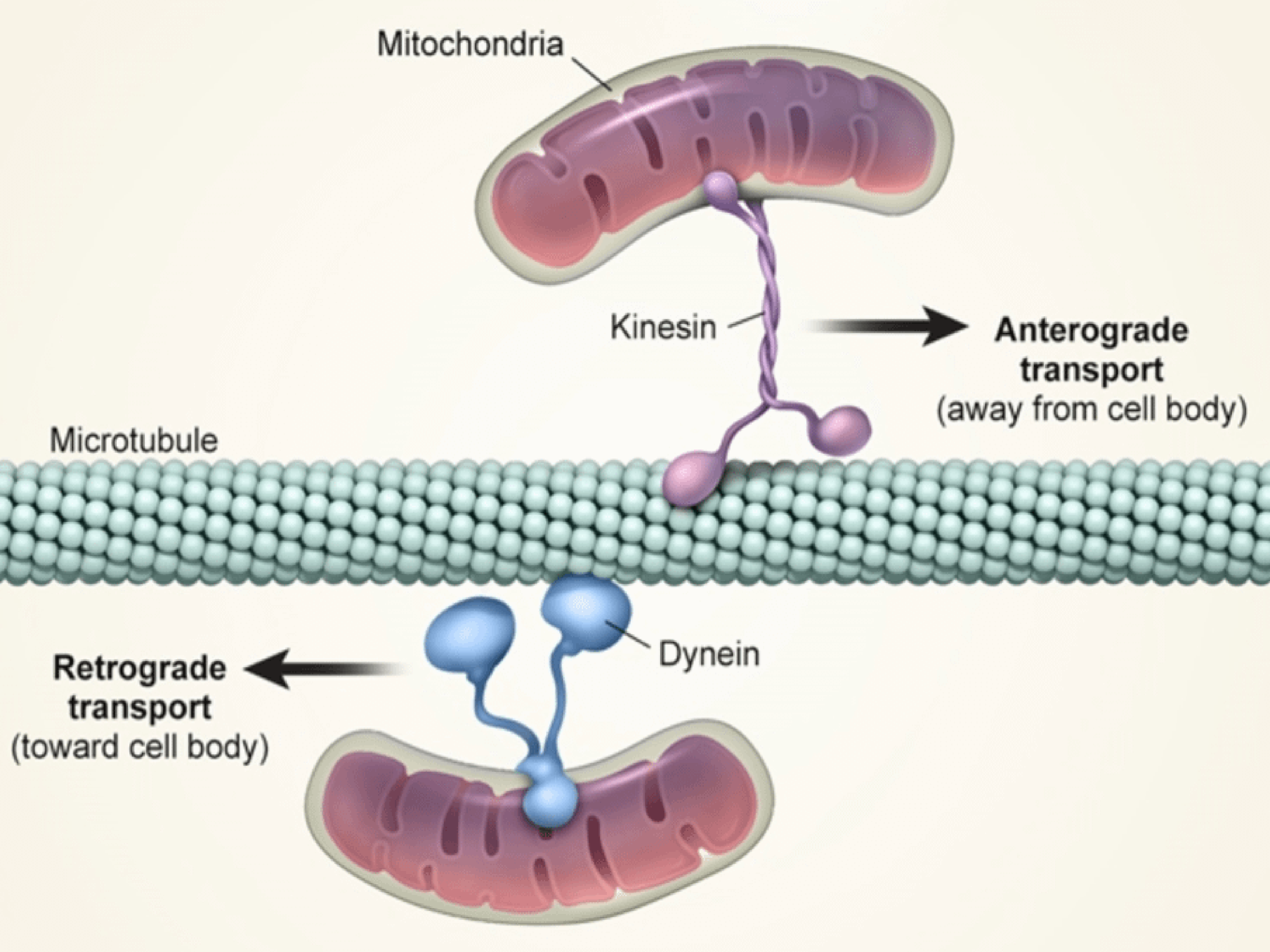

Stabilized microtubule
Free tubulin dimer
Tau-protein


Katanin
MAP
Microtubule
Inorganic Phosphate
p

# 3.E. Motor Proteins (Movement)

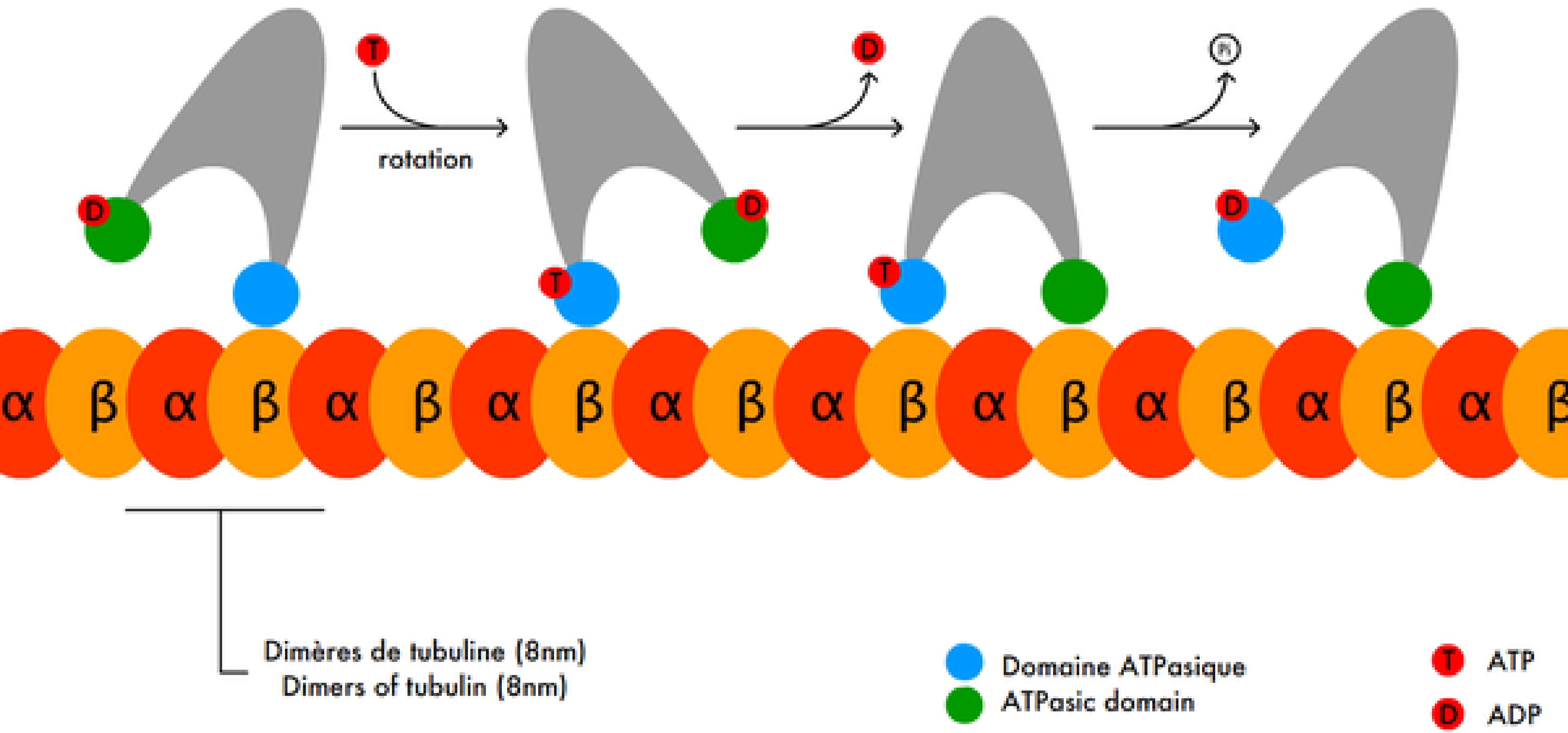


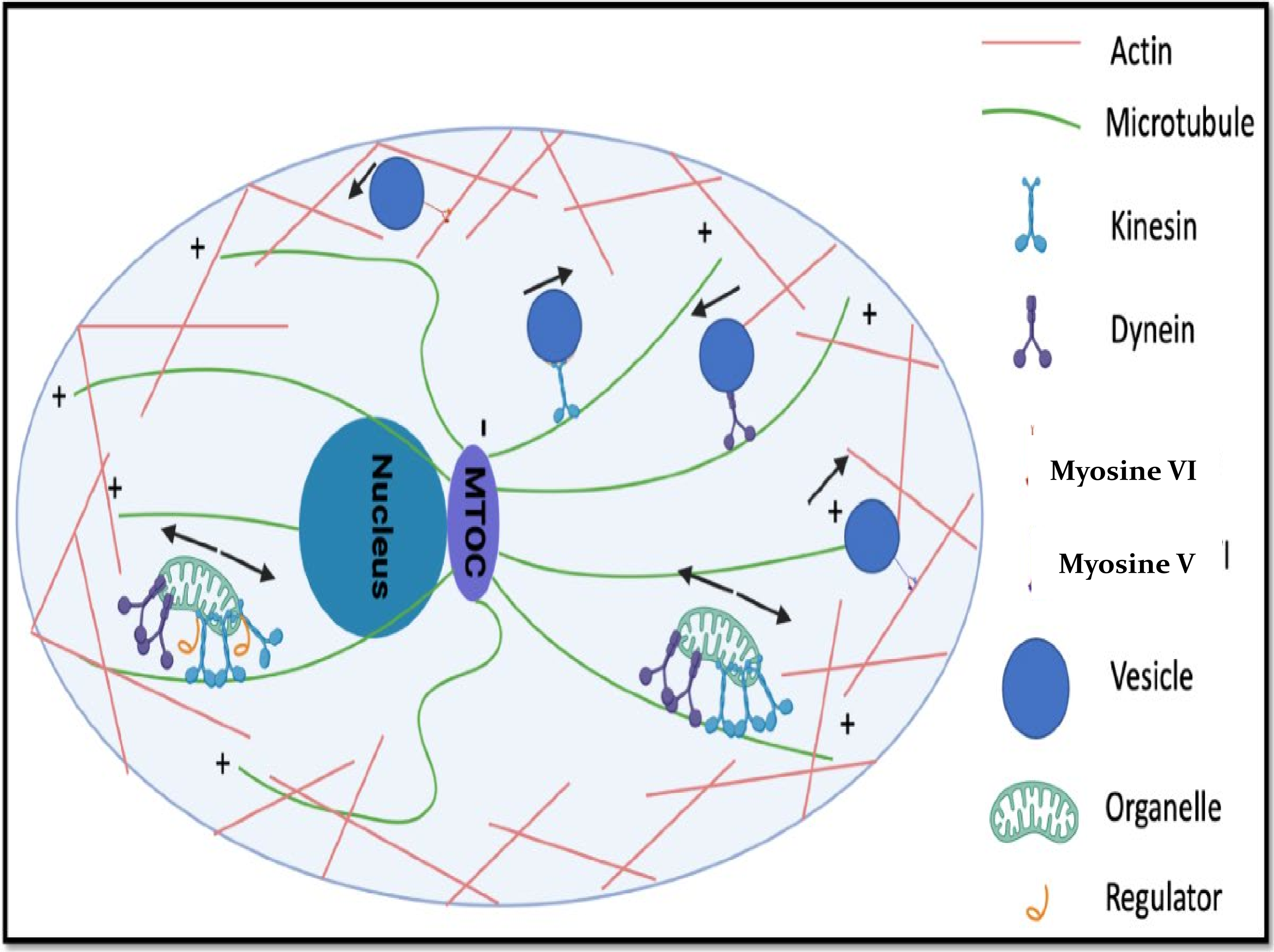

Vesicle
Kinesin
ATP
ADP + $P_i$
Microtubule
Direction of movement
−
+
Direction of movement
Dynein
ADP + $P_i$
ATP
Vesicle

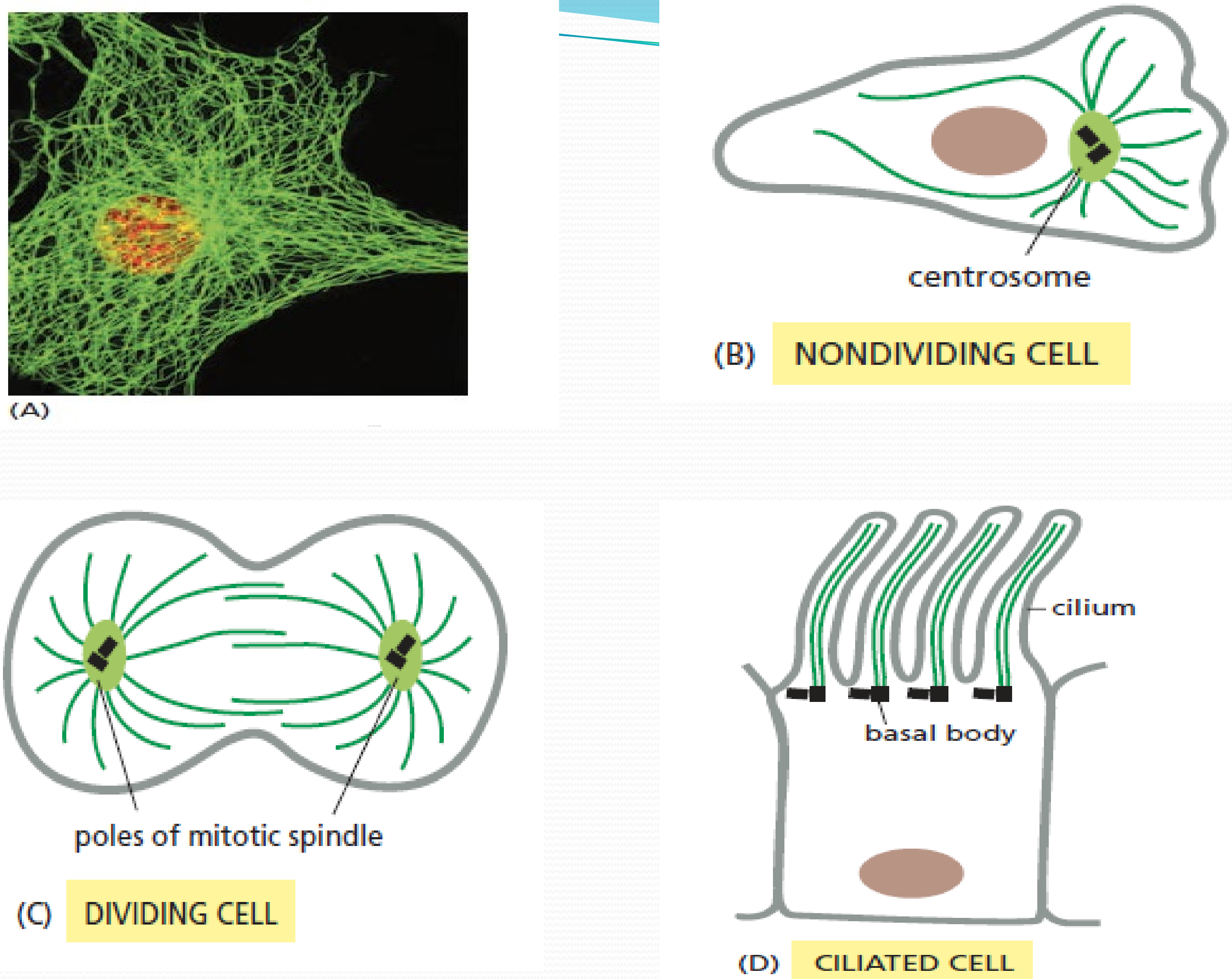


Mitochondria
Kinesin
Anterograde transport
(away from cell body)
Microtubule
Dynein
Retrograde transport
(toward cell body)

# Déplacement d'une kinésine
# Motility of kinesin

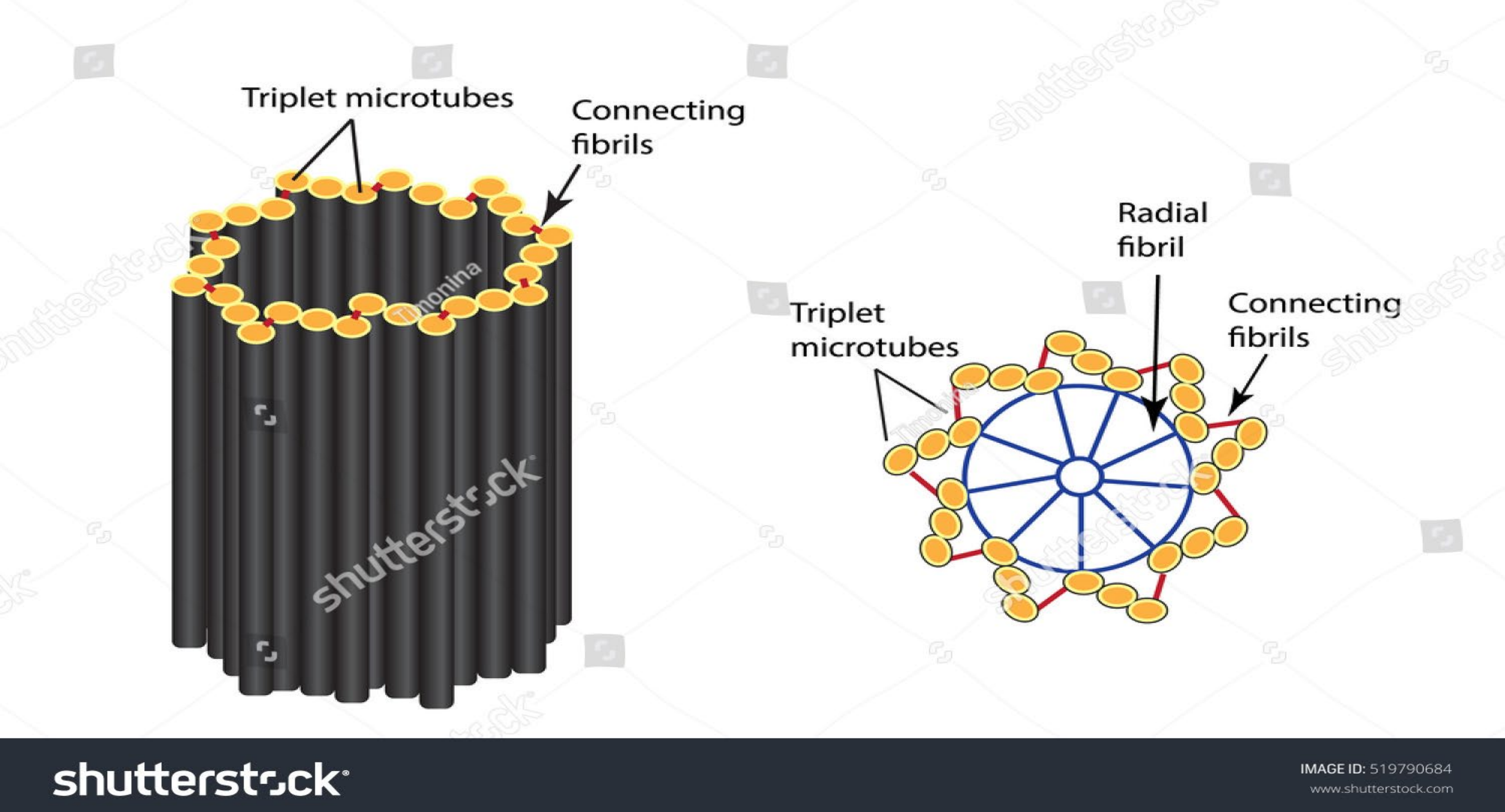


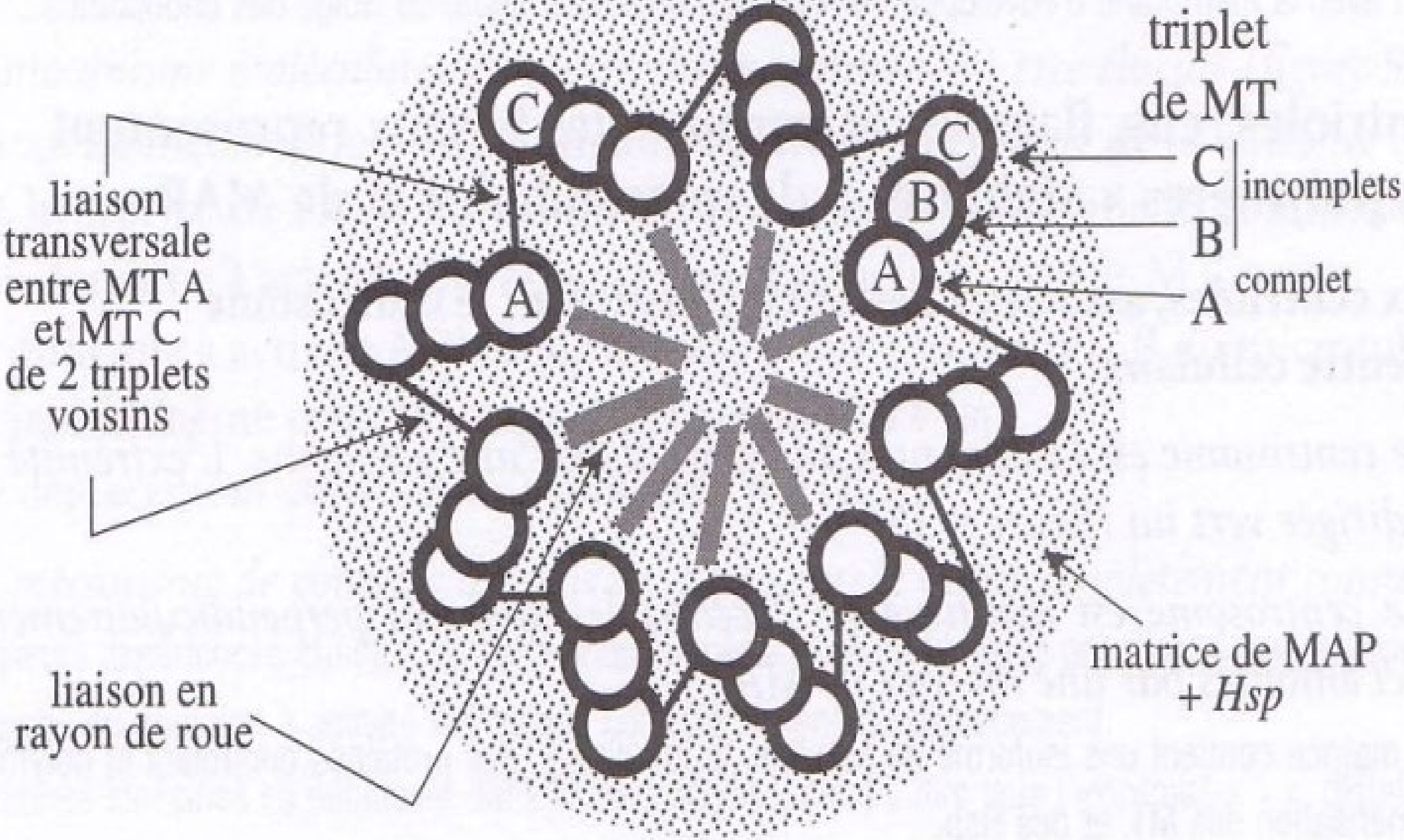

Actin
Microtubule
Kinesin
Dynein
Myosine VI
Myosine V
Vesicle
Organelle
Regulator
Nucleus
MTOC


(A)
centrosome
(B) NONDIVIDING CELL
poles of mitotic spindle
(C) DIVIDING CELL
cilium
basal body
(D) CILIATED CELL

# Stable microtubules

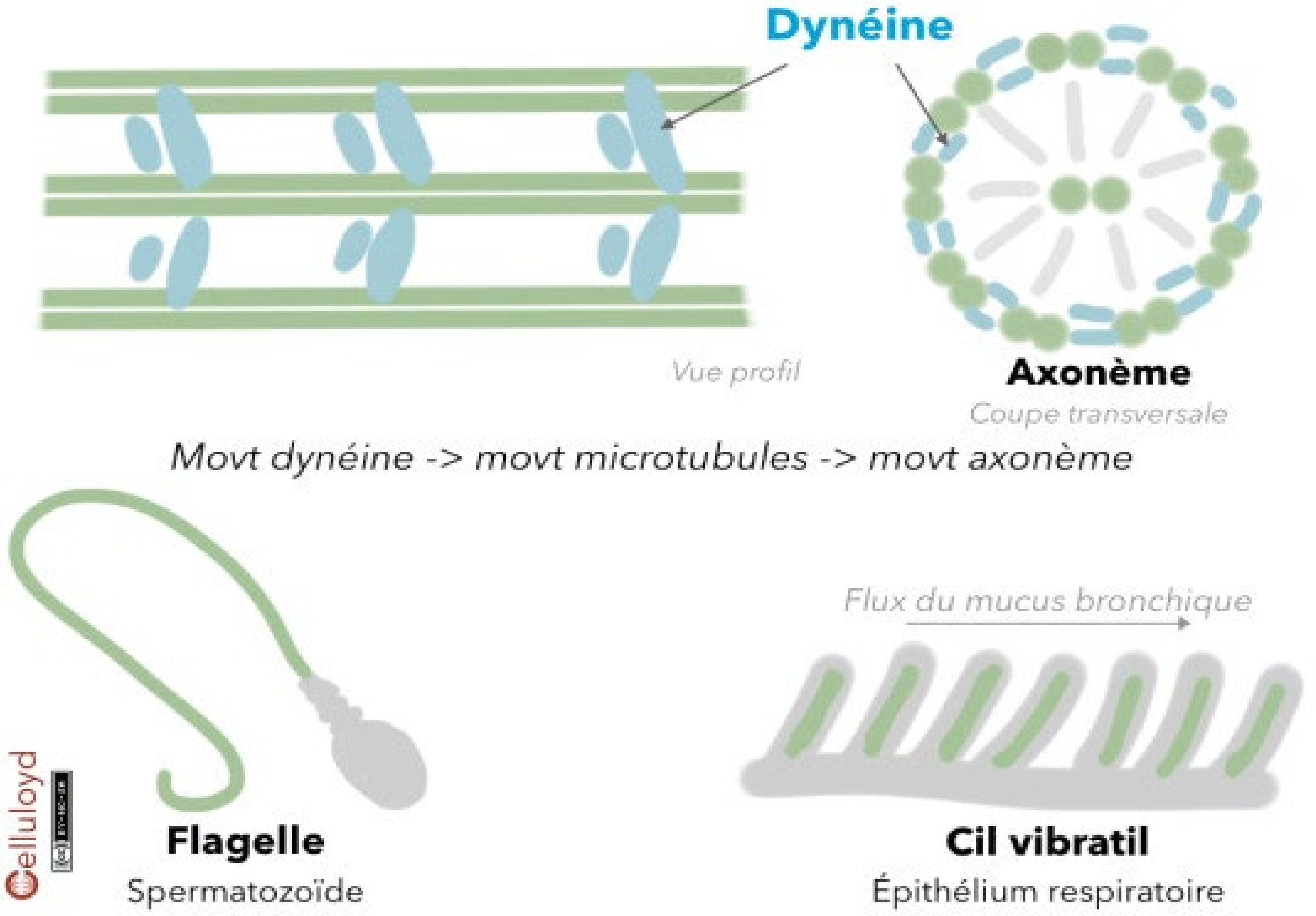
## 1. Centriols






# Cross section of a centriole

# 2. Cillium and flagellum



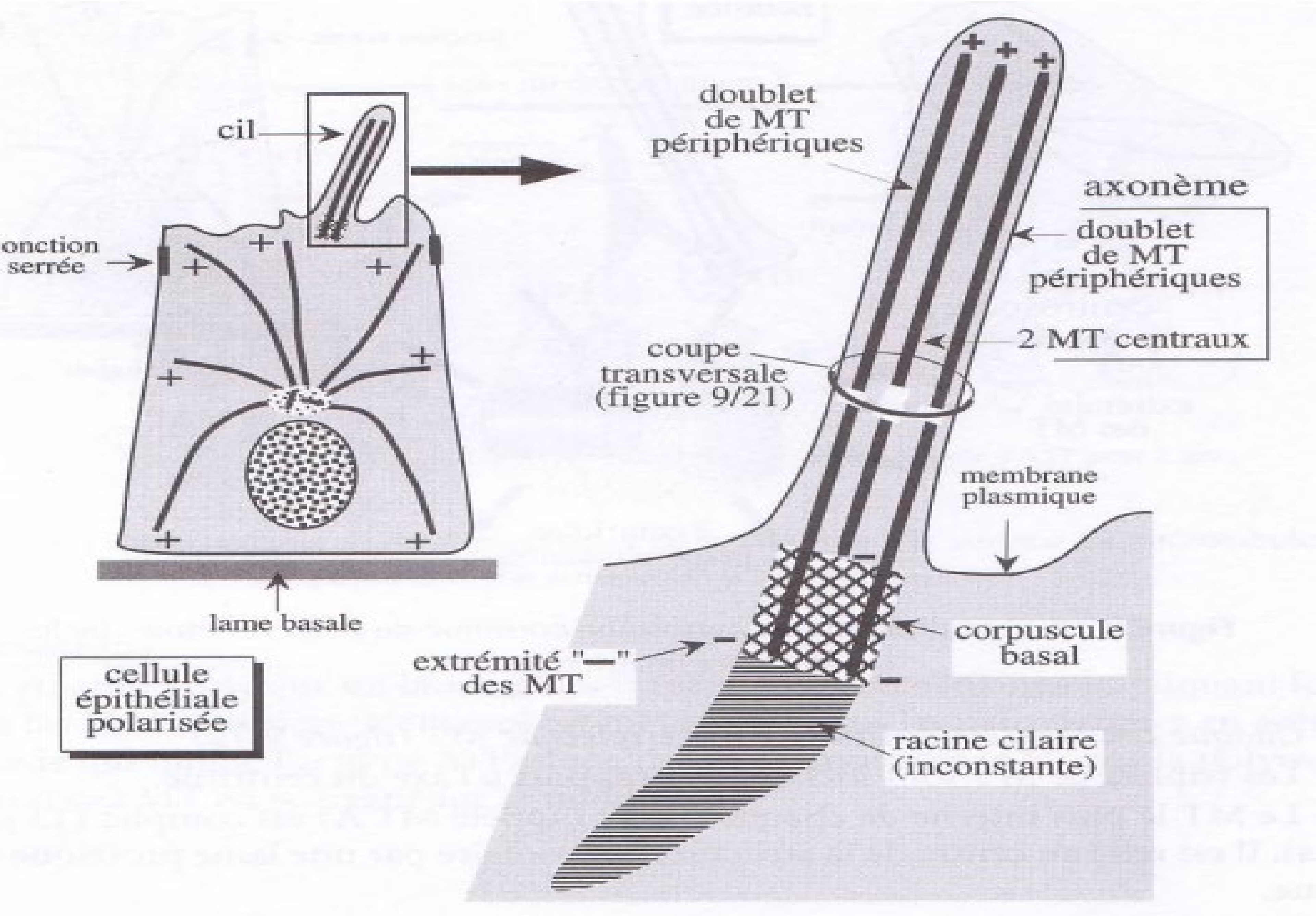
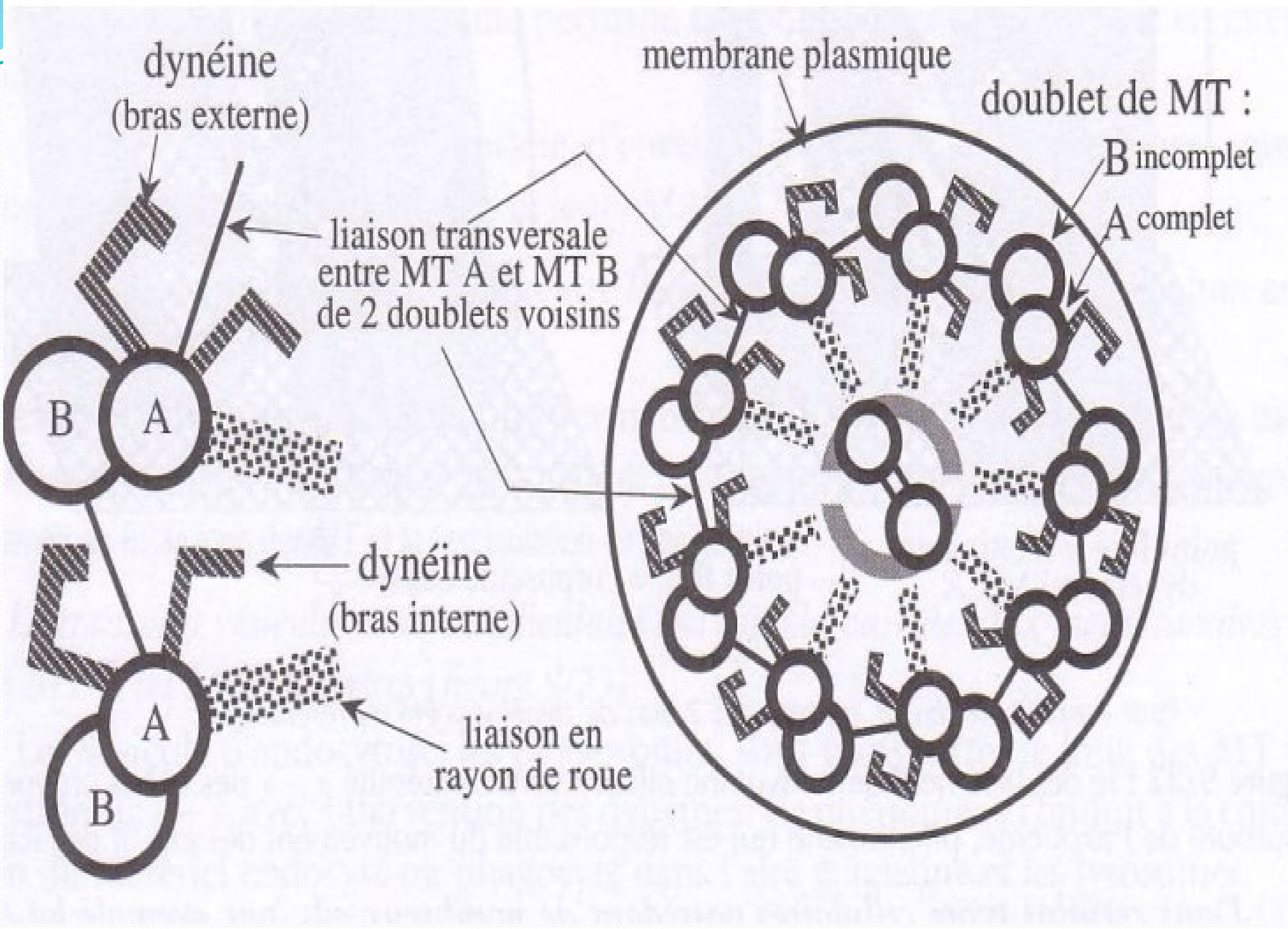
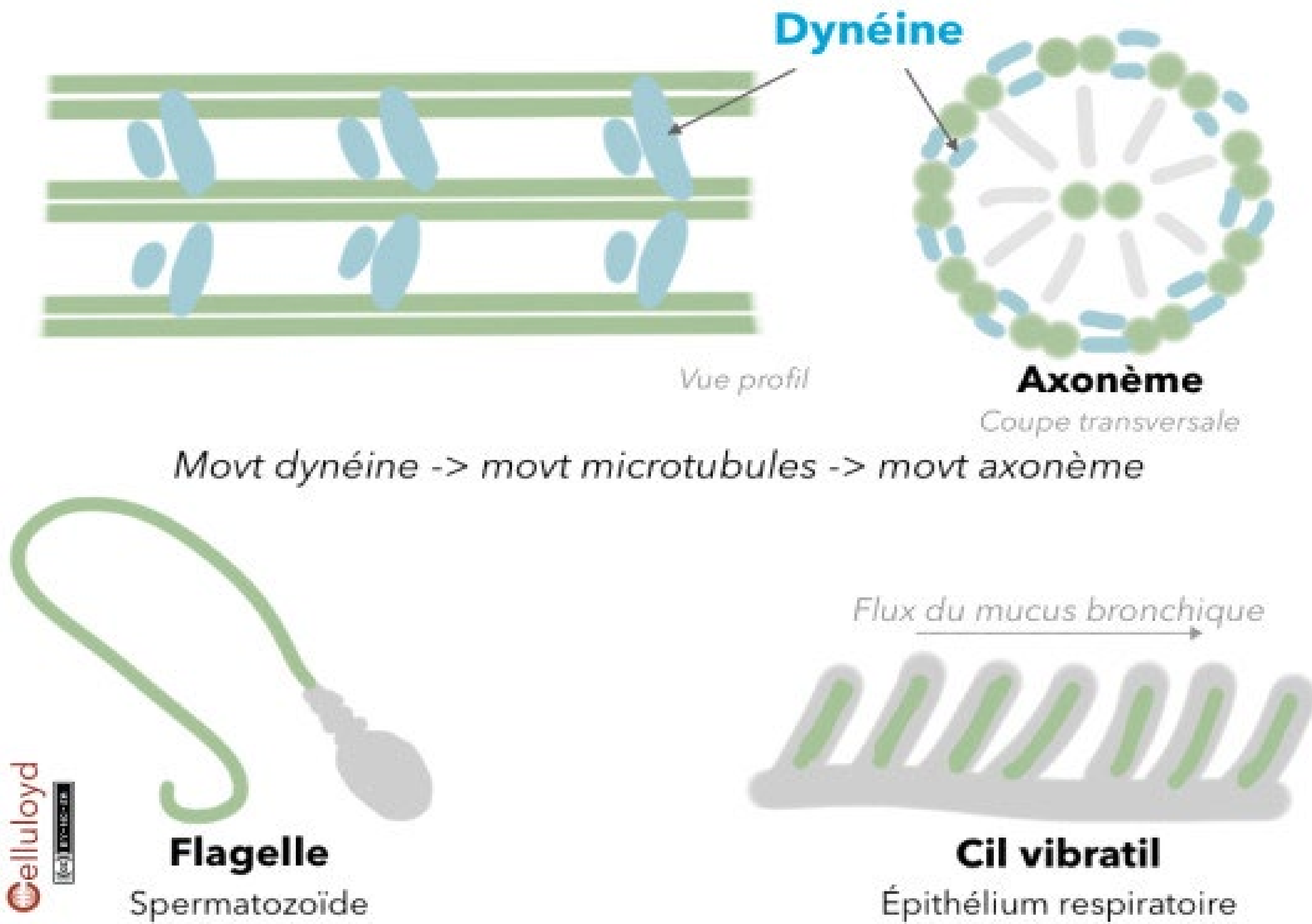
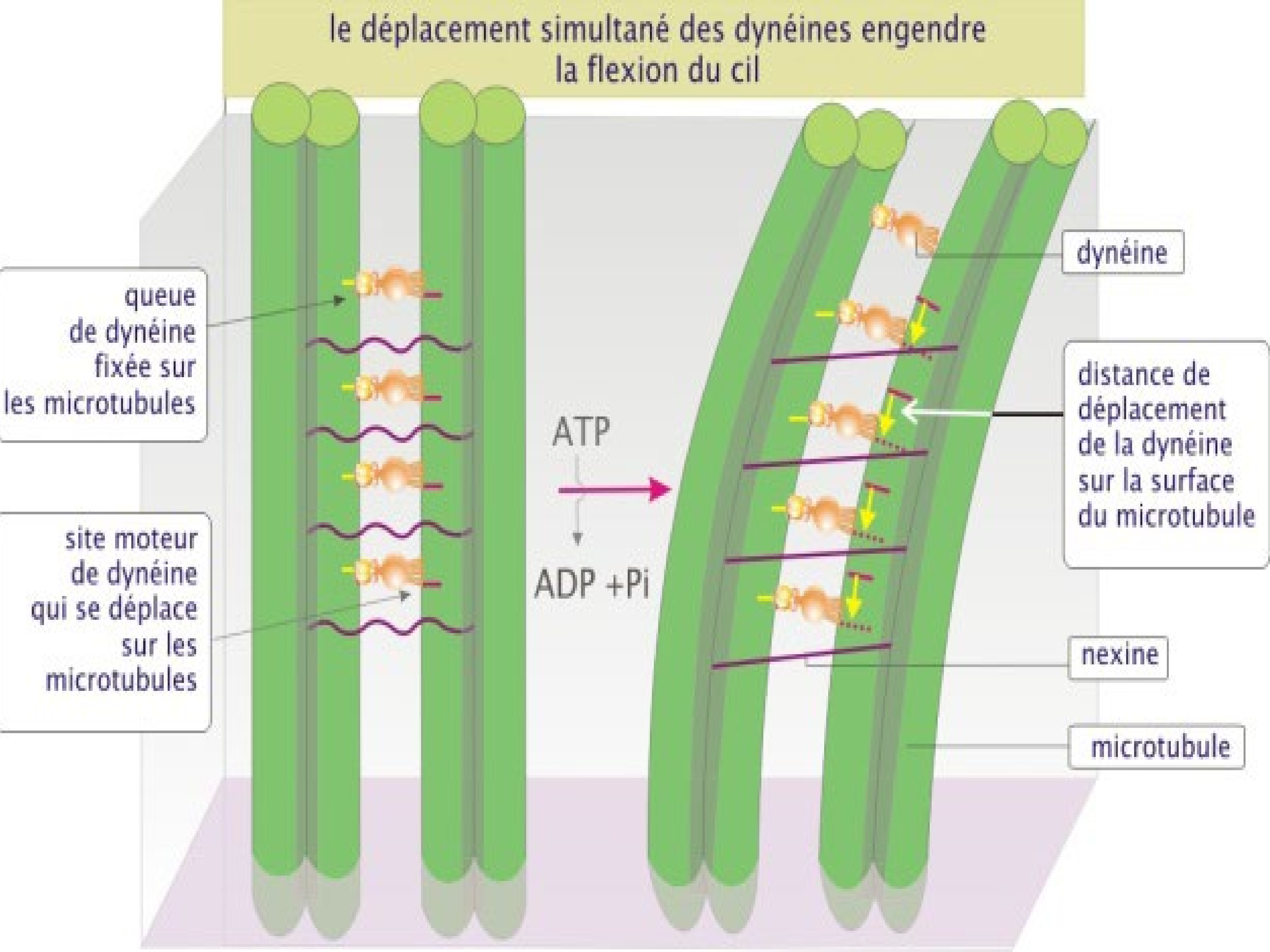



**Figure 9/20 :** le cil


dynéine
(bras externe)
membrane plasmique
doublet de MT :
B incomplet
A complet
liaison transversale
entre MT A et MT B
de 2 doublets voisins
B
A
dynéine
(bras interne)
A
B
liaison en
rayon de roue

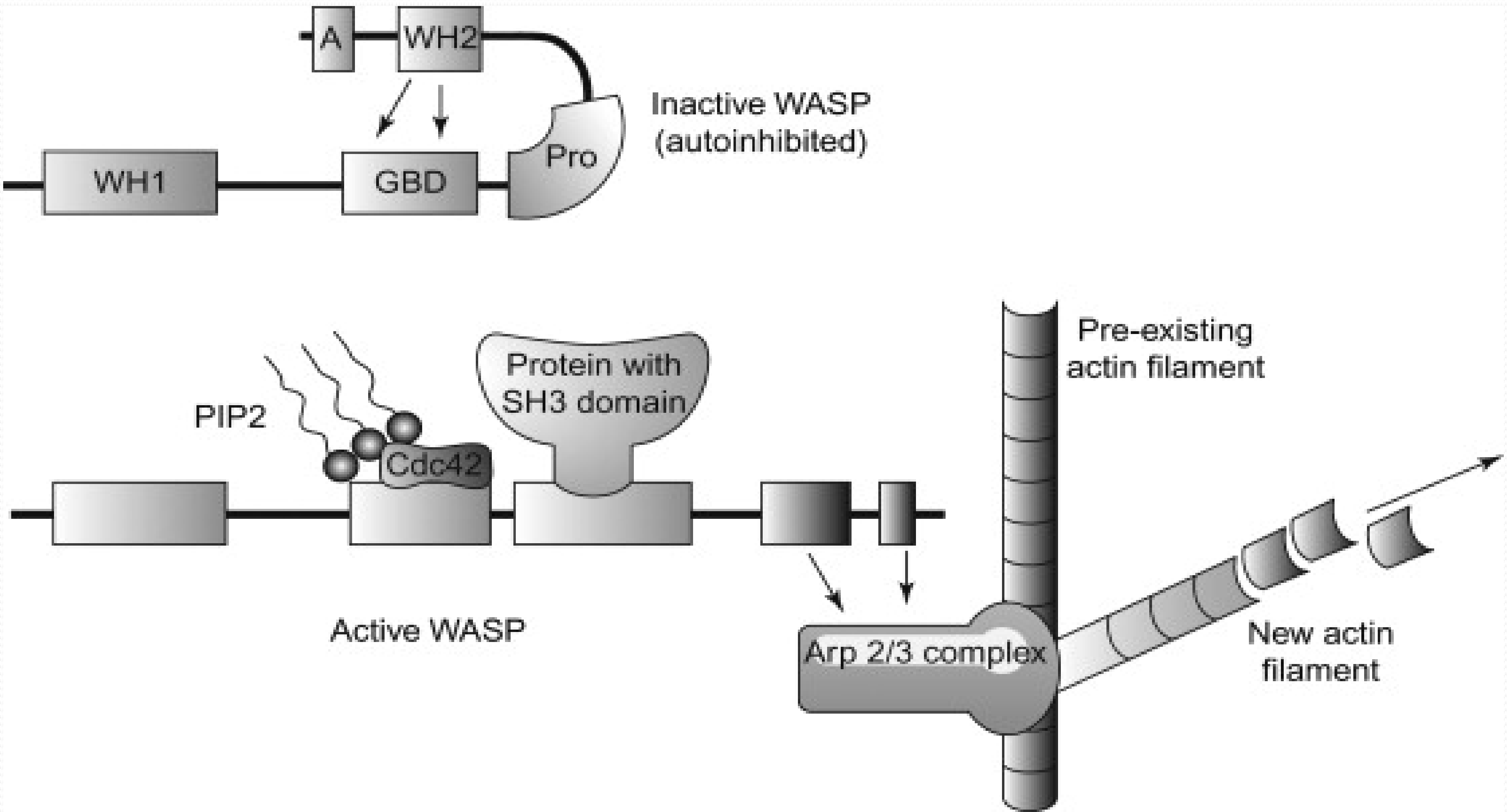

Dynéine
Vue profil
Axonème
Coupe transversale
Movt dynéine -> movt microtubules -> movt axonème
Flux du mucus bronchique
Flagelle
Spermatozoïde
Cil vibratil
Épithélium respiratoire
Celluloyd


le déplacement simultané des dynéines engendre la flexion du cil
queue de dynéine fixée sur les microtubules
site moteur de dynéine qui se déplace sur les microtubules
ATP
ADP +Pi
dynéine
distance de déplacement de la dynéine sur la surface du microtubule
nexine
microtubule

# Some examples of pathologies related to cytoskeleton dysfunction

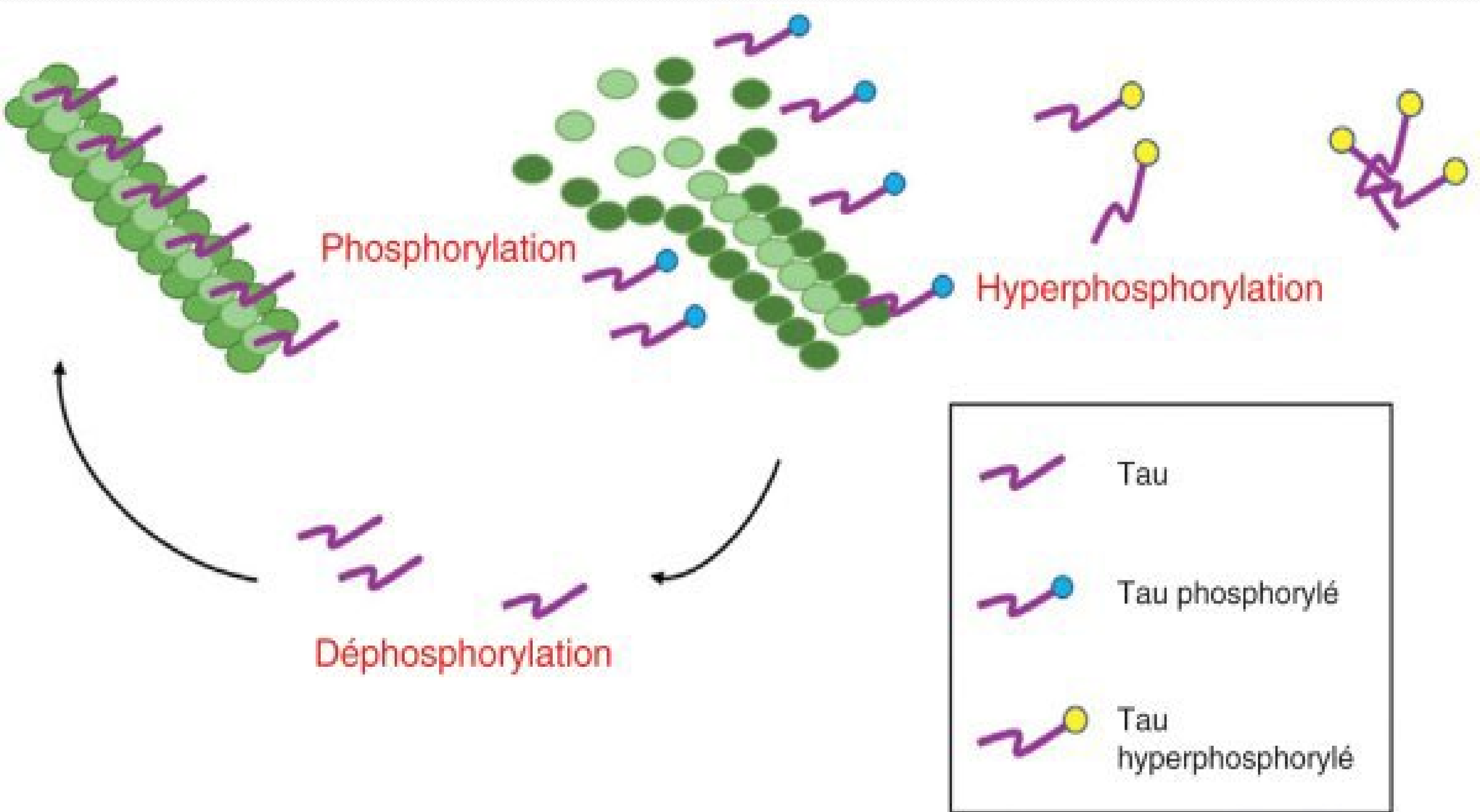
# 1) Pathologies associated with actin dysfuntion

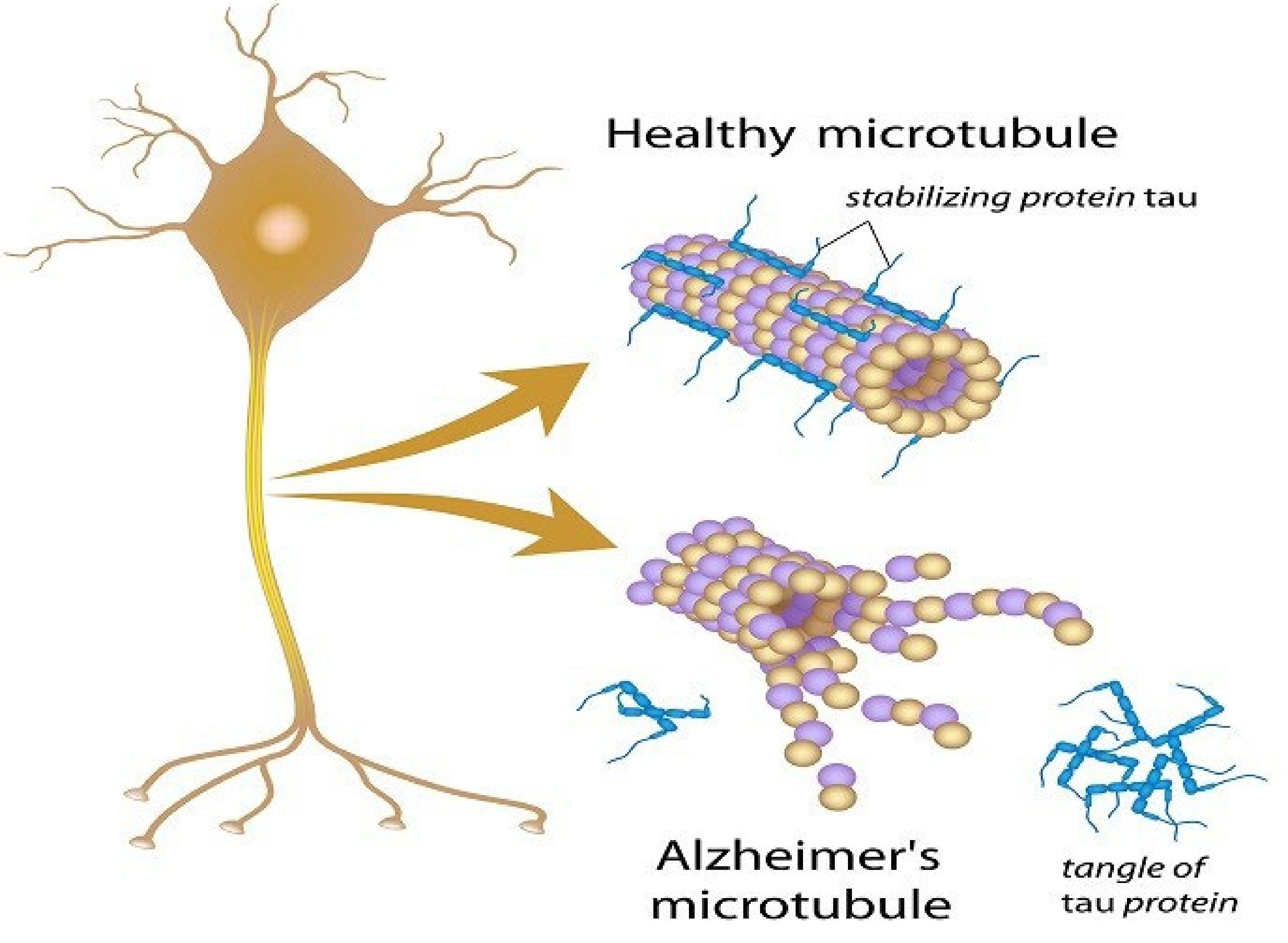
## Exp: Wiskott-Aldrich Syndrome (WAS)



# 2) Pathologies Associated with Microtubule Dysfunction

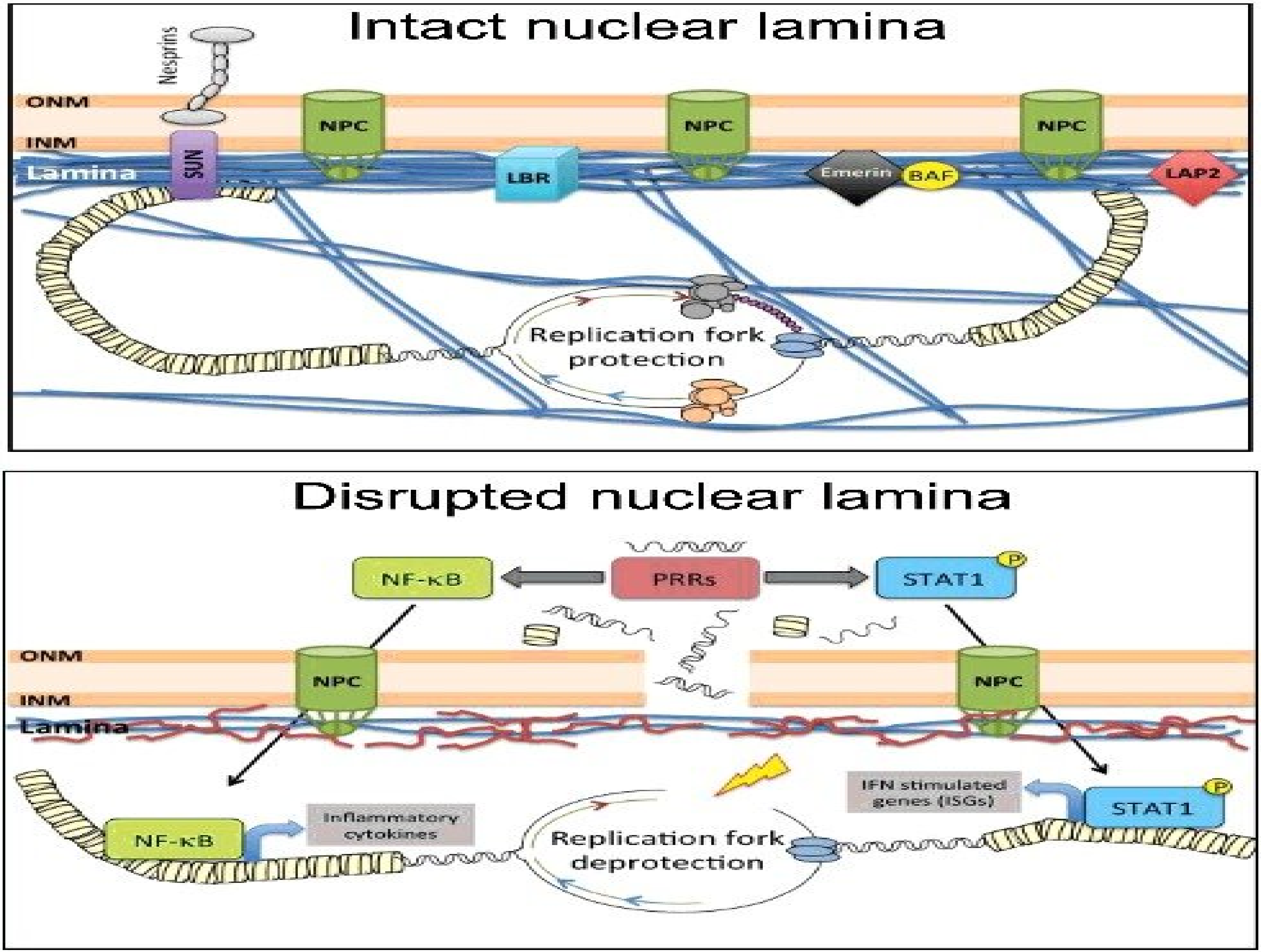
## Exp: Neuro-degenerative diseases (Tauopathies)



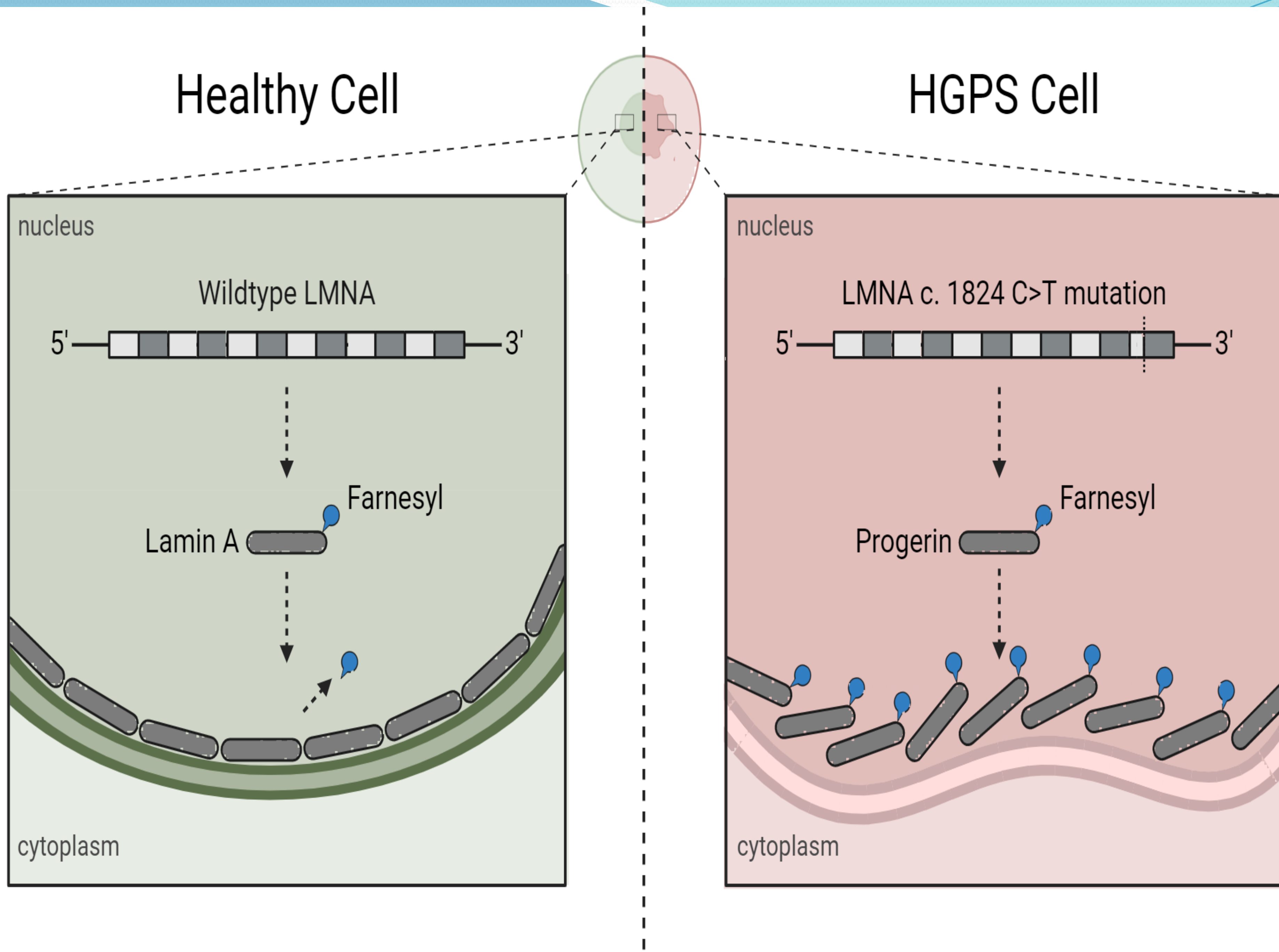

Healthy microtubule
stabilizing protein tau
Alzheimer's
microtubule
tangle of
tau protein

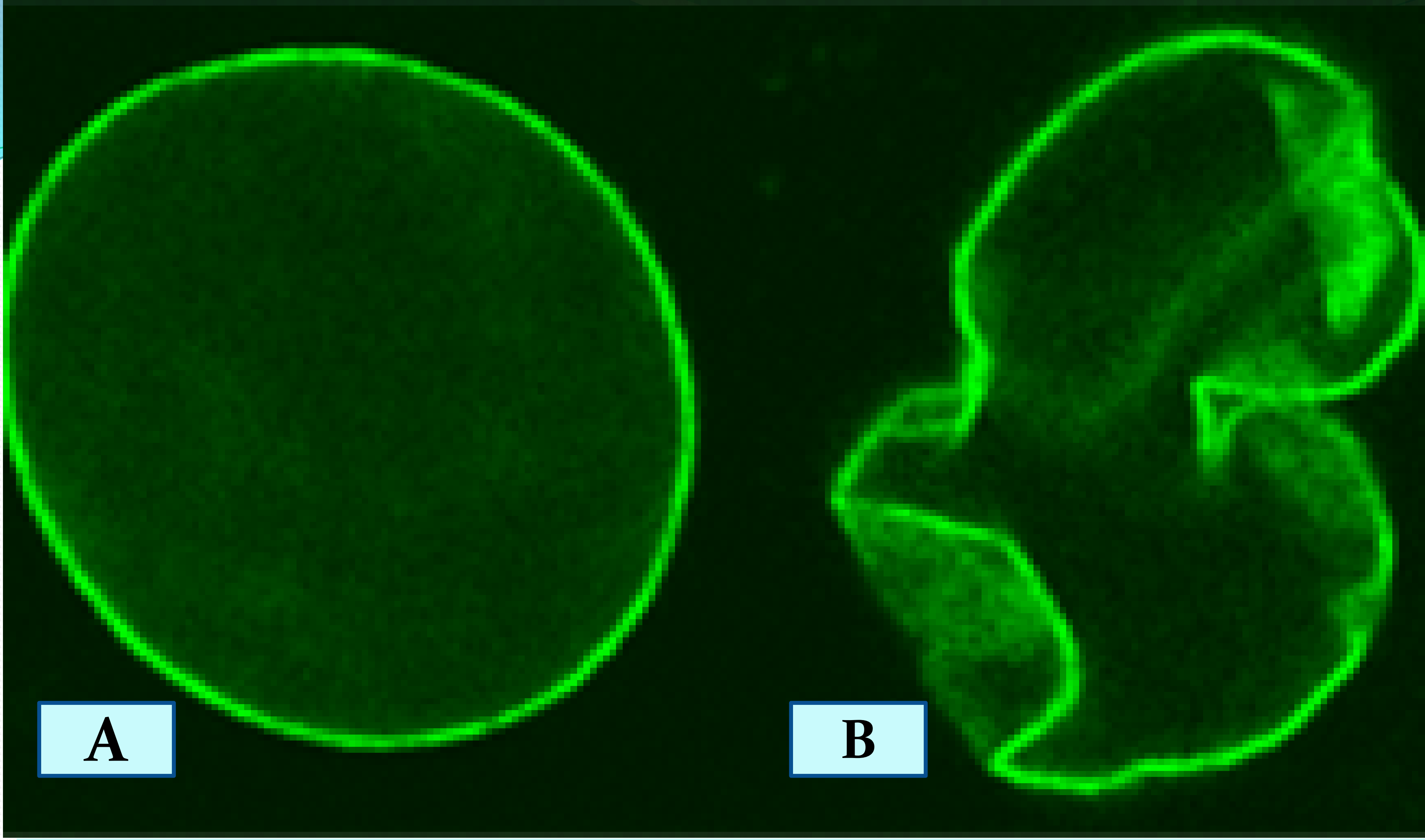
# 3) Pathologies associated with IF dysfonction

**Exp: Progeria or Hutchinson-Gilford Progeria Syndrome (HGPS): Laminopathy**



Intact nuclear lamina
Nesprins
ONM
INM
Lamina
SUN
NPC
NPC
NPC
LBR
Emerin
BAF
LAP2
Replication fork
protection
Disrupted nuclear lamina
NF-κB
PRRs
STAT1
P
ONM
INM
Lamina
NPC
NPC
NF-κB
Inflammatory
cytokines
Replication fork
deprotection
IFN stimulated
genes (ISGs)
STAT1
P


Healthy Cell
HGPS Cell
nucleus
Wildtype LMNA
5'
3'
Farnesyl
Lamin A
cytoplasm
nucleus
LMNA c. 1824 C>T mutation
5'
3'
Farnesyl
Progerin
cytoplasm




**Comparative analysis of nuclear morphology between a normal cell (A) and a Progeria-affected cell (B).**